



INTEGRATED SCIENCE

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8
Class

FOCUS AREAS OF NEP 2020



Created with the new dimension of joyful learning based on **National Education Policy, 2020**.



Each chapter begins with **Put into Focus** includes concept



Key Words is a list of difficult words to help learners to understand the topics.



Cognitive Skills gives descriptive writing tasks to develop the written skills of the learners.



Critical Thinking is the ability to examine learner's thoughts and feelings to make decisions based on their life.



Observing Skills includes learning process whereby students learn with their own senses.





Communication Skills includes discussion which is generally a face-to-face interaction promotes confidence level.



Linking Chain
increases the ability
of students to excel
in two or more
different
subjects.



Activity includes varied activities in which learners construct and demonstrate understanding through an art form.



Computational Thinking provides learners problem solving technique.



Project work is the ability to do effective work together with others by doing some experiments.





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PREFACE

Science is an integral part of our lives. It is, therefore imperative that we help our children to learn the importance of science in such a way that it reflects through their attitude. This series '**Integrated Science**' aims to achieve the objective and to fulfil the guidelines of National Education Policy 2020. The series has been written in such a way as to kindle a genuine interest and motivation in a child to develop a scientific attitude and problem solving abilities from an early age.

This series focuses on the development of cognitive, effective and psychomotor skills through process based, activity-oriented learning, rather than through rote memorization of content-efforts have been made to help children in learning without burdening them, through simple text, child-friendly illustrations, pictures and ample activities. The aim is to encourage them to observe, interact, explore the world around them and learn by doing. Based on the experience gained through exploration, children are further guided and encouraged to **construct knowledge** in order to strengthen the foundation of their learning as well as develop language skills.

Wherever possible, the books of this series instil into children '**Values**' such as the importance of conserving nature, keeping health etc.

Microorganisms

1

- Classification of micro organisms
- Harmful microorganisms
- Where do microorganisms live
- Food preservation
- Useful microorganisms

Small organisms can only be seen through an instrument called **microscope**. The organisms that are visible only through a microscope or cannot be seen with naked eyes are called **microorganisms**.

CLASSIFICATION OF MICROORGANISMS

There are mainly four major groups of microorganisms. They are :

1. Bacteria
2. Fungi
3. Protozoa
4. Algae

Viruses are also considered as microorganisms. They are considered different from the other microorganisms as they reproduce only in the body of the host they reside in, like plants, animals, bacteria, etc.

1. Bacteria

Bacteria are unicellular microorganisms. They are found in mainly four types of shapes. They are:

Rod-shaped (bacilli), spherical shaped (cocci), curved shaped (commas), and spiral shaped (spirilla)



Rod shaped
bacteria



Spherical shaped
bacteria



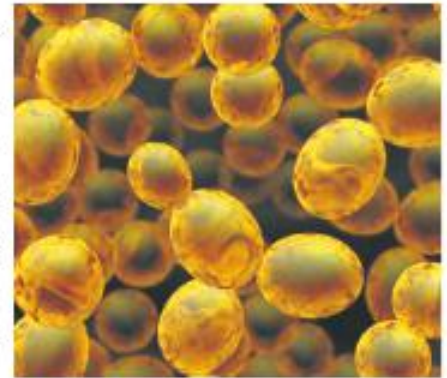
Curved shaped
bacteria



Spiral shaped
bacteria

2. Fungi

They are long thread-like unicellular or multicellular microorganisms. The threads of the multicellular fungi are called **hyphae**. A fungus is neither a plant nor an animal. It is similar to a plant, but it has no chlorophyll and cannot make its own food as a plant can through photosynthesis. The two main classes of fungi are saprophyte and parasite. The saprophytes feed on decaying organic matter, and the parasites feed on/in the body of the living organisms. For example:



Unicellular fungus–Yeast

Multicellular fungi–**penicillium**, **aspergillus**, bread mould, and mushrooms, etc.



(a) Penicillium



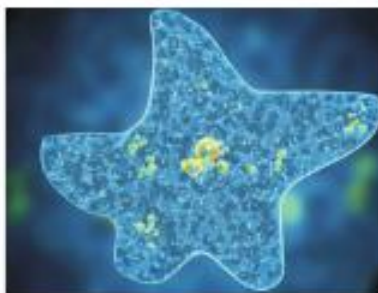
(b) Bread mould



(c) Mushroom

3. Protozoa

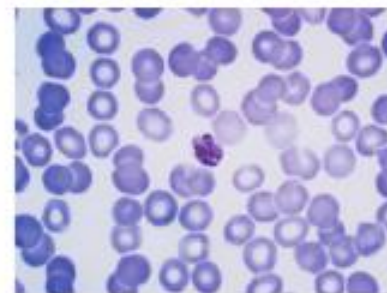
They are unicellular organisms having a size ranging from 2-200 microns. They are heterotrophic in nature, and thus cannot prepare their own food. For example: **amoeba**, **paramecium**, **plasmodium**, etc.



Amoeba



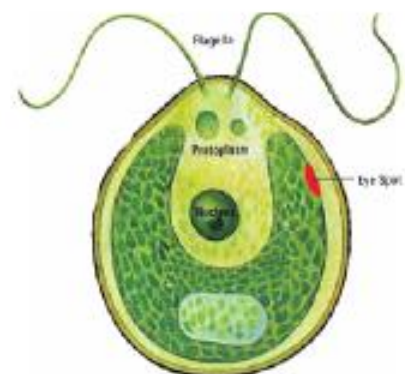
Paramecium



Plasmodium

4. Algae

Algae are aquatic and photosynthetic microorganisms. They can prepare their own food, so they come under the category of autotrophic organisms. They may be unicellular or multicellular. For example: unicellular algae–**chlamydomonas** and multicellular algae **euglena**, **blue green algae**, etc.





Euglena



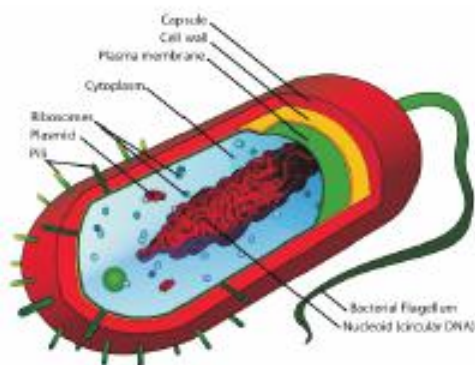
Blue green algae

Viruses

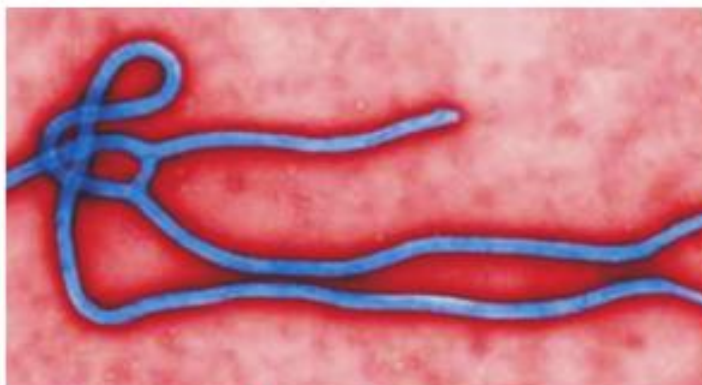
Viruses are the smallest microscopic organisms. As they do not come under the category of cellular organisms, they do not have cell organelles. There are four types of viruses :

1. Plant viruses
2. Insect viruses
3. Bacterial viruses and
4. Mammalian viruses

They are made up of genetic material surrounded by a protein coat. Viruses cannot reproduce on their own. For this, they need to enter a living host cell. Outside the host cell, they do not show any characteristic of living organisms. Common diseases like cold, cough, and influenza (flu) are caused by viruses. Serious diseases like polio, corona and chicken pox are also caused by viruses.



(a) Bacterial virus



(b) Mammalian virus

FACT

Our mouth is inhabited by 500 to 1000 different types of bacteria, 70 to 80 different types of fungi and a few viruses and parasites.

WHERE DO MICROORGANISMS LIVE?

Microorganisms can survive under all types of environment, ranging from icy cold climate to hot springs and deserts to marshy lands. They are also found inside the bodies of animals including humans. Some microorganisms grow on other organisms, while others exist freely. Some are also found in dead and decaying organic matter. Microorganisms like amoeba can live alone; while fungi and bacteria are mostly found in colonies.

USEFUL MICROORGANISM

Some of the uses of microorganisms are as follows:

1. Food Production

Curdling of Milk

The process of the formation of curd from milk is called **curdling of milk**. Microorganisms such as **bacterium lactobacillus** grow in milk and convert it into curd. During the growth, the bacterium lactobacillus produces an acid called lactic acid from the lactose present in the milk.



Activity

Aim: To observe the effect of temperature on curdling of milk.

Materials required: Three bowls, three lids, some milk, a teaspoon and some curd

Procedure: Take three bowls and mark them A, B and C. In the bowl A, pour warm milk. In the bowl B, pour boiling milk and pour cold milk in the bowl C.

Put a teaspoon of curd in all the three bowls and cover them with lids. Observe the changes after 5 to 6 hours.

Observation: Mention your observations against each bowl. Also mention in which bowl curd formation is better and faster and why?

Bowl A	_____
Bowl B	_____
Bowl C	_____

Conclusion: Curd formation is better and faster at suitable temperatures.

Making Cheese

Dairy products like cheese and paneer are prepared by using bacteria like **lactobacillus** and **streptococcus**. A substance called rennet is also added to make the process faster. **Rennet** is a substance usually obtained from the stomach lining of young cattle. Rennet aids in the formation of lactic acid, which makes the milk more acidic.

Making Bread

Yeast is also used in the process of making bread, cakes, pastries and other similar food products.

Have you ever seen the tiny holes on idli and dosa? Such holes are created by the escape of carbon dioxide from the dough from which they are prepared. Yeast is the microorganism which is used to create the puffed-up dough for such items. Yeast's cells release carbon dioxide that actually aid in raising the dough.

Fermentation of Food Products

Some food items such as fish, soyabean, bamboo shoots, etc are fermented and eaten in some countries. Such fermentation is done with the help of the microorganisms.

2. Commercial Use

Fermentation of Natural Sugar to Produce Alcohol

The process of converting sugar into alcohol in the absence of oxygen is called **fermentation**.

Yeast is used for the large scale production of alcohol, wine and beer. For this purpose, yeast is grown on natural sugars present in grains like barley, wheat, rice, and crushed fruit juices, etc. In the absence of oxygen, the yeast cells grow and convert the natural sugar present in grains and fruits into alcohol (ethanol).

Production of Citric Acid and Vinegar

The citric acid which is the most commonly used in the soft drink industries is produced with the help of the **fungus aspergillus**.

Vinegar, commonly known as acetic acid, is produced by the action of aerobic bacteria like **acetobacter** on alcohol.

3. Medicinal Use

Preparation of Antibiotics

The medicines that kill or stop the growth of the disease-causing microorganisms and are produced by microorganisms like bacteria or fungi are called **antibiotics**.

Streptomycin, **tetracycline** and **erythromycin** are some of the commonly known antibiotics which are made from fungi and bacteria. The antibiotics are manufactured by growing specific microorganisms and are used to cure a variety of diseases.

Vaccine

A substance used to produce immunity against diseases in the living body is called a **vaccine**. **Vaccines** are prepared by using the dead or weakened disease causing microbes. Immunity is the ability of the body to fight against diseases. Immunity can be natural or artificial (in form of vaccines). The process of producing immunity by injecting a vaccine into the body is called **vaccination**.

When a disease-carrying microbe enters our body, the body produces a substance called **antibodies**. The antibodies fight and destroy the invader (disease carrying microbe). Several diseases, including cholera, tuberculosis, smallpox and hepatitis can be prevented by vaccination.

4. Sources of Food

Algae and Sea Weeds

Chlorella and many sea weeds are used as food supplements in many countries. They are also the direct source of food for the aquatic animals.

Yeasts

Mushroom is the fleshy, spore-bearing fruiting body of a fungus and rich source of proteins and vitamins. Other yeasts are rich in proteins and vitamins, and are used for the preparation of nutrient supplements.

5. Commensals

Microorganisms which reside inside our body and are beneficial to us are called **commensals**. For example: bacteria such as *Lactobacillus acidophilus* lives in human intestines where it helps in digesting the food we eat, as well as helps in destroying disease-causing microbes too.

6. Agricultural Use

In the chapter 1, you have read about the process of nitrogen fixation. This process involves the fixation of atmospheric nitrogen into nitrogen compounds that are suitable for the plants. Bacteria living in the root nodules of the leguminous plants such as bean, pea and gram help in the process of nitrogen fixation. They are also called **biological nitrogen fixers**. This process helps in increasing the fertility of soil and enriching the soil with nitrogen.

7. Environmental Use

Cleaning the Environment

Some microorganisms like bacteria and fungi help in recycling of waste materials (Organic wastes like dead plants and animals, and faeces) present in the environment. This is done by the process of decomposition which means breaking down of the organic material into simpler organic substances. The microbes that help in the process of decomposition are called **decomposers**.



It should be noted that the non-biodegradable wastes like plastic, tin, polythene, etc are not degraded by the microorganisms. So, they should be disposed separately.

Sewage Treatment

Bacteria are widely used these days for treatment of sewage. Sewage consists of organic wastes like household wastes, human and animal excreta, factory wastes, etc. Sewage is treated in sewage treatment plants in order to make it free from any harmful substances. This treated sewage is then allowed to flow in rivers. During the sewage treatment, a gas called **biogas** is produced which is commonly used as a fuel. This fuel is renewable as well as eco-friendly in nature.

HARMFUL MICROORGANISMS

Microorganisms are also harmful in many ways. Some of the microorganisms cause diseases in human beings, plants, and animals, while others spoil food, clothes and leather.

1. Disease-causing Microorganisms in Humans

Microorganisms that cause diseases are called **pathogens**. The pathogens enter our body through the air we breathe in, the water we drink or the food we eat. Microbial diseases that can spread from an infected person to a healthy person through air, water, food or physical contact are called **communicable diseases**. Examples of such diseases include cholera, common cold, chickenpox, corona and tuberculosis.

Transmission of the pathogens takes place in the following ways:

- Direct contact with an infected person.
- Insects and animals act as the carriers of pathogens. When insects like houseflies sit on the garbage and animal excreta, pathogens stick to their bodies. When these flies sit on uncovered food, they transfer the pathogens onto the food. Whoever eats the contaminated food is likely to fall ill. Another example of a carrier of pathogens is the female Anopheles mosquito which carries the pathogens of malaria.
- Pathogens are released in the air while coughing.

Table 2.1: Some common human diseases caused by pathogens:

Human Diseases	Causative Pathogen	Mode of Transmission
Tuberculosis	Bacterium	Air
Measles	Virus	Air
Chickenpox	Virus	Air and physical contact with the infected person
Polio	Virus	Air and water
Cholera	Bacterium	Contaminated food and water
Typhoid	Bacterium	Contaminated food

Hepatitis A	Virus	Water
Malaria	Protozoan	Bite of female Anopheles mosquito

Preventive Measures

1. Keep the patient and his belongings away from others' contact.
2. Vaccination should be administered at the suitable age.
3. Drinking water should be boiled.
4. Avoid eating uncovered foodstuff.
5. Maintain personal hygiene and good sanitation habits.
6. Use mosquito nets and repellents.
7. Spray insecticides and control breeding of mosquitoes by not allowing the mosquitoes to collect in the surroundings.

2. Disease-causing Microorganisms in Animals

Some microorganisms are responsible for diseases among animals. Some animal diseases caused by pathogens are listed in Table 2.2.

Table 2.2: Some animal diseases caused by pathogens

Diseases	Affected Animal	Causative Pathogen	Mode of Transmission
Anthrax	Cattle	Bacterium	Grazing
Fin rot	Fish	Bacterium / Fungus	Contaminated water / Injury
Foot and mouth diseases	Cattle	Virus	Air and contact with diseased animal
Rinderpest	Cattle	Virus	Air, contaminated water and contact with diseased animal

3. Disease-causing Microorganisms in Plants

Plants too can suffer from different diseases caused by various pathogens. These diseases reduce the yield of crops. They can be controlled by the use of certain chemicals which kill the microbes. Some of the common plant diseases caused by pathogens are listed in table 2.3.

Table 2.3: Some plant diseases caused by pathogens

Plant Diseases	Affected Plant	Causative Pathogen
Root rots	All types of plants	Fungi
Wilt	Potato, Alfalfa, trees	Fungi
Blight	Fruit trees, vegetable crops, etc.	Bacteria
Leaf spot	Beans, peas, cotton, etc.	Bacteria
Yellowing of leaf	Barley, potato and sugar beet	Viruses



Root rots



Wilt



Blight



Leaf spot



Yellowing of leaf

4. Food Spoilage

Food spoilage is the process in which food deteriorates to the point at which it is not edible for humans. Some reasons of food spoilage are:

- Oxidation of food items by air.
- Decomposition by microorganisms like bacteria and fungi.
- Moderately warm temperature
- Moisture
- Air

Signs of Spoilage

Some of the indicators that show that the food has been spoilt by the microorganisms are:

- Foul smell (e.g. cooked food in hot weather smells bad if kept outside the refrigerator for a long time)
- Bad taste or sourness (e.g. milk tastes sour after being spoiled due to the formation of lactic acid by the microorganisms)
- Change in colour (e.g. blue and green moulds on spoilt orange)

Consequences of Consuming Spoilt Food

Microorganisms that grow on food sometimes produce toxic substances. These toxic substances make the food unfit for consumption. Consuming such food items causes an illness

called **food poisoning**. So, it is very important that we should preserve food to prevent it from being spoiled.

FOOD PRESERVATION

The process of preserving food from being spoiled by the microorganisms is called **food preservation**.

Food preservation methods involve the following techniques:

- Temperature control
- Moisture control
- Chemical usage
- Physical methods

In order to achieve a satisfied result, sometimes more than one method is applied to the food product.

Temperature Control

Low Temperature

Low temperature slows down the speed of enzyme reactions and also prevents the growth of most pathogens and microorganisms that spoil food.

Preservation methods under this category are:

(i) Refrigeration

It means storing the food products at the temperature ranging between 0°C to 5°C. Fruits are always stored in refrigerators to prevent them from being spoiled.

(ii) Freezing

It is termed as a technique of storing the food items at -18°C. There are a variety of commercial techniques to freeze food items.

(iii) High Temperature

Heat kills the microorganisms that cause food spoilage. Some of the heat processes also destroy spores.

Preservation methods that involve high temperature include:

(i) Pasteurisation

It is the method that involves the combination of time and temperature which is suitable to destroy active pathogens and spoilage bacteria in a particular food product, e.g., milk is heated to 70 degree celsius for 15 seconds, and then suddenly chilled and stored.

(ii) Sterilisation

Sterilisation is termed as the process of destroying all the microorganisms at a temperature above 100°C. This method is often used for various food items.

(iii) Ultra-heat Treatment (UHT)

A higher temperature than pasteurisation is used in this method. Thus, it involves killing of all the bacteria present in the food items, e.g., UHT milk is produced using this method.

(iv) Ohmic heating

It involves use of high-voltage electric current passed through food to destroy microorganisms causing food spoilage.

Canning

This process also includes removal of oxygen and airtight sealing of containers to avoid post-process contamination. Drinks are usually canned to preserve the carbon dioxide that keeps them fizzy.

Moisture Control

The process of removing moisture is termed as **dehydration**. The growth of microorganisms stops when moisture is removed from the food.

Dehydration involves the following types of preservation methods :

- Traditional methods such as sun drying.
- Commercial methods using hot air including tunnel drying, roller drying and spray drying.
- Accelerated freeze-drying in which the material is frozen under high vacuum so that ice or other frozen solvent will quickly sublime and a porous solid remains uses warm air after food has been quickly frozen.

Chemical Usage

Substances used to check the microbial growth in food items are called **preservatives**.

Salt

Salt has a unique quality that makes water unavailable for microbial growth. Salting is often used to preserve amla, raw mangoes, tamarind, meat, fish etc.

Sugar

Sugar works in the same manner as salt. Food items like jellies, jams and squashes are preserved by adding sugar.

Oil and Vinegar

Oil and vinegar are used in order to lower down the pH below the normal growth range of most microorganisms, thus, microorganisms are not able to survive in such an environment. The method of preserving pickles in oil and vinegar is called **pickling**.

Physical Methods

Some such methods are:

Modified atmosphere packaging

The air around the food product is modified in such a manner that the nitrogen content increases in it and oxygen content reduces which ultimately slows down the growth of many microorganisms. Some fruits and vegetables are preserved by this method.

Vacuum Packing

Vacuum packing involves the removal of oxygen which inhibits the growth of aerobic bacteria but does not prevent anaerobes from growing.

Food Irradiation

In this process, ionising radiation is used to destroy microorganisms. Irradiated food is used for astronauts and patients possessing very weak immune system.

◀ REWIND

- Microorganisms are so small that they cannot be seen with naked eyes. They are classified into four major groups bacteria, fungi, protozoa and algae.
- Viruses are also considered as microorganisms. They reproduce only in the body of hosts like, plants, animals, bacteria, etc.
- Microorganisms can survive under all types of environment.
- Microorganisms are also found inside the bodies of animals including humans.
- Microorganisms are useful in food production, manufacturing medicines, agriculture, cleaning the environment and as commensals and food sources.
- Microorganisms are harmful in many ways. Some of the microorganisms cause diseases in human beings, plants and animals, while others spoil food and cause food poisoning.
- Most of the food preservation methods involve destroying of microorganisms present in the food or stopping them from growing again.
- Food preservation methods involve the techniques such as temperature control, moisture control, chemical usage and physical methods.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

- Which of the following look like plants but do not have chlorophyll?
(a) Fungi ☐ (b) Bacteria ☐ (c) Protozoa ☐
- Which of the following microorganisms causes influenza?
(a) Bacteria ☐ (b) Virus ☐ (c) Protozoa ☐
- Which of the following is another name of vinegar?
(a) Acetic acid ☐ (b) Lactic acid ☐ (c) Carbonic acid ☐
- Which of the following gases is released in sewage treatment?
(a) Carbon dioxide ☐ (b) Biogas ☐ (c) Oxygen ☐
- Which of the following antibiotics was the first one to be discovered?
(a) Tetracycline ☐ (b) Erythromycin ☐ (c) Streptomycin ☐
- Which of the following substances is added to make the process of cheese making faster?
(a) Flour ☐ (b) Milk powder ☐ (c) Rennet ☐
- Which of the following disease is not caused by microorganisms?
(a) Cancer ☐ (b) Goitre ☐ (c) Chicken pox ☐
- Which of the following disease is a plant disease?
(a) Anthrax ☐ (b) Rinderpest ☐ (c) Wilt ☐

2 Fill in the blanks.

- The threads of the multicellular fungi are called _____.
- _____ is used for the large scale production of alcohol, wine and beer.
- The two main classes of fungi are _____ and _____.
- _____ involves the use of high-voltage electric current through food to destroy microorganisms causing food spoilage.
- Food irradiation leads to the deficiency of _____ in the food items.
- Yellowing of leaf is caused by _____.
- During the process of curdling of milk, the bacterium _____ produces an acid called _____ from the lactose present in the milk.
- The citric acid which is most commonly used in the soft drink industries is produced with the help of the fungus _____.
- A substance used to produce immunity against diseases in the living body is called a _____.
- Viruses are made up of genetic material surrounded by a _____ coat.

3 Match the columns (Double matching).

Column A	Column B	Column C
1. yeast	(i) protozoan	(a) rich source of proteins and vitamins
2. amoeba	(ii) fungus	(b) heterotrophic
3. influenza	(iii) bacterium	(c) virus
4. lactobacillus	(iv) communicable disease	(d) curdling of milk

4 Pick the odd one out. Give reasons to support your answer.

1. Bacilli, cocci, commas, aspergillus
2. Polio, chicken pox, sprain, measles
3. Penicillium, aspergillus, bread, mould
4. Pasteurization, canning, refrigeration, ohmic heating.

5 State whether the following statements are 'True' or 'False'.

1. Pathogens are disease-causing microorganisms.
2. Blight is a disease caused by the microorganisms in animals.
3. Vacuum packing kills all the bacteria present in the food items.
4. All the microorganisms are harmful.
5. Chlamydomonas is a multicellular algae.
6. Sugar is used as a food preservative.

6 Define the following terms.

- (a) Pasteurisation
- (b) Canning
- (c) Biological nitrogen fixers
- (d) Food irradiation
- (e) Fermentation
- (f) Antibiotic

7 Answer in a few words only.

1. Name the causative microorganisms and mode of transmissions of the following diseases.

Diseases	Causative pathogens	Mode of transmission
(a) Fin rot	_____	_____
(b) Cholera	_____	_____
(c) Rinderpest	_____	_____
(d) Measles	_____	_____

2. Name two antibiotics.
3. Name two useful microorganisms.
4. Name two methods of food preservation that involve the techniques of temperature control.
5. Name two microorganisms that are used as the sources of food.

8 Answer in one or two sentences.

1. Describe in brief the following microorganisms with examples.
(a) Bacteria (b) Virus (c) Fungi (d) Protozoa
2. What do you mean by UHT?
3. What are communicable diseases? Give examples.
4. How does a vaccine work?
5. Describe in brief how the microorganisms are used in agriculture.
6. Describe the two classes of fungi in brief.

9 Answer in one or two paragraphs.

1. Describe in detail the use of microorganisms in food preservation.
2. What is food spoilage? What are the causes and signs of food spoilage?
3. What do you mean by food preservation? Name the techniques and the methods involved in food preservation.
4. Explain the role of microorganisms in cleaning the environment in detail.
5. Mention the ways by which transmission of pathogens takes place. What preventive measures can be taken to avoid the transmission of pathogens?

NEP QUERY



CRITICAL THINKING

- Complete the given crossword by reading the clues given below.

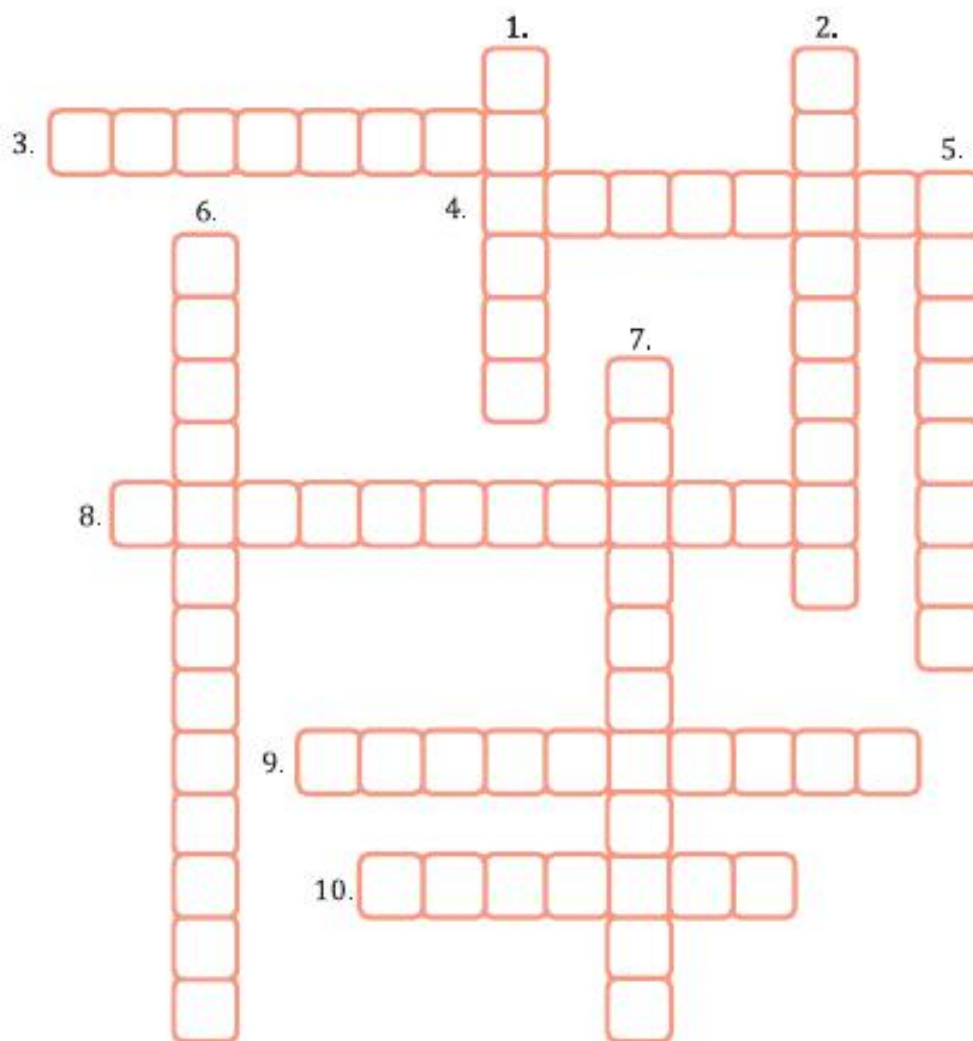
(Self-Awareness)

Across

3. The ability of body to fight against the diseases.
4. They feed on the body of living ones.
8. Converting sugar into alcohol in the absence of oxygen.
9. These body substances destroy disease carrying microbes.
10. It is used to produce immunity towards diseases.

Down

1. Threads of the multicellular fungi.
2. Disease-causing microorganisms.
5. Process in which food deteriorates and becomes unfit for human to eat.
6. Substances used to check the microbial growth in food items.
7. Medicines that kill the growth of disease-causing microorganisms.



COMPUTATIONAL SKILLS

- Make a powerpoint presentation (PPT) on 'Fertilisation in Flowers'.

(Problem Solving)

Crop Production and Management

2

- Agriculture in India
- Increasing crop yield
- Agricultural practices
- Nitrogen cycle
- Crop Protection

The branch of science that deals with growing plants and raising livestock for human use is called **agriculture**. With the human population growing at a fast pace, regular production, appropriate management and distribution of food are the important steps that are required to be taken.

AGRICULTURE IN INDIA

A proper crop production management was initiated for the first time in 1960, with a view to fulfill the food requirements of the increasing population.

Crop Plants

Crop is the term used to describe a plant that is grown in a field on a large scale, e.g., cereal crops, wheat crops and fruit crops. The crops that are grown under the supervision of farmers in large fields are called **cultivated crops**.



Wheat Crops

Products Obtained from the Crop Plants

The products obtained from the crops are called produce. Table 1.1 shows a variety of produce obtained from different crops grown in India.

Table 1.1: Produce obtained from different types of crops grown in India

1. Cereal or grain crops	Rice, wheat, maize, barley, oats
2. Root crops	Carrot, sweet potato
3. Fibre crops	Jute, cotton
4. Pulses or legumes	Gram, peas or beans
5. Plantation crops	Tea, rubber, coconut, coffee
6. Tuber crops	Potato, tapioca
7. Oilseed crops	Soyabean, mustard, groundnut

Crop Plants and Seasons

Based on the seasons in which they are sown, the crops grown in India can be classified as Kharif and Rabi.

Kharif crops are sown in the rainy season by June/ July, and are harvested by September/October. Thus, they are also known as summer season crops. Rice, maize, cotton, jute, pulses, groundnut are the examples of Kharif crops.



Rice



Maize



Cotton



Jute

Rabi crops are sown in the winter season in October or November and are harvested by March/ April. Thus, they are also called winter season crops. Mustard, wheat, potato, barley, pea and gram are the examples of Rabi crops.



Mustard



Wheat



Potato



Barley

AGRICULTURAL PRACTICES

Agricultural practices are those practices which involve the necessary steps to be taken during production of crops:

1. Soil Preparation

The quality and the health of the crops majorly depend on how the soil is prepared, as it is the soil that helps in providing proper nutrients, water and oxygen to the plants, and also helps them to anchor using their roots. There are certain natural agents too which help in making the soil fertile such as microbes, earthworms, snails, etc.

Some of the steps taken during soil preparation are as under:

a. Ploughing or Tilling

In order to make sure that the soil becomes fertile uniformly, ploughing or tilling is done. In this process, the soil is loosened and turned over and over again. The method of loosening the soil is called **ploughing or tilling**.



b. Levelling

Levelling is the process of breaking down the larger lumps of soil, and then levelling it by using a leveler. In this process, the larger lumps of soil left after ploughing are further broken down to smaller pieces. This helps the soil to get mixed easily and makes it more fertile and aerated.

c. Agricultural Implements

Ploughing, tilling or levelling is done with the help of certain agricultural tools such as a plough, hoe, and cultivator.

(i) Plough: A plough is an equipment, mainly made of a wooden shaft called a plough-shaft, attached to a triangular iron blade called ploughshare.

This equipment is mostly used for turning the soil upside down, with the help of the blade attached to the shaft.



(ii) Cultivator: A cultivator is a ploughing tool that has multiple blades.

The cultivators are attached to the tractors to plough the field. This helps in saving both time and energy, and helps in uniform ploughing of the field.

(iii) Hoe: A hoe is a simple farming equipment made of a wooden rod with a curved iron blade attached to it at one end. This equipment is used for either removing the weeds or for loosening the soil.



2. Sowing

Once soil preparation is done, seeds are sowed in the soil. The process of putting seeds in the soil is known as **sowing**.

Some of the most commonly found crops which are grown by sowing seeds include wheat, maize, millet, oats, sorghum, mustard, etc.

Methods

The most common methods of sowing seeds include the following :

(a) Broadcasting : The process of sowing seeds using hands is known as **broadcasting**. In this method, the seeds are sprinkled all across the field, and then are levelled later on using a wooden plank or a harrow. The most common types of crops grown using this method include paddy, wheat, coriander, fenugreek, etc.

(b) Drilling or Line Sowing : In this method, seeds are sown by using a simple agricultural implement such as a **seed drill**. A seed drill is an equipment, consisting of a funnel shaped opening connected to long elongated tubes. These tubes are further connected to a plough.

As the seeds are poured in the funnel shaped opening, they slide down towards the soil through the connected tubes. While the plough makes grooves in the soil, the seeds get deposited inside the soil at the right distance and depth.

Some other implements used in sowing seeds mechanically include, seed-cum-fertiliser drill, zero-till drill etc.

The most common crops grown by using this method include wheat, millet, sorghum, etc.

(c) Transplanting : Some of the crops are first grown in the nurseries and then transferred to the main field. This process is known as **transplantation**.

In this method, the seeds are normally grown for about 3-5 weeks on the nursery beds and then are planted on the main field. However, before planting the seedlings in the main field, the soil of the main field is watered in order to make the field soft for the roots to penetrate through the soil.

Some of the commonly grown crops by this method are paddy, rice, tomato, onion, cauliflower, etc.

3. Application of Manure and Fertilisers to the Soil

a. Manure

Plants utilize the nutrients present in the soil for their growth and development. Gradually, these nutrients present in soil get depleted and need to be replenished. So, these nutrients are added on regular basis in the form of manure and fertilisers. Manure is nutrient rich organic substance obtained from the decomposition of dead plant and animal wastes. Manuring is the process of adding manure to the soil which is essential for the growth and development of the plant. Following are the forms of manure used in fields:

- (i) Farmyard Manure:** It consists of straws, urine, leaves and other farmyard wastes.
- (ii) Green Manure:** It consists of green plants mainly leguminous plants. These plants are ploughed along with the ploughing of soil.
- (iii) Compost :** It is made by dumping plant and animal wastes in pits. Bacteria and fungi

present in the soil help in its decomposition . This decomposed matter is of dark brown colour and rich in nutrients and possess greater water holding capacity.

Table 1.4 : Fertilisers and the nutrients supplied by them:

S.No.	Name of Fertiliser	Nutrients Supplied by them
1	NPK	Nitrogen (N), phosphorus (P) and potassium (K)
2	CAN (Calcium Ammonium Nitrate)	Calcium and nitrogen
3	Diammonium Phosphate	Nitrogen and phosphorus
4	Urea	Nitrogen

Some advantages of fertiliser are :

- It provides a particular nutrient to the soil.
- It is a concentrated source of soil nutrients.
- It is easy to handle, store and transport.
- It is not bulky as compared to manure.

Some disadvantages of fertiliser are :

- Its excessive use causes water pollution.
- Its excessive use makes soil less fertile.

Fertilisers and manure are applied to the soil manually or by using drills.

S.No.	Manure	Fertiliser
1	A natural substance obtained from the decomposition of plants and animal waste.	A man-made chemical substance rich in specific nutrients.
2	Used in large quantities.	Used in small quantities.
3	Provides a lot of humus to soil.	Does not contain any humus.
4	Prepared in fields.	Prepared in factories.
5	Does not cause water pollution.	Causes water pollution as the groundwater table gets contaminated with the salts present in the fertilisers.

4. Irrigation

Water is a basic requirement for the crops to grow.

For a crop to remain healthy, regular watering of the crop is necessary. Water requirements are different for different crops. For instance, water requirements are less for wheat as compared to the water requirements of paddy crops that need constant supply of water.

Rainfall is a major source of water for crops. Owing to the uncertainty of the rainfall, alternative methods have been devised to provide the adequate supply of water to the crops at regular intervals.

The artificial method of watering the plants for assisting in the growth of the plants is called **irrigation**. Main sources of irrigation are wells, tube wells, ponds, lakes, rivers, dams and canals. The two main methods of irrigation are:

a. Traditional Method

In India, the traditional systems of irrigation like the **chain pump**, **dhekli**, **moat** (the pulley system) and **rahat** (lever system) have been in use for centuries. Though these methods are less expensive, they often lead to the wastage of water. Owing to their less efficiency, they are not in use nowadays.



Moat



Rahat

b. Modern Method

Modern methods of irrigation are more inclined towards the use of diesel, biogas, solar energy and electricity for lifting water. The most common modern methods of irrigation are:

(i) **Sprinkler irrigation system:** This irrigation system has an arrangement of vertical pipes with rotating nozzles on the top. These pipes are connected to the main pipeline at regular intervals. Water is allowed to flow through the main pipe under pressure in order to get sprinkled on the crop as if it is raining. It is more useful on the uneven and sandy land where sufficient water is not available.



(ii) **Drip irrigation system:** This irrigation system has an arrangement of pipes or tubes with

very small holes in them to water plants drop by drop just at the base of the roots. It is very efficient as water is not wasted at all. It is good for watering fruit plants and gardens, etc. where there is poor availability of water.

FACT

The Kara Kum Canal in Turkmenistan is the longest irrigation canal in the world. It is over 1,375 kilometres long.

5. Weeding

The undesirable plants that grow among the crops are called **weeds**. **Amaranthus**, **wild oat**, **grass**, etc are some examples of weeds. The process of removing these weeds is known as **weeding**.

Weeds need to be removed as they share the light, space, nutrients and water with the crops growing simultaneously and ultimately affect the growing crops. If not removed, these weeds may prove to be poisonous to animals and human beings if they are harvested along with the crops.

Methods applied for weeding are as follows:

a. Manual Method

Weeds may be uprooted manually by hands or by using implements like a trowel, harrow or hoe.

Though manual weeding is the cheapest way to remove weeds, yet several disadvantages are associated with this method. Some of them are:

- It is a time consuming method.
- Sometimes desired crops may also be removed accidentally.



Amaranthus



Wild Oat



b. Chemical Method

Weedicides are the chemicals that are used to remove weeds. The weedicides such as 2, 4-D are generally sprayed during the vegetative growth of weeds. The major disadvantage associated with the chemical method is that it may cause health problems to the farmers. However, it is recommended that the farmers should cover their mouth and nose while spraying weedicides.

c. Biological Control Method

It involves the use of insects that destroy the growth of weeds without affecting the crops growing in the field.

FACT

Weeds such as wild carrot, cow parsley and celery grow at least 350 seeds on a single flower-head.

6. Harvesting

The process of cutting and gathering the ripened crop is called **harvesting**.

During harvesting, the crops are pulled out or cut close to the ground. In India, most of the harvesting is done manually. A **sickle** is the only tool that is used for this purpose. The sickles are of two types—plain and saw-shaped (serrated). The serrated sickles are used for cutting paddy and wheat crops.



In big fields, some machines such as reapers, threshers or even combine harvester are more in use for harvesting.

However, harvesting with machines also has certain disadvantages as given below:

Harvesting by machines results in shattering of the grain. In case of harvesting with sickles, shattering of the grain is negligible.

Harvesting through machines such as combined harvesters results in loss of hay and thus, yields less fodder.

After cutting, the crop is gathered at one place and is allowed to dry under the sun for a few days.

7. Threshing and Winnowing

The process of beating out the grain from the chaff is known as **threshing**. The traditional methods of threshing include beating the grains manually with sticks, or treading them with animals like bullocks. In case of big fields, threshing is done with the help of threshers. Threshing with threshers is not only cheaper but also quicker and more efficient than the traditional methods.



Threshing



Winnowing

After threshing, the chaff and hay are separated from grains by putting the threshed material

before a good natural breeze. When the threshed material is allowed to fall from a height, the grains being heavier fall straight on the ground, whereas hay and chaff being lighter are blown away and form a separate heap at a small distance from the grain heap. The process of separating the grains from chaff and hay with the help of natural breeze is called **winning**.

8. Storage of Grains

Before the crops are available for consumption, they need to be stored in a suitable environment where they are protected from any damage.

Following are the conditions necessary for a storage area:

- The storage area should be kept clean and dry.
- In order to keep away the pests, pesticides should be sprayed beforehand.
- Periodic inspection of the storage area is necessary to ensure the safety of the grains.

CROP PROTECTION

To protect the crop from being damaged, it is necessary to take immediate steps:

Protection from Pests

Crop pests are those organisms that can damage a crop plant. Pests are some of the most common crop destructors and can come in several different forms. Insects are the most common and destructive, but mammals like mice and raccoons are also considered crop pests in some areas. Birds are other examples of a crop pest since they can damage plants by eating seeds and fruits as well as by hunting for insects hidden in the plants.

Crops are also destroyed by bacteria and fungi and viruses that may cause several diseases to the crops. Diseases may be soil borne, seed borne or airborne. Rust and smut in wheat, blight in potatoes are the common examples of fungal infections. A bacterium causes the disease called **wilt** in plants that blocks the tissues that help in transportation of water.



Rust in wheat



Blight in potatoes

Following steps are taken by the farmers to prevent the damage caused by pests:

- Chemicals like pesticides, fungicides and insecticides are commonly sprayed on the crops. But this method has its own disadvantages. Chemicals cause pollution. Insecticides may even destroy useful insects.
- Putting scarecrows in the fields is the traditional way to scare away the birds.
- Diseased seeds should be separated from the healthy seeds.
- Disease-resistant variety of seeds should be preferred for breeding.

INCREASING CROP YIELD

The crop yield refers to both, the measure of yield of crop per unit area and the seed generation of the plant. In order to increase the crop yield, these methods are being applied in the fields.

1. Mixed Cultivation

In mixed cultivation, a number of crops are grown together in the same season which are all interdependent on each other for their nutritional requirements. For example: in coffee plantations, two types of plants are grown in the same season. They are coffee and tall teak trees to provide shade for the coffee plant.



2. Crop Rotation

Crop rotation is the practice of growing two or more dissimilar crops in the same field one after the other.

A particular crop has a specific requirement of a nutrient. The soil becomes nutrient deficient if a particular crop is grown on the same field year after year. Thus, in order to replenish the soil with nutrients, different crops should be grown alternatively. For example, crops such as wheat and paddy require a lot of nitrogen from the soil. This lost nitrogen can be replenished by the leguminous plants such as pea. The leguminous plants such as peas, lentils, beans, etc are the plants that have special bacteria known as *Rhizobium* in the root nodules that help in restoring the nitrogen content of the soil. The bacteria restore the nitrogen with the help of the process called **nitrogen fixation**. The process has been described below:

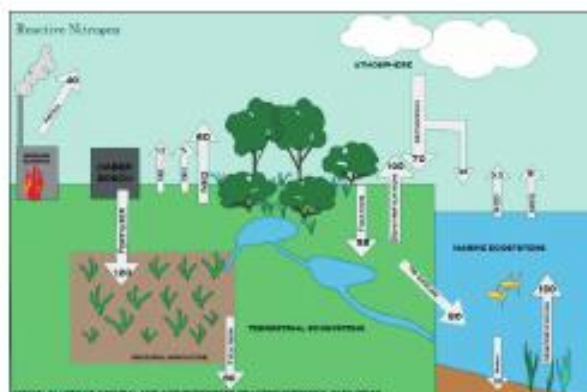
NITROGEN CYCLE

The nitrogen cycle is a natural phenomenon in which the nitrogen is fixed, used by the plants and animals and later returns to the atmosphere.

Nitrogen fixation is the process by which the nitrogen is taken from the atmosphere and is converted into other nitrogen compounds. Nitrogen fixation occurs naturally during lightning or by nitrogen fixing bacteria or blue-green algae.

Below is the process of nitrogen fixation by the leguminous plants:

Once the nitrogen compounds are formed with the help of nitrogen fixing bacteria located in the root nodules of the leguminous plant, they are used by the plants. As the plants serve themselves as food, the nitrogen compounds can be used by animals. When these plants and animals die, the nitrogen compounds present in their bodies are decomposed and returned to the soil. Some of these nitrogen compounds are used by the plants again while others reach the atmosphere.



◀ REWIND

- ▶ Agriculture in India has shown a great development due to advancement in good quality seeds, adequate quantity of medicines, timely application of manure and fertilizers, hybrid plants and modern machines and equipments.
- ▶ There are two types of crops on the basis of seasons in which they are grown. They are kharif and rabi crops.
- ▶ Agricultural practices involved in the production of crops are— soil preparation, sowing, application of manure and fertilizers to the soil, irrigation, weeding, harvesting and storage of grains.
- ▶ In order to protect crops from being diseased, the immediate steps to be taken are— application of antibiotics and fungicidal compounds, use of disease resistant variety of seeds and separation of healthy seeds from the diseased ones.
- ▶ To protect the crops from being damaged by pests, putting scarecrows in the fields and the application of pesticides and insecticides are usually done.
- ▶ Mixed cultivation and crop rotation are the two main processes followed in order to increase the crop production.
- ▶ Animals provide us food in various ways such as milk, meat, fish, eggs, honey, etc.

🔑 ASSIGNMENTS

COGNITIVE SKILLS

(Learning by Doing)

1 Tick (✓) the correct option.

1. Plants show

(a) Heterotrophic nutrition



(b) Autotrophic nutrition



(c) Both (a) and (b)



2. Rhizobium lives in the roots of

(a) Barley



(b) Gram



(c) Mango



3. Chlorophyll traps sunlight and converts it into

(a) Chemical energy



(b) Light energy



(c) Electrical energy



4. The lichen has an association between algae and
 (a) Bacteria (b) Fungi (c) Viruses
5. Nitrogen is supplied to the plants by
 (a) Rhizobium (b) Fertilizers (c) Both (a) and (b)

2 Fill in the blanks.

1. Rabi crops are also known as _____ season crops.
2. The process of raising seeds on nursery beds and then planting them in the main field is called _____.
3. _____ manure consists of green plants, mainly leguminous plants.
4. _____ provides a lot of humus to the soil.
5. _____ is a traditional method of irrigation.
6. 2, 4-D is a _____.
7. Nitrogen fixing bacteria are located in the root nodules of the _____ plants.

3 State whether the following statements are 'True' or 'False'.

1. Fertilisers are the concentrated sources of soil nutrients.
2. Pea is a leguminous plant.
3. Manure is prepared in factories.
4. Amaranthus is a weed.
5. Paddy is sown usually by broadcasting method.
6. Gram is a rabi crop.
7. Roots are able to respire properly if the plants are sown deep.
8. Diammonium phosphate is a commonly known fertiliser.

4 Pick the odd one out. Give reasons to support your answer.

1. Rice, wheat, maize, jute
2. Soya bean, gram, mustard, groundnut
3. Plough, hoe, cultivator, sickle
4. NPK, CAN, compost, urea
5. Peas, beans, paddy, lentils

5 Answer in a few words only.

1. Name two agricultural implements used for the preparation of soil.
2. Why does the soil need to be loosened before sowing seeds in it?
3. Name the organization involved in the production of good quality agricultural seeds.
4. Name the implements used for broadcasting.
5. Name the components of the farmyard manure.
6. Give examples of traditional and modern methods of irrigation.
7. Name a nitrogen fixing bacteria.
8. Give an example of a disease caused by fungus and bacterium in plants.
9. Name one technique used for increasing crop yield.

6 Answer in one or two sentences.

1. What do you understand by cultivated crops?
2. What are agricultural practices? List them in the correct sequence.
3. Describe in brief how the crops are classified according to the seasons in which they are grown.
4. Give two advantages of ploughing the soil.
5. Write a brief note on traditional methods of irrigation.
6. What do you mean by water logging?
7. What is harvesting? Name the implements used for harvesting.

7 Answer in one or two paragraphs.

1. How is the soil prepared for cultivation? What implements are required for this process?
2. Describe different methods of sowing seeds. Which things should be considered while sowing seeds?
3. Differentiate between manure and fertilisers.
4. Write short notes on sprinkler irrigation and drip irrigation methods.
5. What is weeding? How is it done?
6. How are the following agricultural practices carried out?
(a) Harvesting (b) Threshing (c) Winnowing
7. How are the grains stored?
8. What steps can be taken to ensure crop protection?
9. What is crop yield? How can it be increased?
10. Explain nitrogen cycle with the help of a labelled diagram.

NEP QUERY



PROJECT WORK

- Collect different varieties of seeds and find out how and when they should be sown. Make a chart by pasting the individual packets containing these seeds. Write the information about each variety of seeds next to the respective packets of seeds.

(Learning by Experiment)

COMMUNICATION SKILLS

- Organise a debate on the topic 'important source of irrigation in India'.

(Discussion)

Metals and Non-metals

3

- Metals
- Non-metals
- Physical and chemical properties of metals and non-metals
- Uses of metals and non-metals

On the basis of pattern of properties the substances possess, the elements are divided into two broad categories **metals** and **non-metals**. Besides them, there is a third category also called **metalloids** like bismuth, antimony, arsenic and boron. They possess the properties of both metals and non-metals. Let's know more about the metals and non-metals.

METALS

The word 'metal' is derived from the Greek word metallon which means extracted from ground'. Metals can be defined as:

Metals are simple chemical substances with characteristic properties under normal conditions.

For example: copper, iron, silver, gold, aluminium, etc.

Metals are shiny, hard, sonorous, heavy, good conductors of heat and electricity, malleable and ductile.



Copper



Iron



Silver



Gold

Some facts about occurrence of metals

Metals occur in the earth's crust mostly as their compounds. These compounds are called **minerals**. There are certain minerals from which metals are extracted efficiently. These minerals are called **ores**. The technique of extracting metals from their ores is called **metallurgy**.

Aluminium is the most abundant metal in the earth's crust while in the whole earth, iron is the most common, as it forms the core of the earth.

Table 4.1 : Symbols of some common metals

S. No.	Name of the Metals	Symbols	S. No.	Name of the Metals	Symbols
1.	Sodium	Na	6.	Iron	Fe
2.	Potassium	K	7.	Copper	Cu
3.	Calcium	Ca	8.	Gold	Au
4.	Barium	Ba	9.	Silver	Ag
5.	Aluminium	Al	10.	Mercury	Hg

NON-METALS

The elements which lack the characteristic properties of metals like lustre, hardness, malleability, ductility and conductivity, etc are called non-metals.

Some common non-metals are hydrogen, oxygen, carbon, sulphur, phosphorus, etc.



Phosphorus



Chlorine



Bromine



Iodine

Non-metals which are present in a gaseous form like hydrogen, oxygen, nitrogen, etc exist in the atmosphere, while those that exist in a solid form are found in the earth's crust either in a free form or mostly as their compounds like carbonates, nitrates, sulphates, hydrides, etc.

All the gases are non-metals.

Table 4.2 : Symbols of some common non-metals

S. No.	Name of the Non-metals	Symbols	S. No.	Name of the Non-metals	Symbols
1.	Hydrogen	H	6.	Phosphorus	P
2.	Oxygen	O	7.	Chlorine	Cl
3.	Nitrogen	N	8.	Bromine	Br
4.	Carbon	C	9.	Iodine	I
5.	Sulphur	S	10.	Helium	He

PHYSICAL PROPERTIES OF METALS

1. Lustre

Metals possess a peculiar natural shine on their surface. This is called metallic lustre. We can easily see that by observing some metal articles like iron key, aluminium vessel, copper wire, gold and silver ornaments, etc. Gold has a yellow lustre and copper, a brownish. Rest of the metals have white lustre.

2. Physical state

Metals are generally solid at room temperature. Only exceptions are gallium and mercury. They are liquid at room temperature.

3. Hardness

All the metals are hard, but sodium and potassium are so soft that they could be even cut with a knife.

4. Malleability

The property of a substance because of which it can be beaten into thin sheets is called **malleability**.

The substances that possess this property are called **malleable**. Metals are malleable. Non-metals do not have this property. That is why we can see the fine sheets and foils of aluminium, silver, gold, copper or iron but never of carbon, sulphur or graphite.

5. Ductility

It is the property of a substance by which it can be drawn into wires.

The materials that possess this property are called **ductile**. Metals are ductile, i.e. we can draw wires of metals. We can see wires of different metals like copper, silver, aluminium, etc.

6. Sonorous

You must have heard the bells ringing in your school or in temples. What are they made up of? They are made up of iron, brass or some other metals.

Why don't we have bells of plastic, rubber or wood? This is because metals, when struck, produce a ringing sound. This property is limited to metals only. Due to this property, metals are said to be sonorous.

7. Density

Metals generally have high densities. In common language, we can say that metals are heavy. To make it clearer, we can say that same volume of a metal has more number of atoms than that of a non-metal.

8. Conductivity of heat

Metals are good conductors of heat. They get heated up very fast when kept on flame. That's why cooking vessels are made up of metals like iron and aluminium.



Activity

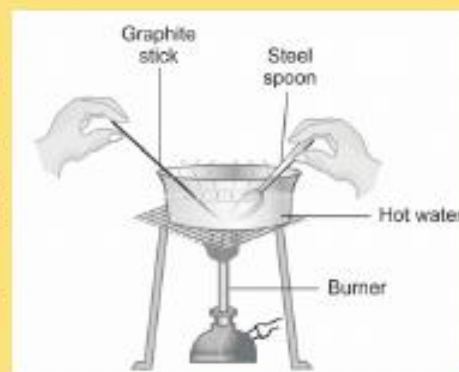
Aim — To demonstrate that metals are good conductors of heat.

Materials required — A pencil, a steel spoon, a bowl, some water and a burner.

Procedure — Take out the graphite stick from the pencil. Take some water in a bowl and keep it over the flame. Immerse one end of both graphite stick and steel spoon in hot water. After some time, touch the other end of both with your hand.

Observation — You will see that steel spoon feels hot on touching while the graphite stick does not.

Conclusion — That proves that metals are good conductors of heat.



9. Conductivity of electricity

Metals are good conductors of electricity. That means they allow the electricity to pass through them. That's why, electric wires are made up of metals like copper and aluminium. Best conductor of electricity among all the metals is silver, followed by copper and aluminium. But silver being expensive, is not used commonly to make electric wires. So, copper is used for this purpose.

10. Melting and boiling points

Metals have very high melting and boiling points. But sodium and potassium are exceptions. Their melting points are very low.

PHYSICAL PROPERTIES OF NON-METALS

1. Physical state

Unlike metals, non-metals can exist in a solid, liquid or gaseous state. Carbon, sulphur, phosphorus, etc. are solids; bromine is liquid; and chlorine, bromine, iodine, helium, etc are gases.

2. Lustre

All the non-metals lack the lustre that metals have, and are dull in appearance. But Iodine and graphite are exceptions. They are shiny.

3. Hardness

Non-metals are soft unlike metals. But the strange thing is that the hardest substance known is diamond, which is the allotrope of carbon, a non-metal.

4. Malleability

Non-metals are non-malleable. They do not flatten into sheets when hammered. Rather, they break down on hammering. These types of substances are called brittle.

5. Ductility

Non-metals are non-ductile. That means they cannot be drawn into wires.

6. Non-sonorous

Non-metals do not produce ringing sound on striking. In other words, they are non-sonorous.

7. Insulator of heat

Non-metals do not allow heat to pass through them. So, they don't get heated up. That's why handles of cooking vessels are made up of non-metals.

8. Insulator of electricity

Non-metals do not conduct electricity through them. Only graphite is an exception.

9. Low melting and boiling points

Non-metals have comparatively low melting and boiling points. But graphite, a non-metal, has the highest melting point of all the elements.

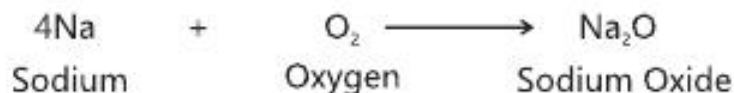
CHEMICAL PROPERTIES OF METALS

1. Metals have 13 electrons in their outermost shell. They can easily lose electrons and make positive ions.
2. Metals are good reducing agents.

3. Reaction with Oxygen

Metals form basic oxides on reacting with oxygen.

- Sodium and potassium are most reactive towards oxygen. These metals react so readily with oxygen that if they are exposed to air, they instantly react with oxygen of air and catch fire. So, they are kept immersed in kerosene oil to avoid their contact with air.



4. Reaction with Water

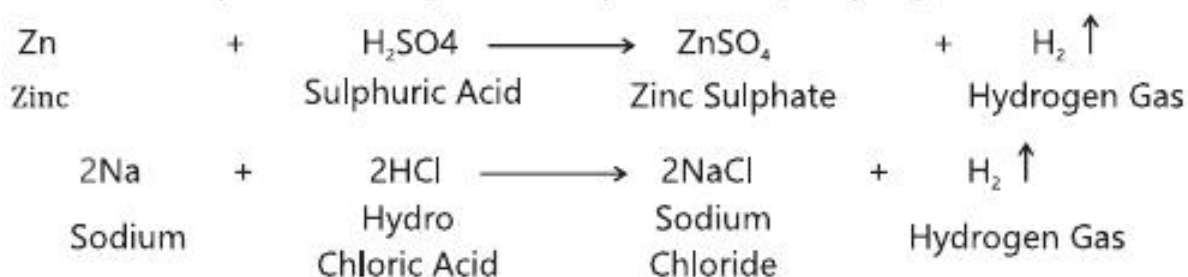
Metals produce their hydroxides or oxides and release hydrogen gas on reacting with water. For Example—



5. Reaction with Acids

Metals react with acids to produce respective salts accompanied by the evolution of hydrogen gas. For example:

Zinc reacts with sulphuric acid to give zinc sulphate and hydrogen gas.



6. Reaction with Bases

Most of the metals do not react with bases. However, some metals like aluminium, lead and zinc react with strong bases like sodium hydroxide to make complex salts and hydrogen gas.

7. Displacement Reactions

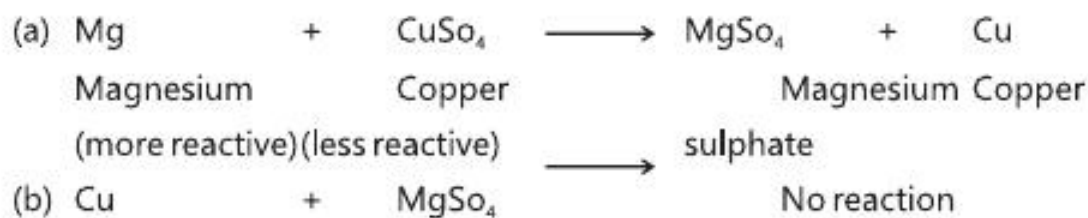
Displacement means 'dislocation'. Displacement reactions are the reactions in which more reactive metal displaces the less reactive metal from its salt.

All the metals are different in their reactivity. Some are very reactive and some are not so reactive. Metals on the top are most reactive and those on the bottom are least reactive. This sequence is called **reactivity series**. This series is as follows:

Table 4.3: The order of reactivity of metals

Metal	Symbol	
Potassium	K	<u>Most reactive</u>
Sodium	Na	
Magnesium	Mg	
Aluminium	Al	
Carbon	C	
Zinc	Zn	
Iron	Fe	
Tin	Sn	
Lead	Pb	
Hydrogen	H	
Copper	Cu	
Silver	Ag	
Gold	Au	
Platinum	Pt	<u>Least reactive</u>

For example,



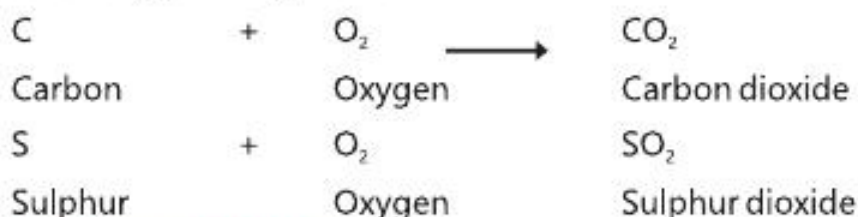
In the above example, in reaction (a), Mg is more reactive than Cu, so it displaces Cu from its salt CuSO₄ and itself forms its salt MgSO₄. Whereas in reaction (b), Cu being less reactive than Mg, is unable to displace Mg from MgSO₄, so no reaction takes place.

CHEMICAL PROPERTIES OF NON-METALS

1. They tend to form negative ions.
2. They are good oxidising agents.

3. Reaction with Oxygen

Non-metals also react with oxygen to produce acidic oxides. But most of the non-metals react with oxygen on ignition.



4. Reaction with Water

Non-metals do not react with water.

5. Reaction with Acids and Bases

Generally, non-metals do not react with acids and bases. Sometimes some complex reactions take place between non-metals and bases.

USE OF METALS

1. Copper and aluminium, are used to make utensils, cooking vessels, electrical wires and appliances.
2. Zinc is used to make dry cells.
3. Mercury is used in an instrument called **sphygmomanometer** used to measure blood pressure as well as in thermometers and barometers to measure temperature and pressure respectively.
4. Iron is used in making bridges, pipes, rods and sheets used in construction work, gates, railings, etc.
5. Various tools, machines and machine parts like screws, nails, bolts, chains are also made up of iron.

6. Zinc is used in the process of galvanisation to prevent articles from rusting.
7. Copper and zinc are important constituents of alloys such as brass, bronze and German silver.
8. Chromium is used in electroplating.
9. Aluminium foil has extensive use in packaging.
10. Aluminium is also used to make parts and bodies of aircrafts and automobiles.
11. Lead is used for making batteries and electrodes.
12. Magnesium, due to its burning with white dazzling light is used to make fireworks.
13. Gold, silver and platinum are called **noble metals** as they do not react easily with anything and are used to make jewellery and other articles.
14. Alloys of various metals are used to make articles of daily use. For example: steel, an alloy of iron, carbon, nickel and chromium is used to make utensils. Brass is an alloy of copper and zinc and bronze of copper and tin.

USES OF NON-METALS

1. Hydrogen and helium are the main constituents of matter of planetary bodies.
2. Hydrogen is a non-polluting efficient fuel.
3. Oxygen is needed for the living beings to breathe in.
4. Oxygen is must for all the burning processes, even inside our body for processes like respiration.
5. Carbon is the essential part of all the living matter.
6. Most of the fuels contain carbon.
7. Sulphur is used for manufacturing sulphuric acid, which is a very important chemical.
8. Sulphur is also used in manufacturing of skin medicines and ointments due to its fungicidal properties.
9. Graphite is used to make pencil leads and electrodes of dry cells.
10. Phosphorus is used in manufacturing matchsticks, fireworks and fertilisers.
11. Chlorine is used as disinfectant of water. It also has bleaching properties and is used to make bleaching powder.
12. Iodine is important to prevent a disease called 'goitre'. So it is consumed as iodised salt.
13. Nitrogen is the important constituent of proteins. It is very important for plants also.
14. Nitrogen is also used to preserve blood.
15. Helium is used by deep divers to respire under water, along with oxygen.

◀ REWIND

- Elements are divided into three broad categories called metals, non-metals and metalloids.
- Metals possess specific properties like lustre, malleability, ductility, high melting and boiling points, sonority, conductivity, etc.
- Non-metals lack these properties but graphite and diamond show exceptional behaviour.
- Metals form basic oxides on reacting with oxygen, while non-metals form acidic oxides.
- Metals react with water to produce hydroxides and hydrogen gas. Non-metals do not react with water.
- Metals also react with dilute acids to produce H_2 gas but non-metals generally do not.
- Displacement reactions are the reactions which involve displacement of a less reactive metal from its salt solution by the more reactive one.
- Metals and non-metals are very significant for human beings, plants and industries.
- An alloy is a combination of metals or a combination of one or more metals with non-metallic elements.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

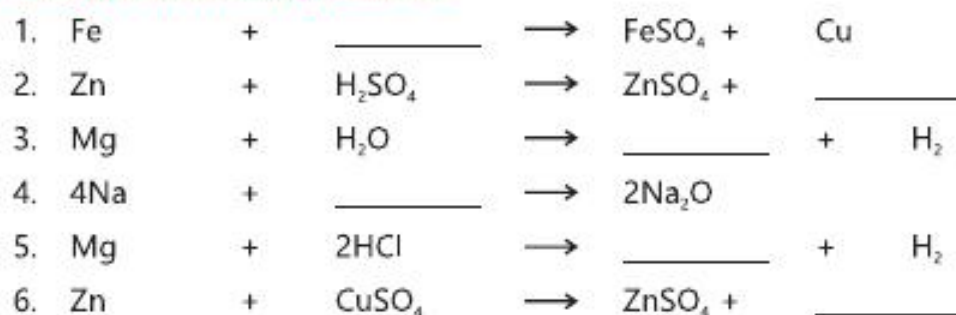
(Learning by Doing)

- | | | | | | |
|---|--|----------------|--|----------------------|--|
| 1. Materials that possess properties of both metals and non-metals are called | | | | | |
| (a) Half-metals | | (b) Metalloids | | (c) Metallites | |
| 2. Which of the following is not a metal? | | | | | |
| (a) Zinc | | (b) Lead | | (c) Graphite | |
| 3. Which state of matter is essentially a non-metal? | | | | | |
| (a) Solid | | (b) Liquid | | (c) Gas | |
| 4. The symbol of chlorine is | | | | | |
| (a) C | | (b) Cl | | (c) Cr | |
| 5. A malleable substance among the following is | | | | | |
| (a) Carbon | | (b) Sulphur | | (c) Copper | |
| 6. Sodium produces H_2 gas on reacting with | | | | | |
| (a) Water | | (b) Acid | | (c) Both (a) and (b) | |
| 7. Which of the following is capable to displace zinc from the solution? | | | | | |
| (a) Al | | (b) Fe | | (c) Ni | |

2 Match the columns (Double matching).

Column A	Column B	Column C
1. metals	(i) a fuel	(a) low melting point
2. non-metals	(ii) highest boiling point	(b) conduct electricity
3. graphite	(iii) malleable	(c) constituent of living matter
4. copper	(iv) mostly soft	(d) constituent of brass
5. carbon	(v) electric wires	(e) ductile

3 Complete the following reactions.



4 Which combination of reactants will give products?

- | | | | |
|-------|----|---|---|
| (i) | K | + | CaSO ₄ |
| (ii) | Mg | + | ZnSO ₄ |
| (iii) | Zn | + | Al ₂ (SO ₄) ₃ |
| (iv) | Ca | + | BaSO ₄ |
| (v) | Cu | + | FeSO ₄ |

5 Answer in a few words only.

1. Give two uses of graphite.
2. Name two noble metals.
3. Name a non-metal that is placed with metals in reactivity series.
4. What is the nature of metallic oxides?
5. Give an example of sonorous metal.
6. Name two metals that can be cut with a knife.
7. Name a lustrous non-metal.
8. Name two liquid metals and non-metals.
9. Name three metals that are the best conductor of electricity.
10. What are brittle substances?

6 Answer in one or two sentences.

1. Define malleability.
2. Define ductility.
3. Why are bells of temples made up of metals rather than non-metals?

- How do metals react with dilute acids? Give two examples.
- What are displacement reactions? Give an example.
- Which metal is used in fireworks and why?
- Which non-metals show exceptional behaviour?
- How do non-metals react with oxygen?
- What are alloys? Give the components of two alloys.
- What do you understand by the terms mineral, ore and metallurgy?

7

Answer in one or two paragraphs.

- How are metals useful to us?
- State various uses of non-metals.
- Compare the physical and chemical properties of metals and non-metals.
- Give reasons for the following:
 - Chlorine is used as disinfectant of water.
 - Cookwares are made up of metals, but their handles of non-metals.
 - You can see foils of aluminium but not of graphite.
 - Noble metals are used to make jewellery.

NEP QUERY



CRITICAL THINKING

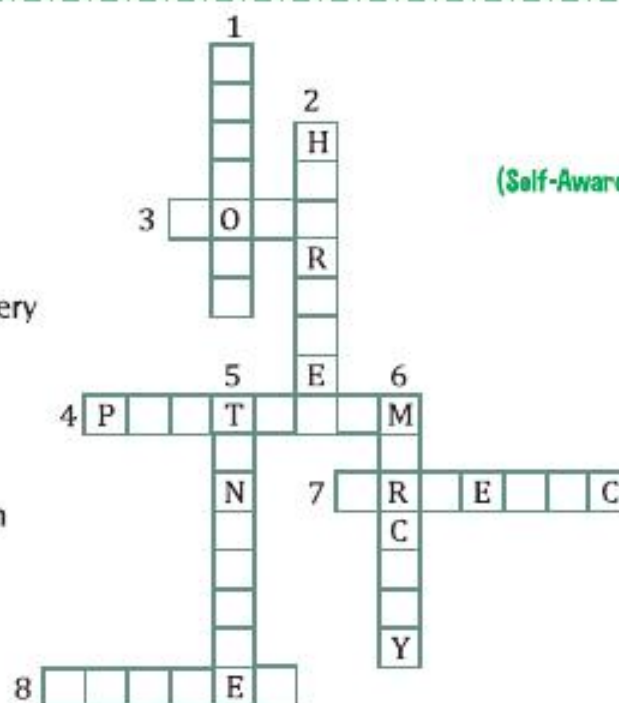
- Crossword Puzzle

Across

- A noble metal
- Another noble metal mostly used to make jewellery
- A metalloid
- A metal used to make electric wires

Down

- The hardest substance
- A gas that forms water on combining with oxygen
- The metal with the highest melting point
- A liquid metal



(Self-Awareness)

COMMUNICATION SKILLS

- Discuss that can you store lemon pickle in on aluminium utensil. Take help from your teacher to find the answer.

(Discussion)

PROJECT WORK

- How can you show experimentally that metals are good conductors of electricity? Do this activity with the help of a battery, a bulb, a switch, wires of some metals like iron, copper, aluminum.

(Learning by Experiment)

Synthetic Fibres and Plastics

4

- Synthetic Fibres
- Plastic : A Hazard to environment
- Types of Synthetic Fibres
- Plastics

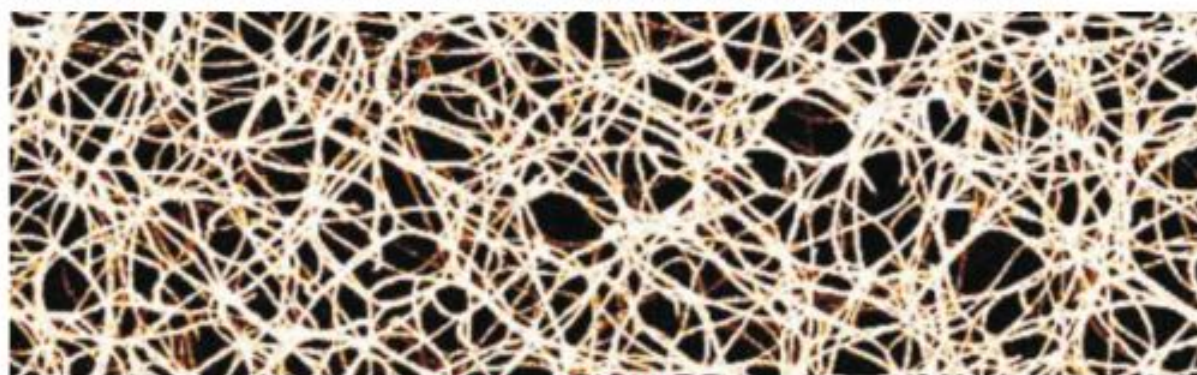
Fibres may be natural or man-made (synthetic). Natural fibres are obtained from natural sources like plants and animals. For example: cotton, silk, wool, etc. Synthetic fibres are manufactured by chemical reactions of different chemical substances in factories. Some commonly used synthetic fibres are rayon, nylon, polyester, etc.

SYNTHETIC FIBRES

Raw materials for preparation of synthetic fibres are obtained from substances like coal, petroleum and natural gases. Synthetic fibres are made up of polymers. 'Poly' means 'many' and 'mer' means 'unit'.

Polymers are compounds that are made up of same, small, repeating units, joined together through bonds in a linear pattern.

Their structure can be understood by observing a chain or a bead necklace.



Polymer of synthetic fibre

TYPES OF SYNTHETIC FIBRES

1. Rayon

Rayon was the first synthetic fibre to be produced. It is produced by the chemical processing of wood pulp, which is a natural substance. Therefore, it is considered to be a semisynthetic fibre. That means it is neither truly natural nor entirely synthetic.

Rayon, synthesised for the first time in England was known as 'viscose'.

Rayon resembles silk in its appearance, but is cheaper than silk. It is also known as **artificial silk**.



Properties of Rayon

1. Rayon is a versatile fibre.
2. It resembles silk in look and comfort.
3. It can be dyed in different colours.
4. It is soft, smooth, cool and comfortable.
5. It can absorb water and sweat, so it is comfortable to wear in summers also as a substitute for cotton.
6. Rayon does not insulate heat.
7. Rayon is a biodegradable fibre.

FACT

Grasim India is the largest producer of rayon in the world.

It holds approximately 24% of total rayon market.

Uses of Rayon

1. Rayon is used to make apparels like shirts, blouses, jackets, undergarments, ties and socks.
2. Rayon is also very useful to make furnishings and upholstery like bedsheets, bedcovers, curtains, cushion covers, etc.
3. Rayon is also used in many industries making surgical products and tyre cords, etc.

2. Nylon

This was the first true synthetic fibre that is truly made up of chemicals. It is a polymer of amides. It was used as a supplement of silk when silk got deficient during World War II for many military applications like making parachutes and vehicle tyres.

It is one of the strongest and most durable fibre.

It was made in 1931 for the first time from coal, air and water.

Properties of Nylon

1. Nylon shows variation in lustre. It may be very shiny, can have moderate lustre or may even be dull.
2. It is very durable and elastic.
3. It shows high tensile strength, i.e, capacity to bear weight.

4. Nylon is resistant to moths, insects and fungi.
5. It is light and wrinkle free, easy to wash and maintain.
6. It shows low absorption of water, so dries up quickly.
7. It melts on heating.

Uses of Nylon

1. Nylon is widely used for making clothes, bridal veils, packaging paper, carpets, ropes, pipes, musical strings, stockings, etc.
2. It is also used in many mechanical parts like screws and gears.
3. It is used in combs, toothbrushes, etc.
4. Ropes, tents, fishing nets, parachutes are also made of nylon due to its high tensile strength.
5. Its elasticity makes it suitable for making socks and stockings.

3. Polyester

As clear from its name, poly + ester, it is a polymer of ester. Ester is a sweet-smelling compound formed by the reaction of organic acids and alcohol. The raw material for polyester is petroleum. Polyester is also a very strong and wrinkle free fibre. It is easy to wash and maintain. Terylene is a common variety of polyester.

Polyester could be blended with natural fibres to enhance its properties. The common varieties of polyester are :

1. **Terrycot or polycot** — It is made by blending polyester and cotton and is used in dress materials, shirts, skirts, etc.
2. **Terrywool or polywool** — It is made by blending polyester and wool and is used to make suit lengths.
3. **Terrysilk** — It is made by blending polyester and silk. Sarees and dress materials are made of it.
4. **PET (Polyethylene terephthalate)** — It is used to make bottles and jars used in the kitchen.

Properties of Polyester

1. Polyester is combustible at a high temperature.
2. It shows low absorption of water and dries quickly.
3. Polyester shrinks minimum on washing.
4. It is durable.
5. It retains its colour on washing.
6. It is light weight, elastic and wrinkle free and regains its shape when creased.
7. It is a good insulator of heat.

Uses of Polyester

1. It is used in making home furnishings and apparels like shirts, pants, jackets, hats, bedsheets, carpets, etc.
2. Its elasticity makes it useful for making conveyor belts and safety belts.
3. Recording tapes of cassettes and floppy disks are also made up of polyester.
4. PET bottles and jars, holograms, filters, etc. also involve the use of polyester.
5. Polyester is used for finishing on guitars and pianos.

4. Acrylic

Nylon and rayon are the synthetic fibres that resemble silk, whereas acrylic is the synthetic fibre that resembles wool in its texture and properties. That's why it is used to make sweaters, shawls, socks, etc. It is a soft, warm, wrinkle free and shrinkage free fibre. It can be dyed into a variety of colours. It is cheaper than wool.

Advantages of synthetic Fibres

If we compare natural fibres and synthetic fibres, synthetic fibres are more advantageous in many aspects than natural fibres. Some of them are as follows :

1. Synthetic fibres are strong and durable.
2. They are maintenance free, easy to wash and wear. They do not get wrinkled.
3. They absorb less water and are quick to dry, that's why swimming costumes and beach wears are made from synthetic fibres.
4. They possess shine and lustre similar to that of natural fibres, but they are less expensive and affordable.
5. They do not shrink.
6. They are moth and insect resistant.
7. We don't need to depend upon crops or animals to yield them.

Disadvantages of Synthetic Fibres

1. Synthetic fibres catch fire easily and they melt on heating. On melting, they stick to body that causes severe burns. So, they are not suitable to wear at places where flame or heat is used.
2. They do not absorb water and sweat so, in hot and humid conditions, they stick to body due to sweat and make us feel uncomfortable.
3. Sometimes, wearing synthetic clothes in summers can cause skin allergies due to deposition of sweat.
4. They are not good for environment.
5. They use a huge amount of our exhaustible resources like petrochemicals for their production.

PLASTICS

We can see so many objects around us that are made up of plastics; objects like containers, buckets, chairs, tables, baskets, bottles and the list is endless. Even polythene bags we use every day are plastics. We are surrounded by and dependent on plastics for so many things. Let us know what exactly the plastic is!

Plasticity

Plasticity is the property of materials by which they can be moulded into any shape.

Plastics are those substances which are mostly synthetic in nature, obtained mainly from petrochemical sources and can be moulded into different shapes. Plastics do not react easily with air and waste so they do not get oxidised and don't corrode easily. Plastics do not conduct heat and electricity. It is non-biodegradable.

Plastics are also polymer in nature. Polythene is a polymer of ethene.

On the basis of effect of heat, plastics may be classified in two categories :

1. Thermoplastics
2. Thermosetting plastics

1. Thermoplastics

Thermoplastics are those which soften on heating and harder on cooling. Again, they can be heated and moulded into another shape, or we can say, they do not lose their plasticity on repeated heating and cooling, for example, polythene and PVC (Polyvinyl chloride). They are commonly used to make containers, bottles, shoes, handbags, etc.

2. Thermosetting plastics

These plastics, when moulded once, cannot be softened again and lose their plasticity. Most common examples of this type are bakelite and melamine. Bakelite is a good electrical insulator, that's why it is used to make electrical switches, plugs and other electrical fittings. Melamine has a hard and smooth surface and is used to make table tops and crockery.

Table 3.1 : Types of plastics and their uses

S. No.	Name of some plastics	Uses
1.	PET	Fibres, textiles, soft drink bottles, jars, films, microwave packagings
2.	Polyethylene	Supermarket bags, bottles
3.	Polyvinyl chloride	Plumbing pipes, shower, curtains, window frames, floorings

PLASTIC : A HAZARD TO ENVIRONMENT

You must be aware of campaigns and missions that are being carried out all over the world in opposition to plastics. When plastic is such a versatile material, why is it being protested against worldwide? The answer lies in its non-biodegradability. Plastics take more than a 100 years to decompose. So, they are turning our earth into a heap of long-lasting wastes. Disposal of plastic is the main problem.

1. If these are disposed in drains, they lead to choking of drains and disturb the drainage system, causing unhygienic conditions and waterborne diseases.
2. Waste food is dumped in plastic bags. Animals swallow these plastic bags along with their food and get harmed. A large number of aquatic animals also die each year due to polythenes.
3. If plastics are burnt, they evolve poisonous gases. These gases are harmful to the environment.
4. If plastic wastes are submerged in soil, they inhibit reaching of water deep down the soil. So, they affect absorption of water by plants, as well as recharging of groundwater.
5. Manufacturing of plastics involves consumption of a large amount of petroleum which is our exhaustible natural resource.
6. Many colouring agents and other additives are added to plastics to enhance their properties. These chemicals have a negative impact on health if food items are kept in them for a long time.
7. Landfill areas are also being piled up with plastic wastes. Many microorganisms are produced in these garbages, which produce greenhouse gases like methane which contributes to global warming.

Strategies for Plastic Waste Management

1. Plastic bags should be replaced with paper and jute bags which are biodegradable materials. People should carry their own carrier bags when they go to the market.
2. The government should take strict steps to ban plastics. Many states like Jammu & Kashmir, West Bengal and Himachal Pradesh have already banned the use of plastic bags and bottles. Himachal Pradesh government has taken a cabinet decision to ban plastic all over the state since 15th August, 2009 under HP Non-biodegradable Garbage Act, 1995.
Australia, Ireland, China, Austria are among those countries which have banned plastics.
3. Plastic materials should be picked up by rag pickers and recycled to make other useful products like fuel oil.

4. Implementation of effective civic practices by citizens is the most important thing. We can do at least this much :
 - (a) Using garbage bins at public places to dispose plastic bags, bottles and containers.
 - (b) Avoid throwing plastic garbage in water bodies.
 - (c) Follow the government's rules and regulation regarding plastic management .
 - (d) Creating awareness among people and encouraging them to use alternatives of plastics.
5. Practising 3R's, i.e., Reduce, Reuse and Recycle should be encouraged.

◀ REWIND

- Synthetic fibres are made up of polymers.
- Some synthetic fibres are regenerated from natural sources like rayon and some are completely made up of chemicals, like nylon.
- Rayon is a semisynthetic fibre known as artificial silk.
- Nylon is the first truly synthetic fibre.
- Polyester is another synthetic fibre. PET is one of its types used to make bottles and jars.
- Acrylic resembles wool.
- Synthetic fibres are strong, durable, maintenance free and economical.
- On the other hand, these fibres are not suitable for humid conditions and near heat.
- Plastics are of two types thermoplastics and thermosetting plastics.
- Plastics are hazardous to environment.

🔑 ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

1. Plastics that retain their plasticity on repeated heating are called

(a) Thermosetting plastics	☐	(b) Thermoplastics	☐	(c) Thermal plastics	☐
----------------------------	--------------------------	--------------------	--------------------------	----------------------	--------------------------
2. A semisynthetic fibre is

(a) Nylon	☐	(b) Polyester	☐	(c) Rayon	☐
-----------	--------------------------	---------------	--------------------------	-----------	--------------------------
3. Synthetic fibres are

(a) Durable	☐	(b) Wrinkle free	☐	(c) Both (a) and (b)	☐
-------------	--------------------------	------------------	--------------------------	----------------------	--------------------------
4. Synthetic fibres are made up of

(a) Cellulose	☐	(b) Protein	☐	(c) Chemicals	☐
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5. Which of the following is a characteristic of plastics?

(a) Electrical conductors

(b) Heat conductors

(c) Non- biodegradable

6. Rayon is obtained from

(a) Wood Pulp

(b) Plastic

(c) Plants

2 Who am I?

1. I am a fibre, neither truly synthetic nor purely natural. Who am I?
2. I am a very strong fibre used to make parachutes. Who am I?
3. I am the first synthetic fibre. Who am I?
4. I am a type of fabric made by mixing polyester and cotton. Who am I?
5. I don't come from any animal, but I can keep you warm. Who am I?

3 Give two examples of each.

1. Synthetic fibre
2. Types of materials that can be an alternative to plastic bags
3. Thermosetting plastics
4. Materials used to make bottles
5. Varieties of polyester
6. Countries that have banned plastics

4 Fill in the blanks.

1. Synthetic fibres do not absorb _____.
2. Plastics do not conduct _____ and _____.
3. Bakelite is a _____ plastic.
4. Synthetic fibres have a high _____ strength.
5. Terylene is a variety of _____.
6. Synthetic fibres _____ on heating.

5 Answer in a few words only.

1. What was the rayon first named?
2. Why is rayon called semisynthetic?
3. Write two qualities of synthetic fibres.
4. Name the fibre that is the polymer of an amide.
5. What are hot air balloons made up of?
6. Give the names of two types of plastics.

7. Which fibre was the supplement of silk during World War II?
8. What is tensile strength?

6 Answer in one or two sentences.

1. Define thermoplastics and thermosetting plastics.
2. What variations could be done with polyester?
3. How is rayon fibre made from wood pulp?
4. Write down the properties of rayon.
5. Write the uses of nylon.
6. What are the advantages of plastics?
7. What is an ester? How is it significant for making fibres?
8. Comment on reactivity of plastics.

7 Answer in one or two paragraphs.

1. Describe the advantages and disadvantages of synthetic fibres.
2. Write down the properties and uses of polyester.
3. What are the disadvantages of plastics?
4. Explain the properties of plastics.

NEP QUERY



ACTIVITY

- Write an activity to show that synthetic fibres are stronger than the cotton fibre.

(Creativity)

ACTIVITY

- Make a poster on 'Say no to plastics'.

(Art Integration)

COMMUNICATION SKILLS

- Organise a debate on 'plastic pollution'.

(Discussion)

PROJECT WORK

- Paste the samples of various synthetic fibres in your scrapbook and explain their properties and uses.

(Learning by Experiment)

Combustion, Flame and Fuels

5

- Combustion
- Flame
- Combustible and Non-Combustible Substances
- Fuels
- Control of fire

COMBUSTION

Combustion is an oxidation process. This is the process of burning a substance in the presence of oxygen to get heat and light energy from it.

More precisely it can be defined as:

Combustion is a sequence of exothermic chemical reactions between fuel and an oxidant, accompanied by production of heat and change in chemical composition of the substance being combusted.

Energy of food in our body is also released by the process of combustion known as respiration.

Combustion and Burning

Though, combustion and burning seem to be one and the same thing at a glance, but there is a little difference. In fact, burning is the most common type of combustion. Burning necessarily involves production of flame and light energy, along with the heat energy, whereas combustion may or may not involve flame and production of light. For example: respiration is a combustion process, which does not produce flame or light, and so it is not burning.

COMBUSTIBLE AND NON-COMBUSTIBLE SUBSTANCES

Combustible Substances

The substances which can undergo the process of combustion are known as **combustible substances**.

For example: wax, paper, wooden stick (thin), cloth, cotton, kerosene, LPG, etc.



Wax



Paper



LPG



Cloth



Cotton



Kerosene

Non-combustible Substances

The substances which cannot undergo the process of combustion are known as non-combustible substances.

For example, iron, steel, glass, etc.



Iron



Steel



Glass

Types of Combustion

Combustion may be classified into two ways:

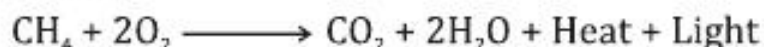
1. On the basis of extent of combustion
2. On the basis of manner of combustion

1. On the Basis of Extent of Combustion

Combustion may be complete or incomplete as per the extent of combustion.

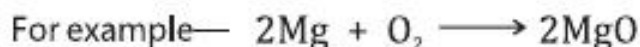
(a) Complete Combustion

In this type of combustion, combustible substance burns completely, without leaving any residue. Ash or smoke are not produced in this type of combustion. Mostly gases, like hydrogen and hydrocarbons undergo this type of combustion. Hydrocarbons produce carbon dioxide, water and heat on combustion.



(b) Incomplete Combustion

Most of the elements do not burn completely and produce ash and smoke along with their oxides.



2. On the Basis of Manner of Combustion

(a) Rapid combustion

When the combustible material burns instantly and produces a large amount of heat and light, the combustion is said to be rapid combustion. For example, on turning on the knob of a gas stove, LPG starts burning immediately to produce heat and light.

(b) Spontaneous combustion

If a material burns suddenly without the supply of heat or friction externally, the process is known as **spontaneous combustion**. For example: white phosphorus can catch fire on 35°C . So, in summers, when temperature exceeds 35°C , it catches fire automatically. That's why it is kept immersed in water. Sodium also catches fire on coming in contact with air. So, it is kept immersed in kerosene oil.

Forest fires and coal fires may also occur spontaneously.

(c) Explosion

When during the process of combustion, tremendous amount of heat light and sound are evolved with a blast or force, along with the production of large amount of gases, the combustion is known as **explosion**, For example: fireworks and bombs.

Conditions Required for Combustion

Three things are necessary for combustion to take place.:

1. Presence of combustible substance
2. Presence of supporter of combustion (Oxygen)
3. Attainment of ignition temperature

1. Presence of combustible substance

Combustion cannot take place if the substance to be burned is non-combustible. Only combustible substances can undergo combustion.

2. Presence of supporter of combustion (Oxygen)

Oxygen is supporter of combustion. In absence of air or oxygen, combustion is not possible. That's why we need to breathe in oxygen. It helps in combustion of food through the process of respiration in order to produce energy.

In case of unwanted fire, we cover the site of fire with a non-combustible material like sand, soil or blanket, so that its contact with air breaks off and it does not get enough supply of oxygen and gets extinguished.

3. Attainment of Ignition Temperature

There is a certain minimum temperature for each combustible substance, below which it cannot burn. This is called the **ignition temperature**.

This is also called **kindling temperature**. Ignition temperature is different for different substances. Some substances have a very low ignition temperature like phosphorus, sodium and kerosene and catch fire immediately, on providing a small amount of heat to them. Such substances are called **inflammable substances**. Matchstick has such low ignition temperature that a very little energy produced by friction (i.e. by rubbing it against matchbox), makes it ignite.

CONTROL OF FIRE

We know about three necessary conditions for combustion to take place:

1. Presence of combustible material
2. Presence of oxygen
3. Ignition temperature

Absence of any of these three conditions will extinguish the fire. So, to extinguish the fire, we can either—

- (a) remove the combustible substance
- (b) cut off the supply of oxygen
- (c) lower down the temperature of burning substance below its ignition temperature

Since it is not possible to remove combustible substance from the place of fire. We mainly try to

- (a) cut off oxygen supply
- (b) lower down the temperature
- (c) add a non-supporter of combustion like CO_2

(a) Cutting off the Supply of Air

For this, we mainly use the following ways:

- (i) Covering the burning object with a safety blanket
- (ii) Throwing sand or soil on the fire
- (iii) Throwing water on the fire

Water vapours surround the burning object and cuts off its contact with air.

(b) By Lowering Down the Temperature

Water is very effective and most easily available extinguisher. It not only cuts off the oxygen supply to the site of fire, but also helps in lowering the temperature. When the temperature becomes lower than the ignition temperature, the fire gets extinguished.

Water could be thrown manually on small fires or through fire brigades on big fires. But, it could not be used in case of fire of electrical short circuits, as water can conduct electricity and give an electric shock to a person who is trying to extinguish fire.

Water is also not effective in case of fires of kerosene, petrol and oils, because all of these are lighter than water and float over the water surface and keep on burning.

(c) Adding a Non-supporter of Combustion like Carbon dioxide

Carbon dioxide has a unique quality to extinguish fire. It is heavier than oxygen and makes a blanket around the fire and thus cuts off its contact with oxygen.

FLAME

We generally identify combustion or burning by the presence of flame. The part of burning we see is a flame.

A flame is the gaseous, visible and luminous part of a fire.

All the combustible substances do not produce flame. Wax, paper, cloth, LPG, kerosene, etc produce a flame on burning while coal does not. It simply glows on combustion.

Flame is produced only when a substance vaporizes and undergoes combustion.

In the burning of a candle, first the molten wax of the candle rises up in the wick and gets vaporized. Now, these vapours burn to produce a flame.

The colour of flame may be different for different substances and in different amount of oxygen.

A gas produces a blue flame in the presence of adequate supply of oxygen whereas in case of insufficient supply of oxygen, it burns with a yellow flame.

Flames may be luminous and non-luminous. The flame, a candle produces is a luminous flame which is yellow in colour and is produced by the presence of unburnt carbon particles in it (incomplete combustion of carbon) whereas the blue flame of LPG is the non-luminous one which is produced due to complete combustion of carbon.

Colours of flame produced by some substances are given below:

Candle Flame

If we study the structure of a candle flame, we will observe three distinct zones in it.

1. Dark Zone

This is the innermost zone of a flame. It is black in colour. It is the coldest part of a flame.

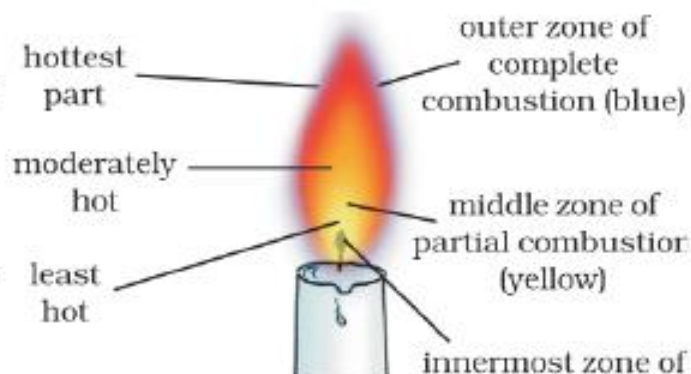
2. Luminous Zone

It is the middle part of a flame. It is moderately hot.

This part looks luminous and yellow.

3. Non-luminous Zone

This is the outermost part of a flame. This is the hottest part of a flame. It looks non-luminous and blue in colour.



FUELS

Fuels are the substances that provide us energy on burning. Coal, petroleum, LPG, etc are some common fuels.

Types of Fuels

Fuels may be classified on the basis of their physical state. So, fuels are of three types:

1. Solid fuels such as wood, coal, cowdung, etc.
2. Liquid fuels such as petrol, kerosene, diesel, oil, etc.
3. Gaseous fuels such as hydrogen, methane, LPG, CNG, biogas, water gas, producer gas, etc.

Calorific Value of Fuels

Calorific value is the amount of heat produced when 1 kilogram of fuel is completely burnt in sufficient supply of oxygen.

For example,

4.5 kg of a fuel burns completely to produce 1,80,000 KJ of heat. Its calorific value can be calculated as

$$\begin{aligned}\text{Total mass of fuel} &= 4.5 \text{ kg} \\ \text{Total heat produced} &= 1,80,000 \text{ kJ} \\ \text{Heat produced from 1 kg of fuel} &= 1,80,000 \text{ kJ} / 4.5 \text{ kg} \\ &= 40,000 \text{ kJ/kg} \\ \text{So, its calorific value is} &= 40,000 \text{ kJ/kg.}\end{aligned}$$

Here, we see that the unit of calorific value is kJ/kg (kilojoules per kilogram).

LPG is also a good fuel for domestic use.

Calorific value of some fuels is listed below.

Name of fuel	Calorific value (kJ/kg)
Cowdung cake	6,000 - 8,000
Wood	17,000- 22,000
Coal	25,000 - 33,000
Petrol	45,000
Kerosene	45,000

Diesel	45,000
Methane	50,000
CNG	50,000
LPG	55,000
Biogas	35,000 - 40,000
Hydrogen	1,50,000

Characteristics of a Good Fuel

Besides having high calorific value, a good fuel must possess some other characteristics also. These are:

1. Their ignition temperature must be low, so that they can catch fire easily.
2. A good fuel should not produce much smoke or other harmful substances as they lead to environmental pollution.
3. A fuel must be cheap, economic and affordable.
4. A fuel should be easy to transport.
5. It must be easy to handle and store.
6. It should burn at a moderate rate, neither too fast, nor too slow.

Effects of Combustion of Fuels on the Environment

Burning of fuels at a large scale is affecting our environment very badly.

1. Fuels having carbon and unburnt carbon particles add smoke and soot to the air. These particles when inhaled cause respiratory diseases.
2. Many harmful gases are released like carbon dioxide and carbon monoxide which are greenhouse gases which lead to global warming.
3. These gases are also harmful for lungs and can even kill a person by suffocation.
4. Acid rains are also a side effect of harmful gases produced by fuels. They affect our buildings, monuments and aquatic life.
5. Overexploitation by mining of coal has made our mountains hollow. As a result they become weak and unable to bear the pressure of heavy rains and end up causing floods and landslides.
6. For obtaining wood, forests are being cut on a large scale. It has also affected the annual rain pattern.

Thus, fuels are very useful to us, but their excessive and careless use can be harmful as well.

◀ REWIND

- Combustion is a process of burning a substance in the presence of oxygen to produce heat and light.
- Some substances can undergo combustion while others cannot.
- Combustion may be complete or incomplete, rapid, spontaneous or explosive.
- The lowest temperature at which a combustible substance catches fire and burns on its own is called ignition temperature.
- Three conditions are necessary for combustion-combustible substance, oxygen and ignition temperature.
- Fire can be controlled by eliminating any of these three from the site of fire.
- Fuels provide us energy by combustion.
- Burning of fuels excessively may be harmful.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

1. A good fuel should be
(a) Cheap ☐ (b) High in calorific value ☐ (c) Both (a) and (b) ☐
2. Which of the following is not necessarily associated with combustion?
(a) Presence of oxygen ☐ (b) Production of heat ☐ (c) Flame ☐
3. Substances which give off heat and light after burning are called
(a) Fossil ☐ (b) Fuel ☐ (c) Flame ☐
4. Combustion is a
(a) Physical process ☐ (b) Chemical process ☐ (c) Reversible process ☐
5. Rusting is an example of
(a) Chemical reaction ☐ (b) Oxidation reaction ☐ (c) Combustion reaction ☐
6. The coolest part of a flame is the
(a) Innermost part ☐ (b) Middle part ☐ (c) Outermost part ☐
7. 2 kg of a fuel produces 1,80,000 kilo joules of heat on burning. The calorific value of this fuel in kJ/kg is
(a) 1,80,000 ☐ (b) 3,60,000 ☐ (c) 90,000 ☐

2 Fill in the blanks.

1. CNG is a _____ fuel.
2. Substances that can undergo combustion are called _____.
3. A material starts burning at its _____ temperature.

- _____ does not give flame on combustion.
- Burning of phosphorus at room temperature is an example of _____ combustion.
- Fuel produces _____ and _____ energy on combustion.

3 State whether the following statements are 'True' or 'False'.

- Wood is a combustible substance.
- LPG is not a clean fuel.
- Oxygen is used to extinguish fire.
- CO₂ is required for burning.
- LPG produces yellow flame on combustion.
- Increased consumption of fuels is harmful.

4 Answer in a few words only.

- Name two combustible and two non-combustible substances.
- Name a physiological process that includes combustion.
- Write down the products of complete and incomplete combustion.
- Name a fuel that is clean, has high calorific value and is good for domestic use.
- What is the luminous visible part of a fire called?
- Which part of a candle flame appears yellow?
- What happens when magnesium is burnt completely?
- Name three conditions required for combustion.
- What do you understand by inflammable substances?
- Name a substance used in fire extinguisher.

5 Answer in one or two sentences.

- Give two conditions when water cannot be used to extinguish fire.
- Explain the various types of combustion.
- What is ignition temperature?
- What is calorific value?
- What are the characteristics of a good fuel?
- What are the ill effects of fuels on environment?
- How does a substance produce flame?
- How does water help in extinguishing a fire?

6 Differentiate between.

- Combustion and burning
- LPG and coal

3. Coal and wood
4. Dark zone and non-luminous zone of a candle flame
5. Combustible substances and non-combustible substances.
6. Oxygen and carbon dioxide.
7. Complete combustion and incomplete combustion.

7 Answer in one or two paragraphs.

1. Explain how CO_2 is able to control fires.
2. Describe the structure of flame with a well labelled diagram.
3. How can we extinguish a fire? Explain.
4. Give reasons for the following.
 - (a) LPG burns immediately on coming in contact with a lighted matchstick while coal takes time.
 - (b) If clothes of a person catch fire, we cover him with a blanket.

NEP QUERY



PROJECT WORK

- ♦ Visit a nearby fire station. Collect information about the working of fire brigade and write a short report on it.

(Learning by Experiment)

ACTIVITY

- ♦ Make a model of the structure of candle flame.

(Art Integration)

Coal and Petroleum

6

- Natural resources
- Petroleum
- Fuels
- Natural Gas
- Fossil Fuels
- Conservation of fossil fuels

All the things that are required to fulfil one's needs are called resources. So, air, water, fuels and plants all are resources. Some resources occur in nature. They are called **natural resources**; like sunlight, soil, air, water, forests, etc. Some resources are man-made; like cement, machines, plastics, etc.

NATURAL RESOURCES

Natural resources may be living natural resources or non-living natural resources. Living resources include plants and animals while non-living resources include air, sunlight, soil, water, etc.

Resources may also be classified as exhaustible natural resources and inexhaustible natural resources. Exhaustible natural resources get spent up in a due course of time by continuous consumption like forests, coal and petroleum. Inexhaustible natural resources include air and sunlight, which never get spent up no matter how much we consume them.

Some of the exhaustible resources get replenished by natural processes in a certain period of time. Such resources are called **renewable resources** like water and forests. On the other hand, some resources get produced at such a slow rate or have such a limited amount of them that once they are used up, they are not replenished by nature. They are called **non-renewable resources**; for example: coal, petroleum, natural gas and minerals.

FUELS

We need energy for various activities like heating, cooking, boiling, running automobiles and generating electricity. We get this energy from fuels. Fuels give energy by burning. But anything that burns is not a fuel. Rather, 'anything that burns to produce energy in an efficient manner is called a **fuel**.'

Wood, cowdung, charcoal, coal, petroleum, natural gas, CNG, LPG are commonly used fuels. All of these fuels contain carbon.



Coal



Charcoal



Petroleum

FOSSIL FUELS

Natural fuels that are made up by burial of living organisms deep down under the earth for over a long period of time, are called **fossil fuels**. For example: coal, petroleum and natural gas.

FACT

- 90 million barrels of petroleum are consumed each day.
- World coal consumption is more than 5.3 billion tonnes per year.

Fossil fuels may be divided into three categories:

1. Solid — Coal
2. Liquid — Petroleum
3. Gas — Natural gas

Coal

Coal is one of the main sources of energy worldwide.

The word 'coal' is derived from the old English term col that means 'the mineral of fossilized carbon'. Coal is a black-coloured hard substance that mainly contains carbon. But other minerals are also present in it in fractions. They are hydrogen, oxygen, sulphur and nitrogen. Some other minerals are present in parts per millions (ppm).

Coal is extracted from coal mines through mining. Surface mining and underground mining are two main techniques of mining.

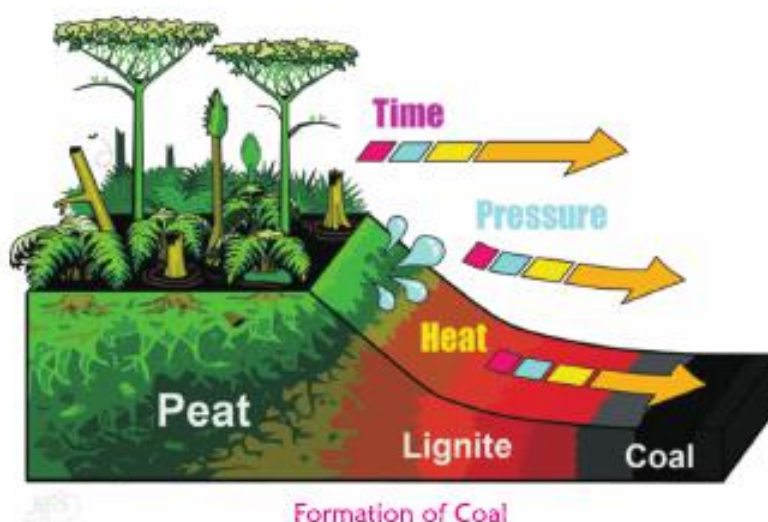
Formation of Coal

Millions of years ago, trees, plants, ferns and forests got buried below the rocks, soil and sand. It is believed that microorganisms and bacteria got developed in that buried up mass. Under the anaerobic conditions, those bacteria decomposed the organic matter. The soil that was deposited over this matter was also fertile, so more plants grew on it. These new plants also died and further got buried in by soil and rocks. This process went on for millions of years.

Meanwhile, the earthquakes and volcanoes also contributed in destroying the large areas covered with plants, vegetation and forests, and ground them up in soil. Thus, the older dead organic matter was pushed deeper down in the earth and new dead organic matter was buried over that.

This led to the conditions of high pressure and heat. These conditions along with the anaerobic conditions, turned the carbon enriched organic matter of wood into coal. This is a very slow process and completes in centuries.

This slow process of conversion of wood into coal is called **carbonisation** and the time period taken to complete this process is called **carboniferous age**.



Varieties of Coal

Depending upon the amount of carbon content, coal may be of four types :

1. **Peat:** This is the most inferior and softest form of coal. Its carbon content is very low. It has a large amount of moisture and is not much suitable to be used as fuel. Its carbon content is 20—25%.
2. **Lignite:** Lignite is comparatively harder than peat, but still is very soft. Its carbon content is more than that of peat. It also has a very low amount of moisture. Its carbon content is 60—70%.
3. **Bituminous:** Its carbon content is even higher, as it is free from moisture and other impurities. That's why it may be used as fuel, but it produces much pollution on burning. Its carbon content is 65—85%.
4. **Anthracite:** This is the highest grade coal. This is also known as hard coal. It has maximum amount of carbon. It produces a very little smoke and burns for a longer time. Its carbon content is 92—98%.

Destructive Distillation of Coal

Destructive distillation is the process of heating a substance strongly in absence of air to obtain other useful substances from it. Destructive distillation of coal is done at $1,000^{\circ}\text{C}$ in absence of air to obtain many other useful organic and inorganic substances from it. They are

(i) Coke

On heating the coal strongly, the volatile impurities escape out from it and solid, pure carbon residue is left behind. This is coke. It is tough porous and greyish black in colour. This is 98% pure carbon. It is very useful as fuel, as it produces a very little smoke.

(ii) Coal Tar

It is a black-coloured, thick, viscous liquid.

We can get benzene, toluene, naphthalene, phenol, etc from coal tar. The pitch left behind is used for construction of roads.

(iii) Coal Gas

It is a mixture of gases. These gases are mainly hydrogen, methane and carbon monoxide. Coal gas works as an efficient fuel for industrial purposes.

(iv) Other Products

Ammoniacal products are also obtained during destructive distillation of coal, which are used as fertilisers.

PETROLEUM

This is the liquid form of fuel. The word 'petroleum' is derived from *petra* meaning 'rock' and *-ileum* meaning 'oil'. That means, it is the oil that is obtained from rocks. It is also known as crude oil and found trapped between the layers of impervious rocks under the ground.

Petroleum is a dark coloured viscous fluid and like all the other oils, it is lighter than water. That's why it floats over water surface. It is drilled out through oil wells.

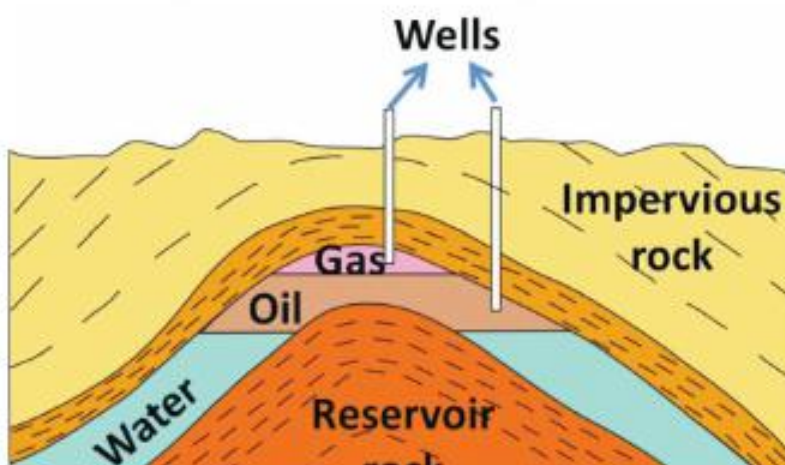
Petroleum is a mixture of different hydrocarbons. It is made up mainly of carbon and hydrogen compounds, but also contain significant amounts of nitrogen, sulphur, oxygen and some other metals like nickel, vanadium etc. It undergoes fractional distillation to yield petrol, kerosene, diesel and other chemicals used in manufacture of plastics and pharmaceuticals.

A list of various products obtained during refining of petroleum is given below :

Fraction	Boiling point range	Uses
Petroleum gas	up to 20°C	Fuel (LPG)
Petrol (Gasoline)	30°C to 120°C	Fuels for vehicles
Naphtha	130°C to 150°C	Chemicals for plastics and insecticides
Kerosene (Paraffin oil)	180°C to 210°C	Jet fuel
Diesel oil	270° C to 340°C	Automobile fuels
Lubricating oil	Above 350°C	Polishes and Waxes
Fuel oil	Above 500°C	Fuel for ships and power stations

Formation of Petroleum

Petroleum is formed by the burial of aquatic plants and animals below the sea bed. The marine zooplankton (animals & plants) died thousands of years ago and settled down in the bottom of sea. Soil, sand and rocks got deposited over it. Microorganisms decomposed this organic matter in anaerobic conditions and is converted into a waxy material called **kerogen**. Kerogen, with the action of heat and pressure, further got converted into liquid and gaseous hydrocarbons. This process is called **catagenesis**. The liquid hydrocarbons thus formed infiltrated through ground and reached up to non-porous or impervious rocks and got trapped between the layers of these rocks. Gaseous hydrocarbons (natural gas) remained above it.



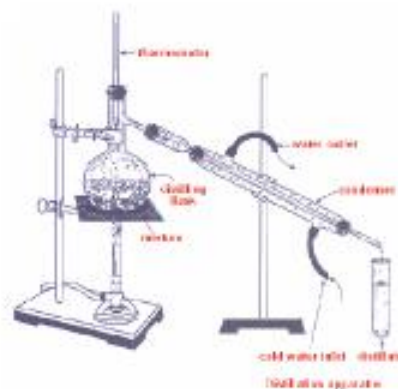
Petroleum has such a great significance that it is named as 'black gold'. Petroleum industry involves exploration, extraction, refining and transporting.

Refining

Petroleum is a mixture of various hydrocarbons which have different number of carbon atoms in their molecules. Separation of different constituents of petroleum through fractional distillation is called refining of petroleum.

Principle of fractional distillation

Fractional distillation is done in a tall column, called **fractionating column**. Crude oil is heated up to 4000°C in a furnace. The vapours thus formed are made to enter the bottom of the fractionating column. Here, the hydrocarbons of highest boiling point (above 500°C) condense as fuel oil. The vapours further rise up in the column and different fractions of it condense at different heights such that the fractions having higher boiling points condense on lower side and those of lower boiling points on the upper side of the column. Hydrocarbons having 14 carbon atoms are gases and escape out of the column.



NATURAL GAS

During the decomposition of marine organic matter millions of years ago, gaseous and liquid hydrocarbons were produced. Liquid hydrocarbons seeped down and got trapped between the

layers of impervious rocks. Gaseous hydrocarbons remained as a separate layer over the crude oil layer. It is known as natural gas. This is a mixture of various gases, but its main component is methane.

Natural gas is a very effective fuel used in industrial and domestic purposes. It is a pollution free fuel. It is transported through pipelines. It is stored under high pressure as Compressed Natural Gas or CNG.

CNG is used for power generation and for running automobile engines. It also has a significant use in manufacturing of many chemicals. On strong heating, it decomposes into carbon and hydrogen.

CONSERVATION OF FOSSIL FUELS

We should take smart steps to conserve these resources in the following ways :

1. We should switch off fans, lights, coolers and other electrical devices when not in use.
2. We should use energy efficient methods while consuming our resources. For example: we can save domestic fuels like LPG by
 - (i) cooking food on low flame
 - (ii) covering our cooking vessels while cooking
 - (iii) soaking the pulses before cooking
 - (iv) cutting the vegetables in small pieces
 - (v) using pressure cookers
3. Devices that consume less energy should be preferred like CFLs and LED lights.
4. Instead of use of individual vehicles by every single person, use of public transport and carpool should be encouraged to save petrol.
5. Optional resources of energy like tidal or solar energy should be developed and encouraged.
6. Instead of generating electricity by burning coal or petroleum, energy of water reservoirs like dams should be brought in use.
7. Public awareness programmes should be conducted in order to conserve energy.

◀ REWIND

- We use natural and man-made resources to meet our requirements.
- Fossil fuels are exhaustible natural resources.
- Coal, petroleum and natural gas are main fossil fuels.
- Coal is formed by burial of terrestrial organic matter deep under the earth.
- Coal is mainly of four types peat, lignite, bituminous and anthracite.

- Destructive distillation of coal can produce, coke, coal tar and coal gas.
- Petroleum is formed by the action of heat, pressure and microorganisms on buried marine animals and plants in the sea bed.
- Different components of petroleum are separated by refining.
- Natural gas is found over the layer of petroleum underground.
- Fossil fuels are limited in nature and are being consumed at a large rate, so we need to use them judiciously and conserve them.



ASSIGNMENTS

COGNITIVE SKILLS

(Learning by Doing)

1 Tick (✓) the correct option.

- | | | | |
|--|------------------------------------|--------------------------------|----------------------|
| 1. A less power consuming device is | (a) Bulb | (b) CFL | (c) Fan |
| 2. Natural gas mainly consists of | (a) Methane | (b) Ethane | (c) Propane |
| 3. The softest quality of coal is | (a) Anthracite | (b) Bituminous | (c) Lignite |
| 4. Petroleum is mainly formed by the burial of | (a) Terrestrial plants and animals | (b) Aquatic plants and animals | (c) Both (a) and (b) |

2 Fill in the blanks.

- Coal was first discovered in _____.
- Coal may be of _____ types.
- The _____ obtained from coal tar is used in construction of roads.
- The rate of consumption of fossil fuel is _____ than the rate of its production.
- Fossil fuels are _____ natural resources.

3 State whether the following statements are 'True' or 'False'.

- The full name of CNG is 'Coal and Natural gas'.
- Carpool is an effective way to conserve fuel.
- Anything that burns is a fuel.
- Fossil fuels are found deep inside the earth.
- Coal does not contain any other element than carbon.
- Earthquakes and volcanoes play an important role in the formation of fossil fuels.

4 Answer in a few words only.

1. Name the component of coal tar used to make moth repelling balls.
2. Name the substance obtained by destructive distillation of coal that contains maximum amount of carbon.
3. Name the process of separating the components of petroleum.
4. What is kerogen?
5. At what temperature is destructive distillation of coal done?
6. What is the process of conversion of wood into coal called?
7. What do you understand by carboniferous age?

5 Answer in one or two sentences.

1. Write about the varieties of coal.
2. Write the classification of resources.
3. Write a short note on coke.
4. State about the components of petroleum.
5. How is refining of petroleum done?
6. How does natural gas exist in nature?

6 Answer in one or two paragraphs.

1. How is coal formed?
2. Explain the process of formation of petroleum.
3. Write about the various products obtained by destructive distillation of coal.

NEP QUERY



CRITICAL THINKING

- ♦ 9 words selected from this chapter are hidden in the following word mesh. Can you find them out?

(Self-Awareness)

P	A	N	T	C	M	C	N	D	P
E	E	A	K	P	N	R	B	I	U
A	S	T	E	v	C	O	K	E	N
T	G	U	R	S	O	C	M	S	O
L	I	R	O	O	A	P	Q	E	R
C	F	A	S	S	L	V	W	L	X
B	H	L	E	T	U	E	F	Y	Z
L	I	G	N	I	T	E	U	L	C
R	T	A	E	T	O	M	E	M	R
V	U	S	W	I	A	E	L	A	K

COMMUNICATION SKILLS

- ♦ Organise a debate in your classroom on the topic 'should the world stop using fossil fuels?'

(Discussion)

OBSERVING SKILLS

- ♦ Make a list of various materials used by us in daily life and classify them as natural and man-made.

(Emotional Thinking)

Cell-Structure and Functions

7

- Cell—the basic unit of life
- Classification of Organisms on the Basis of Cell Number, Cell Shapes and Cell Size
- Cell structure and function
- Discovery of the cell
- Prokaryotic and Eukaryotic cells
- Cell division and growth

A variety of organisms are found on this planet. Have you ever thought why these organisms possess certain similarities and differences?

Some of the features that differentiate these organisms are—:

Size: Some organisms are microscopic in size (e.g., microorganisms like bacteria, fungi, etc), while others are macroscopic (e.g., plants and animals).

Appearance: Different organisms possess different appearances (e.g. a giant elephant, a small rat, a fat human being, a small baby, etc).

Colour: When it comes to colour, we find that even the human beings possess different types of skin colour. Similarly, other organisms also have different colours which differentiate them from each other.

Habitat: Some organisms are aquatic, while others are terrestrial. There may be aerial or arboreal organisms too.



Animal in Habitat

CELL - THE BASIC UNIT OF LIFE

The only similarity between all the organisms is that all are made up of cells.

The cell is the basic structural and functional unit of all the organs of the living organisms. Cells may be compared to bricks of a building. Buildings though have different designs and shapes but all of them are made up of bricks. Similarly, in the living world, organisms differ from each other but all are made up of cells. Further, these cells together form complex living structures in the living beings, which the non-living bricks are not able to do.

All the living organisms are made up of one or more cells. Cells help the organisms to grow and survive. They also help the organisms to respond to the external environment and stimuli. A cell depicts all the characteristics of a living thing that include respiration, nutrition, response to stimuli, etc.

The egg of a hen represents a single cell and is big enough to be seen with naked eyes.

DISCOVERY OF THE CELL

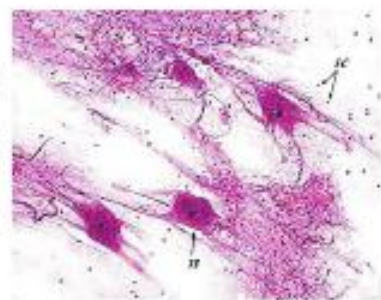
An English botanist Robert Hooke, in 1665 examined the slices of a dried section of the cork (a part of the bark of a tree) under a microscope. He observed small chambers in the slice of cork. These small chambers appeared to be like those present in a honeycomb. Therefore, he named this structure as honeycomb structure. Hooke then called these small chambers cells. (The word 'cell', means 'a little room' in Latin).

In 1839, the German botanist Matthias Jacob Schleiden and the German physiologist Theodor Schwann, together proposed the cell theory. The cell theory comprises of the following postulates:

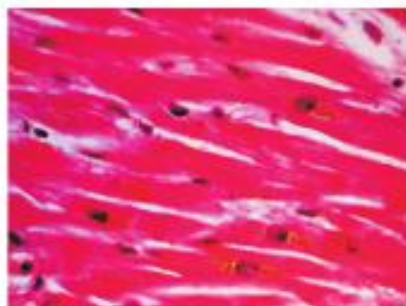
1. All the living organisms are composed of cells. It is the basic structural and functional unit of life.
2. All the cells are produced by the division of pre-existing cells. This phenomenon is called reproduction of cells, i.e., new cells are formed from the old cells.
3. All the basic chemical and physiological functions like growth, repair, movement, immunity, digestion, respiration, etc are carried out by cells.
4. The structure of all the living organisms is determined by the organization of these small units called cells.

Organization of Cells in Organisms

We have learnt that all the living organisms are composed of cells. All the cells that are similar in shape, size and function are grouped together to form a tissue. Therefore, a tissue is a group of similar cells performing a specific function. Nerve tissues and muscle tissues are examples of tissues.



Nerve Tissue



Muscle Tissue

An organ is a structure that contains more than one type of tissues and is visible to the naked eyes. For example, the heart, the stomach, the brain, etc are the organs present in animals, and the roots, the stem, and the leaves, etc are the cellular structures found in plants.

Usually, a number of organs work together to perform a specific function. A collection of organs working together for a particular function is called an organ system.

The organ systems perform various functions. Some such functions are listed in the table 8.1.

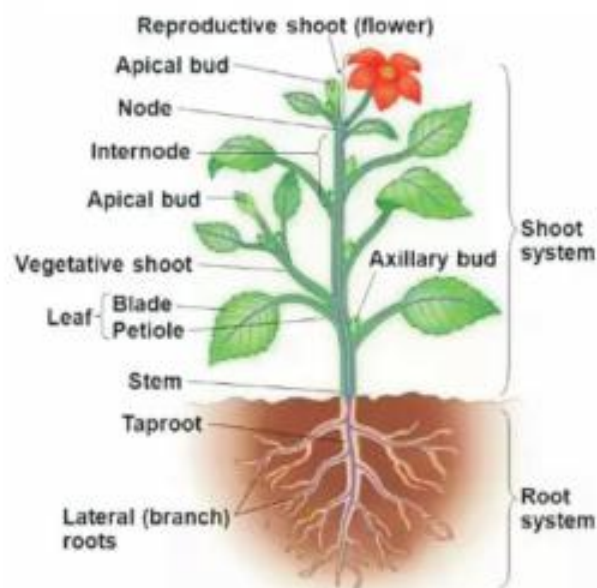
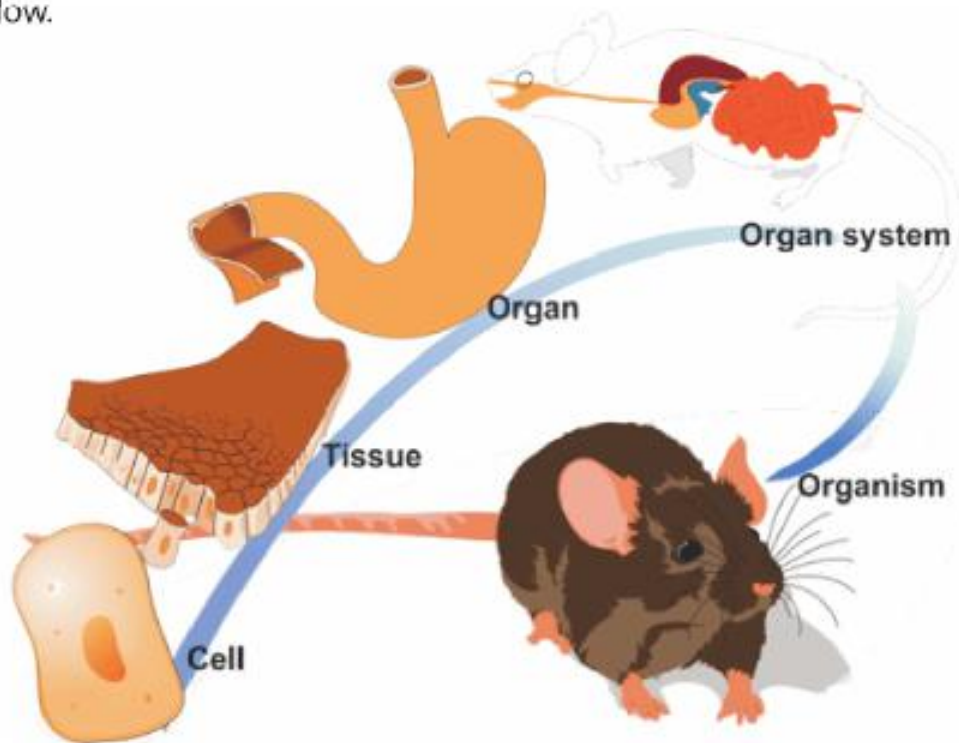


Table 8.1: Various vital processes/functions performed by different organs and organ system of plants and animals respectively

Process	Function	Organs in Animal	Organs in Plants
Movement Growth	All living beings move All living things increase in size	Muscle and skeletal system Digestive system	Plants as a whole do not move Xylem and phloem
Reproduction	Producing living beings of the same kind	Reproductive system	Flowers
Sense Excretion Nutrition	Response to any stimuli Removal of waste products from the body Intake and break down of food to provide energy for all other life processes	Nervous system Digestive, urinary and respiratory systems Digestive system	Plants do not sense Leaves Storage organs, leaves, etc.

The different organ systems working together further constitute to form an organism. See the figure given below.

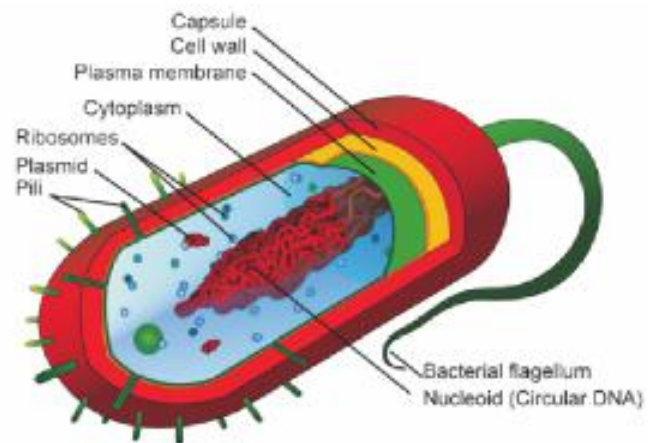


CLASSIFICATION OF ORGANISMS ON THE BASIS OF CELL NUMBER, CELL SHAPES AND CELL SIZE

Cell Numbers

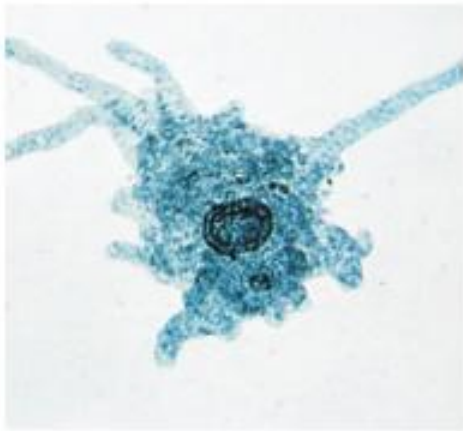
All the living organisms vary in shape and size because of the number of cells present in them. Organisms like amoeba, paramecium, bacteria, etc, which consist of only one cell, are called **unicellular organisms**. A single-celled organism also performs all the necessary functions like respiration, growth, digestion, movement, etc, like those performed by a multicellular organism.

Organisms made up of more than one cell are called **multicellular organisms**. You will be surprised to know that the human body has trillions of cells which vary in their shape and size. A variety of functions are performed by different groups of cells. The dog, fish, cat, mango tree, elephant, sunflower plant, and other large organisms are some examples of multicellular organisms.

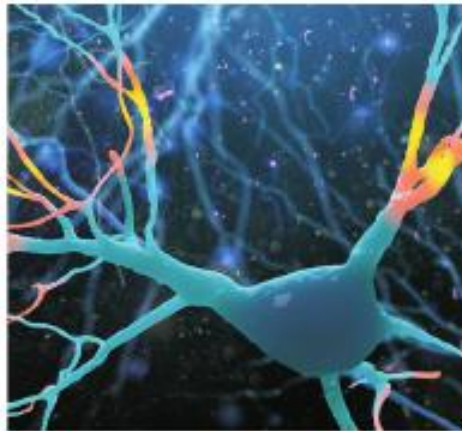


Cell Shapes

Cells exist in different shapes. The shapes can be round, irregular, polygonal, pointed and rectangular; depending upon the function that a particular cell performs.



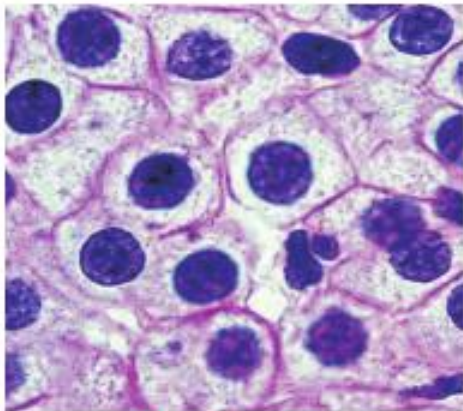
Amoeba cell



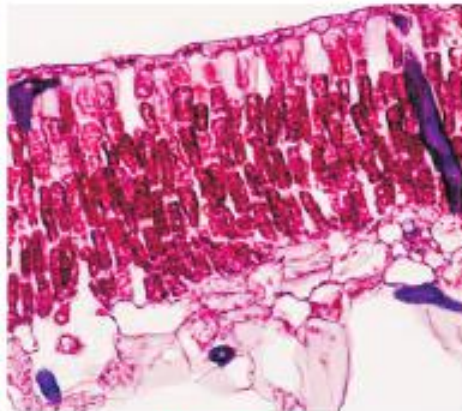
Nerve cell



Red blood cell



Human skin cell



Mesophyll cell



Guard cells

Cell Sizes

Most of the cells are so small that they cannot be seen with naked eyes. Cell size may vary from a micrometre to a few centimetres.

Bacteria are believed to be the smallest cells. These cells generally range in size from 0.1 to 0.5 micrometre.

An ostrich egg is the largest cell amongst all. It measures about 170 mm × 130 mm and is about 25,000 times bigger than a red blood cell. This cell is big enough to be seen with naked eyes.

Human nerve cells are the longest cells.

Viewing Cells

An instrument called **compound microscope** is generally used to view a cell. The object to be viewed under the microscope is termed as a **specimen**. In order to hold the specimen in the correct position, a thin sheet of glass called **microscopic slide** is used. Another much thinner glass slide, called the **coverslip**, is used to cover the sample to make its view much clearer. Moreover, it also prevents the direct contact of the microscope with the specimen.

Below is the procedure to be followed while using a microscope:

1. The mirror of the microscope should be first cleaned to make the view clear.

2. Place the slide in an appropriate position.
3. Adjust the lenses (eyepiece and objective) with the help of your lab assistant.
4. Observe the sample through the eyepiece.

CELL STRUCTURE AND FUNCTION

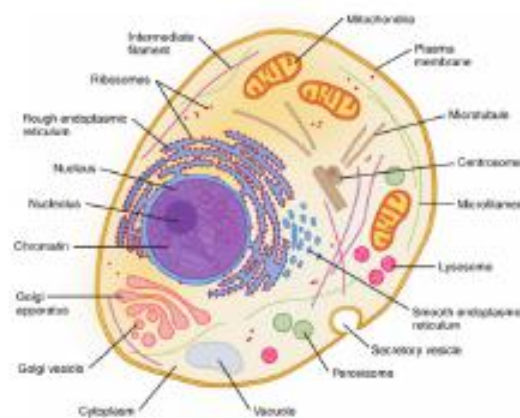
Cells may vary in their structure. Let us now learn about the basic parts of the cell.

(A) Cell Membrane or Plasma Membrane

Cell membrane or plasma membrane is a thin, delicate membrane surrounding the cytoplasm. The cell membrane or plasma membrane performs the following functions:

- It separates the cells from one another and also separates the cells from the surrounding medium.
- It gives a definite shape to the cell.
- Being porous, it allows the movement of substances from both inside and outside the cells.
- This helps in regulating the movement of materials through the cells.

In addition to the cell membrane, there is an outermost thick layer in plant cells called **cell wall** which is made up of a non-living and a rigid substance called the **cellulose**. This additional layer surrounding the cell membrane is required by the plant cell for its protection against temperature variations, atmospheric moisture and high speed winds. Plant cells need protection because they are unable to move from their place in such situations. The cell wall also provides rigidity to the plant cell.



(B) Cytoplasm

The jelly-like substance present between the cell membrane and the nucleus is called **cytoplasm**. There are various other tiny components present in the cytoplasm. They are called the **cell organelles**.

Some of the cell organelles are mentioned below:

Mitochondria

They are called the power houses of the cell because they act as the source of energy production. They are the organelles that act like the digestive system of the cells because they take in the nutrients, break them down and produce energy for the cell. The process of creating cell energy is known as **cellular respiration**.

Mitochondria are very small and rod-shaped. The number of mitochondria in a cell depends on what the cell needs to do. For example, a cell like the muscle cell, which needs more energy to perform its activities, has more number of mitochondria. Both the plant and animal cells possess mitochondria.

Golgi Bodies

They are responsible for the secretion of chemical substances like enzymes, hormones and proteins. Hence, they are called the 'post office' of the cell. They are made up of vesicles (small vessels), tubules (long tube-like structures) and vacuoles (empty spaces). They are present in both plant and animal cells.

Ribosomes

They are the small granular structures scattered in the cytoplasm. They are the protein builders or the protein synthesisers of the cell, therefore, they are also called the 'protein factories of the cell'.

Centrosome

It is located in the cytoplasm, usually close to the nucleus. It initiates and regulates cell division. This structure is present only in animal cells.

Plastids

Plastids are the largest of all the cell organelles, but they are present only in plant cells.

There are three types of plastids found in plant cells. They are chloroplasts, chromoplasts and leucoplasts

Chloroplasts

The process of photosynthesis takes place with the help of chloroplasts, which contain a green coloured pigment called chlorophyll. Chloroplasts thus, help plants in making their food; therefore they are also referred to as the 'kitchen of the plant cell'.

They are situated in those parts that receive sufficient light, like the leaves, which are green in colour.

Chromoplasts

They are non-green coloured plastids. They may be orange, yellow or red in colour. They are found in petals of flowers and fruits. They contain two pigments which are responsible for their colour. These two pigments are carotene and xanthophyll.

Leucoplasts

They are colourless plastids as they do not contain any pigment. They are present in those parts which are not exposed to light, such as roots and seeds. These plastids help in the storage of food.

Lysosomes

They are small spherical sac-like structures present in the cytoplasm of animal cell. They hold the digestive enzymes that are created by the cell and thus help in digestion. When the cells absorb food, lysosomes attached to it release their digestive enzymes.

These enzymes then break down the food into simple sugar and proteins. During the breakdown of cell structures, when the cell gets damaged, lysosomes may burst and the enzymes eat up their own cells. Therefore, they are called as suicidal bags of a cell.

Endoplasmic Reticulum (ER)

This complex organelle helps in the synthesis, storage and transportation of materials within the cells. Both plant and animal cells have endoplasmic reticulum.

Vacuoles

Vacuoles are single-layered membrane bounded by sac-like vesicles present in the cytoplasm. A plant cell has a larger vacuole compared to an animal cell. Vacuoles store excess water, useful minerals, pigments and many other substances.

(C) Nucleus

Nucleus is a dense round body found in the centre of an animal cell and mostly on the periphery of the plant cell. The nucleus controls all the activities in a cell like digestion, movement of substances within the cell, etc. It also controls the process of cell division (reproduction). Therefore, nucleus is also known as the '**brain of the cell**'.

Nucleus consists of four major parts. They are:

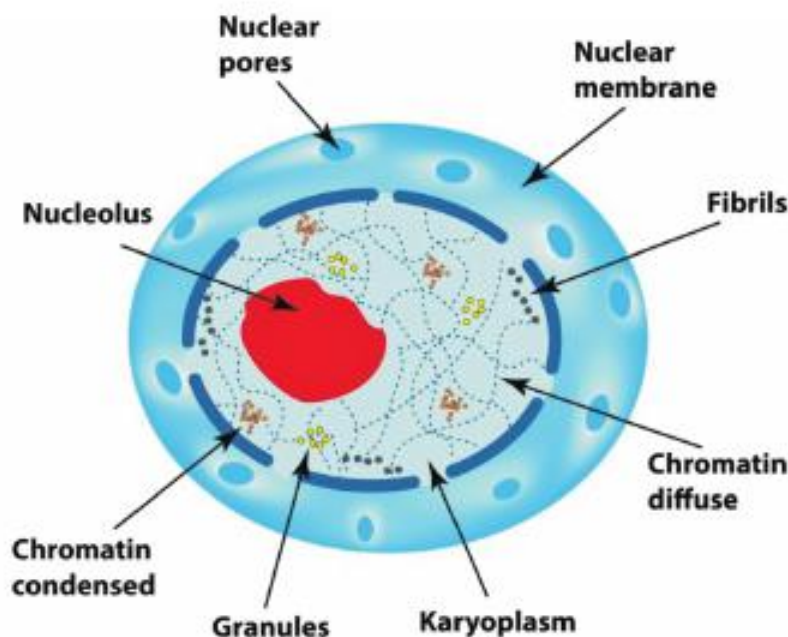
- | | |
|---------------------|----------------|
| 1. Nuclear membrane | 2. Nucleoplasm |
| 3. Nucleolus | 4. Chromatin |

The nucleus consists of a jelly-like liquid called **nucleoplasm** surrounded by a plasma membrane called **nuclear membrane**. The nuclear membrane consists of small pores and spaces that allow certain substances like proteins to flow in and out of the nucleus.

A spherical body present at the centre of the nucleus is called the **nucleolus**. It is made up of proteins and RNA (Ribonucleic acid).

Thread-like structures present in the nucleoplasm are called as **chromatin**. A chromatin is made up of DNA (Deoxyribonucleic acid), RNA (Ribonucleic acid) and proteins. DNA and RNA are the nucleic acids present inside the cell and contain the basic hereditary units called **genes**. As the cell divides, the chromatins condense to form chromosomes.

These chromosomes carry genes which are responsible for the transmission of characters from one generation to the next, i.e. from the parents to the offspring.



PROKARYOTIC AND EUKARYOTIC CELLS

On the basis of the nucleus, cells are divided into two types eukaryotic cells and prokaryotic cells.

Cells that have a well-defined nucleus are known as **eukaryotic cells**. This term has been derived from the combination of the two words 'eu' meaning 'true' and 'karyon' meaning 'nucleus'. Animals cells and plant cells are eukaryotes.

Cells that do not have a well-defined nucleus are known as **prokaryotic cells**. This term has been derived from two words 'pro' meaning 'primitive' and 'karyon' meaning 'nucleus'. Bacteria, blue-green algae and viruses are considered as prokaryotes.

CELL DIVISION AND GROWTH

The growth of the body is directly proportional to the cell growth. The growth of the cells lead to the increase in the size of the cell. After reaching a certain size, cell division takes places. Cell division is a phenomenon by which a fully grown cell divides into two daughter cells. This cell division continues, thus increasing the number of the cells in the body. As the number of the cells increases in the body, the body grows gradually.

The cell division is also responsible for healing wounds by replacing the dead and damaged cells with the new ones.

The outermost layer of the skin is covered with dead cells. An average adult carries around about 2 kg of dead skin. Billions of tiny fragments of the skin are lost every day. Every time you run your finger on a dusty table, you shed a lot of old skin.

Conditions like cancer are caused when cells divide in an imbalanced number.

◀ REWIND

- Cells were first observed by Robert Hooke in 1665.
- In 1839, cell theory was proposed by German scientists Schleiden and Schwann.
- Organisation of cells in an organism occurs in the following manner:
similar cells → tissues → organs → organ systems → organisms
- Cells are found in different shapes such as irregular, long and branched, spindle, round or spherical, etc.
- Cell size may vary from a micrometre to a few centimetres.
- In a human body, the smallest cells found are the blood cells and the largest cells are the nerve cells.
- An ostrich egg is the largest cell among all.
- Cell membrane or plasma membrane, cytoplasm and nucleus are the three basic components of a cell.
- Mitochondria, golgi bodies, ribosomes, centrosomes, plastids and lysosomes are some of the cell organelles.
- Nucleus consists of nuclear membrane, nucleoplasm, chromatin and nucleolus.
- On the basis of the nucleus, cells are divided into two types eukaryotic cells and prokaryotic cells.
- After reaching a certain size, the cells start dividing themselves.



ASSIGNMENTS

COGNITIVE SKILLS

1

Tick (✓) the correct option.

(Learning by Doing)

- The structural and functional unit of life of all the organisms is called
(a) Cell ☐ (b) Tissue ☐ (c) Organ ☐
- Which of the following scientists discovered the cell first time by examining the slices of a dried cork?
(a) Robert Hooke ☐ (b) Isaac Newton ☐ (c) Schleiden ☐
- Which of the following structures is called the brain of the cell?
(a) Mitochondria ☐ (b) Cytoplasm ☐ (c) Nucleus ☐
- Which one of the following is called the power house of the cell?
(a) Mitochondria ☐ (b) Chromosomes ☐ (c) Nucleus ☐
- Which of the following is called the kitchen of the plant cells?
(a) Chloroplast ☐ (b) Chromoplast ☐ (c) Leucoplast ☐
- Which of the following is not present in an animal cell?
(a) Chloroplast ☐ (b) Cell wall ☐ (c) Peripheral nucleus ☐

7. Which of the following are the longest animal cell?

(a) Muscle cell

(b) Nerve cell

(c) Platelet cell

8. Which of the following is not a unicellular organism?

(a) Sunflower plant

(b) Amoeba

(c) Paramecium

2

Fill in the blanks.

1. A group of similar cells performing specific functions is known as _____.
2. The irregular shape of an amoeba is due to the formation of _____, which helps it to move and capture food.
3. _____ is a long and branched cell, which helps to control and coordinate the working of different parts of the body.
4. _____ and _____ are enclosed within the cell membrane.
5. The thread like structures present in the nucleus of the cell is called as _____.
6. _____ cells do not have plastids.
7. In addition to the cell membrane, there is a _____, protecting the plant cell.
8. On the basis of cell numbers, organisms are classified into _____ and _____ organisms.
9. The nucleus of the cell is surrounded by a membrane called _____.

3

State whether the following statements are 'True' or 'False'.

1. Some cells have same functions to perform while others don't.
2. New cells are formed on their own.
3. All the organisms (living and non-living) are composed of cells.
4. Robert Hooke postulated the cell theory.
5. The organisation of cells determines the structure of an organism.

4

Answer in a few words only.

1. Name the scientists who proposed the cell theory.
2. Give three examples each of unicellular and multicellular organisms.
3. Name the three basic components of the cell.
4. What is chromatin made up of?
5. Name three different types of plastids present in the plant cell.
6. Name the animals' organ systems and the organs involved in performing the following functions:
(a) Nutrition (b) Excretion (c) Growth

5

Answer in one or two sentences.

1. Why is the cell called the basic structural and functional unit of an organism?
2. What are cell organelles and where are they found?

3. Where are chromosomes present in the cell?
4. How are cells organised in an organism?
5. Mention the functions of the following:

(a) Cell wall	(b) Chloroplasts	(c) Ribosomes	(d) Endoplasmic reticulum
(e) Nucleus	(f) Vacuole	(g) Golgi bodies	
6. Describe in brief how the organisms are classified on the basis of the cell number they exhibit.
7. What do you understand by cell division?
8. Differentiate between

(a) Cytoplasm and nucleoplasm	(b) Chloroplast and chromoplast
(c) Protoplasm and cytoplasm	(d) Prokaryotes and eukaryotes

6

Complete the following sentences.

1. Mitochondria are called the power house of the cell because _____
2. Nucleus is known as the brain of the cell because _____
3. Endoplasmic reticulum helps in _____
4. Plant cells have an additional cell wall above the cell membrane so as to _____
5. Mitochondria act as the digestive system of the cell because _____
6. Leucoplasts are colourless as _____

7

Answer in one or two paragraphs.

1. Differentiate between 'plant cell' and 'animal cell' with the help of a labelled diagram.
2. How did the discovery of cell take place? Explain.
3. Write a note on the 'cell theory'.
4. What is the procedure of viewing cells? Write the procedure to be followed while using a microscope?

NEP QUERY



COMMUNICATION SKILLS

- Conduct a group discussion on the topic, "How the microscope has increased our understanding of diversity of life on the earth". (Discussion)

OBSERVING SKILLS

- List some professions where a microscope is used on daily basis. Research the use of microscope in any of those professions. Write a detailed project report on the particular profession. (Emotional Thinking)

Conservation of Biodiversity

8

- What is biodiversity
- Conservation of Biodiversity
- Threats of biodiversity
- Deforestation
- Reforestation
- Role of wildlife in biodiversity
- Endemic species
- Migration

WHAT IS BIODIVERSITY?

The term 'biodiversity' has been derived by combining the terms 'bio', and 'diversity' which together refer to 'the variety of living organisms present on the earth'. Biodiversity, as a term, includes all the forms of living organisms, such as the plants, the animals and the microorganisms. The distribution of the living organisms is not even on the earth and varies from place to place and form to form.

CONSERVATION OF BIODIVERSITY

Conservation of biodiversity refers to 'the activities performed for conservation or protection of the various forms of living organisms on the earth'. The plants and animals found in a particular area are termed flora and fauna respectively. Table 7.1 shows the comparison of flora and fauna.

Table 7.1: Comparison of flora and fauna

Flora	Fauna
The plant life found in an area is called flora. Examples: mango, apple, teak, etc.	The animal life found in an area is called fauna. Examples: tiger, leopard, elephant, etc.

Biodiversity is an asset to us. It maintain ecological balance, soil fertility, climatic changes and boosts tourism. It helps in maintaining the adequate number of each form of living organism.

THREATS TO BIODIVERSITY

Some of the most significant factors posing a threat to biodiversity include the following:

Destruction of Habitats Due to Overpopulation and Over Consumption

Major imbalance in biodiversity is caused due to the depletion of the natural habitats of both

plants and animals. With the increase in human population, the demand for vacant land for living, production of agricultural products, firewood and timber harvesting, etc, has increased tremendously resulting in destruction of the natural habitats of both animals and plants. This in turn has contributed in disturbing the ecological balance as a whole.

Pollution Caused by Industrialization

With the increase in urbanisation, a number of pollutants have added to the earth's atmosphere with time. This has majorly disturbed the ecological balance of the earth by bringing climatic changes in the earth's atmosphere.

Pollution, therefore, acts as a big threat to the conservation of biodiversity, as it majorly affects the ecosystems existing on the earth's surface.



Overgrazing

Overgrazing is another factor that affects biodiversity. Overgrazing is caused due to sudden increase in the number of animals and leads to slow depletion of one of the forms of plants consumed in bulk by the animals.



Poaching

Poaching means 'killing animals illegally'. Even protected areas are sometimes not completely safe for the wild animals as the poachers have been found to have access to these areas as well. Though the government has banned poaching, yet some efforts still need to be made to prevent such illegal actions.

DEFORESTATION

Forests are biologically rich areas offering a variety of natural habitats to both plants and animals. They play a **crucial** role in the survival and well being of the living beings. Nowadays, forests are being cut to fulfil the increased demands of food, timber, agricultural land, etc. Clearing of earth's forests on a massive scale for various human needs is called deforestation.

Causes of Deforestation

Causes of deforestation **triggered** by humans are as follows:

- Urbanisation is a major factor causing deforestation.
- Building houses and factories has led to wiping out of forests, leading to deforestation.
- Clearing land for cultivation, in order to meet the increasing demand of the growing population is another factor that has majorly contributed towards deforestation.
- Cutting down trees for obtaining products like fuel, paper, wood, etc has also led to reduction of trees in the forests.
- Overgrazing by animals also leads to deforestation in the long run.

Natural reasons for deforestation are listed below:

- Forest fires
- Floods and droughts
- Earthquakes, etc

Consequences of Deforestation

Deforestation has a greatly negative impact on the environment. It disturbs the natural ecological balance. The most negative effect of deforestation is the loss of habitat for large number of species. Forests also serve as a home for rare animals and plants.

Apart from these general effects, there are several other losses caused by deforestation.

1. Deforestation is one of the major contributing factors for global warming. As the level of carbon dioxide continuously increases in the atmosphere due to excessive cutting down of trees, global warming increases perpetually.
2. Deforestation also impacts the climate by increasing the ambient temperature and decreasing the rainfall. Gradually, it leads to droughts.
3. Deforestation contributes in soil erosion, for in the absence of vegetation, the uppermost layer of the soil becomes loose, leading to erosion of soil in fast currents of water or air.
4. Deforestation results in desertification in which the fertile land gradually gets converted into deserts. Desertification happens as a result of soil erosion. Fast currents of water and soil, wipe away the top fertile layer of soil, exposing the lower less fertile and rocky layer and thus, the fertile lands get converted into deserts.

5. It increases the risk of natural calamities like droughts and floods.

REFORESTATION

Reforestation is the best way to conserve our forest and hence helps in conservation of biodiversity.

Reforestation can be defined as the process in which more and more trees are planted in the barren land. In this process, it is ensured that larger number of trees are planted than those being cut down. There are a number of advantages associated with the process of reforestation.

1. This process helps in preventing soil erosion, by keeping the uppermost fertile layer of the soil intact.
2. As forests serve as natural habitats for most plants and animals, the process of reforestation helps in securing the natural habitat of both plants and animals.
3. Reforestation helps in reducing global warming.
4. The trees planted in reforestation help in the formation of more clouds as they transpire larger amount of water in the atmosphere.

ROLE OF WILDLIFE IN BIODIVERSITY

Wildlife includes all non-domesticated animal species living in natural habitats. These wild animals form an important link in the food chain and play a vital role in various nutrient cycles like carbon, oxygen, water, etc.

The Impact of Wildlife Extinction

Wildlife refers to animals that are untouched by human factors. But domesticating wild plants and animal species by humans for their benefit has affected wildlife very badly. Thus, humans play a tremendous role in wildlife extinction. But natural factors like climate change and pollution cannot be avoided too. Let us understand what exactly the word 'extinction' means.

Millions of years ago, dinosaurs existed but their presence is no more on the earth today. A species is called extinct when it stops existing.

Causes of Extinction of Wildlife

Some of the causes of the extinction of wildlife are as follows:

1. Loss of natural habitats caused due to excessive cutting of trees or deforestation.
2. Hunting, poaching, and killing of animals for fun or for the purpose of obtaining their by-products like hide or meat.
3. Natural calamities like forest fires, storms, droughts or floods.
4. Sometimes animals are introduced to places new to their natural existence. Here, they have to compete not only for food and shelter but also for their survival.

5. Excessive use of the natural resources by their increased consumption in the form of food, etc, may also lead to extinction of some of the species. For example, fish, prawns, crabs, etc when consumed in excess lead to extinction of many of their species in the long run.
6. Construction of dams, buildings, etc lead to deforestation and that results in the extinction of some of the species in the long run.
7. Lack of knowledge of wildlife laws and lack of awareness among people keep them in ignorance and thus, become a threat to the naturally existing plants and animals.

Threatened Species

These species require immediate protection and conservation so that they do not fall under the category of endangered, vulnerable and rare species. These threatened species are considered to be the protected species of the country where they are found. Some of the threatened and protected species of Indian wildlife are listed below:

Python, snow leopard, peafowl, musk deer, horned pheasant, sambar, chinkara, gharial, etc.

ENDEMIC SPECIES

These are the species of plants and animals whose habitat is limited to a particular area. They are geographically isolated species not found naturally anywhere else in the world. Pachmarhi Biosphere Reserve nests some of the endemic species of flora and fauna. Some such species are:

Flora—sal and wild mango

Fauna –flying squirrels, Indian giant squirrels and bison

Laws for protection of Forests and Wildlife in India

The government of India has passed some laws in order to protect and conserve our most precious flora and fauna. One such law is the Wildlife Protection Act which was passed in the year 1972 and was amended in the year 1991. Under this act, many protected areas have been established in different states of the country. Forest Conservation Act was passed in the year 1980 and in the year 1986, Environment Protection Act was passed by the government.

Protected Areas Established Under Wildlife Protection Act

Wildlife Sanctuaries

A sanctuary is an area where killing or capturing of any species of animals is strictly prohibited. Hence, this area provides protection to the endangered species. In sanctuaries, animals live in their natural habitats where any kind of human activity is restricted. In India, there are 515 wildlife sanctuaries that protect some of the threatened species like pink-headed duck, Indian elephant, marsh crocodile, blackbuck, etc. Some of the famous wildlife sanctuaries are Bharatpur bird sanctuary, National Chambal wildlife sanctuary, Vedanthangal Bird sanctuary Chinnar wildlife sanctuary.

National Parks

These are the areas reserved by the government for wildlife conservation and protection. In India, there are 98 national parks which are protected for their natural beauty and for scientific interests. Like Periyar national park (Kerala) some of the national parks where animals live in their natural habitats are given in Table 7.4.

Table 7.4: Some national parks and the wild animals which they protect

National Parks	Wild Animals
• Corbett National Park, Uttarakhand • Kaziranga National park, Assam • Bandipur National park, Karnataka • Kanha National park, Madhya Pradesh • Gir National park, Gujarat • Desert National park (The desert), Rajasthan	Bengal tigers One-horn rhinos Elephants Swamp deer (Barasingha) Asiatic lions Great Indian bustards, neelgai, chinkaras and black bucks

Table 7.5: Differences between wildlife sanctuaries and national parks

Wildlife Sanctuaries	National Parks
Wildlife sanctuaries are meant for the protection of particular species like Blackbuck, chinkara etc	National parks are the habitat for particular wild animals like rhinoceros, lion, etc.
The boundaries of wildlife sanctuaries can be increased.	The boundaries of national parks cannot be increased.
Wildlife sanctuaries are 515 in number.	Number of national parks in our country is 98.
Area of wildlife sanctuaries is from 0.61sq km to 7,818sq km.	Area of national park varies from 0.04 sq km to 3162 sq km.
Harvesting timbers, cultivation, collection of forest products are allowed with permission.	These activities are restricted here.
Private ownership may be allowed.	No human habitation is permitted in the main area.

Biosphere Reserves

These are the protected areas meant for conservation of wildlife along with the culture of that area. These are much larger in area as compared to sanctuaries and national parks. A biosphere reserve consists of many protected areas like national parks, sanctuaries, lakes and mountains. In India, there are 18 biosphere reserves for balancing conservation of biodiversity and human use of an area. Some of the biosphere reserves are:

- Nanda Devi Biosphere Reserve
- Pachmarhi Biosphere Reserve
- Simlipal Biosphere Reserve
- Sundarbans Biosphere Reserve

Zoological Parks

Zoos are the places where animals and birds are kept in an artificial habitat for public view. This is done for educating the people regarding the importance and role of animals in the ecosystem. Zoos are found in almost every city.

Botanic Gardens

These are the areas where all the rare and threatened plants are conserved. They also serve as seed banks as seeds are preserved here in controlled conditions. National Botanic Gardens in Howrah, Kolkata and botanic gardens of Ooty and Bengaluru are known for preserving rare species.

Red Data Book

IUCN (International Union for the Conservation of Nature) also publishes a catalogue, called the Red Data Book. It keeps a record of all the endangered species of plants and animals. It gives the statistics of population of species so that special measures can be taken for the conservation and rehabilitation of these species. Each Red Data Book usually deals with a specific group of animals and plants (e.g. reptiles, mosses, insects) etc.

Project Tiger

This project was launched on 1st April 1973 by the government to ensure the survival and maintenance of the depleting tigers in our country. Project Tiger has shown significant increase in the number of tigers over the years. In the year 1972, the number of tigers had got reduced to about 1,800 only, which used to be about 5,000 earlier in the 70's. This had happened due to increased incidents of poaching and hunting during that period of time. However, since the day poaching has been prohibited, and projects like 'Project Tiger', have been initiated, a significant increase in the number of tigers has been observed.

MIGRATION

It is the seasonal movement of animals in groups. Mainly birds and fish migrate from one habitat to another for feeding and breeding purposes. Birds migrate due to climatic changes,

especially for laying eggs at a safer place. Migration of animals takes place at the same time and at the same place every year, for better food and breeding options.

For example,

Siberian cranes migrate from Siberia to Bharatpur Bird Sanctuary (Rajasthan) every year. Similarly, Arctic Terns migrates from north pole to south pole every year.

Some of the advantages and disadvantages of migration are listed below:

Advantages

Through migration, birds and animals are able to:

- Avoid harsh climatic conditions.
- Avoid lack of food during winters.
- Enhance their chances of survival.

Disadvantages

Apart from the benefits associated with migration there are number of drawbacks too associated to migration. Some of them are:

- Exposure to predators and harsh weather conditions.
- Storms or contrary winds sometimes blow away many species of birds.

◀ REWIND

- Conservation of biodiversity is essentially required to maintain an ecological balance on the earth.
- Some threats to biodiversity are overpopulation, industrialisation, overgrazing and poaching.
- Nowadays, forests are being cut to fulfil the increased demands of food, timber, agricultural land, etc.
- Many human activities and natural factors are responsible for deforestation.
- Deforestation causes global warming, climatic changes, soil erosion, desertification and increased risks of natural calamities.
- Reforestation helps in preventing soil erosion, securing natural habitat for wildlife, reducing global warming and maintaining the cycle of rainfall on the earth.
- Recycling waste paper saves trees, time, energy and water which are required to manufacture paper.
- Various reasons behind the extinction of wild animals are loss of natural habitats, poaching, natural calamities, excessive use of natural resources, construction and lack of awareness among the people.

3. How many biosphere reserves are there in India?
4. When was 'Project Tiger' launched?
5. Give one example each of an endemic species of flora and fauna.

4 Answer in one or two sentences.

1. Define the following terms:
(a) Biodiversity (b) Red Data Book (c) Reforestation
(d) Extinction (e) Botanic gardens (f) Ecosystem
2. Why should we conserve biodiversity?
3. Protected areas are sometimes not completely safe for the wild animals. Why?
4. What are the advantages of reforestation?

5 Answer in one or two paragraphs.

1. Differentiate between
(a) Flora and fauna
(b) Wildlife sanctuaries and national parks
(c) Biosphere reserves and wildlife sanctuaries
(d) Endangered species and extinct species
2. Name a few national parks and the animals they protect.
3. What are the factors that pose a threat to biodiversity?
4. What is deforestation? What are its causes and consequences?
5. Define the term migration. Write some advantages and disadvantages of migration.

NEP QUERY



COMMUNICATION SKILLS

- Find out the details of all the protected areas in your state. Gather information about the species protected in such protected areas. Write a report on the basis of your research. (Discussion)

PROJECT WORK

- Write a project report on the topic 'Deforestation'. In your report, describe the effects of deforestation on wild animals, environment, cities, villages and forthcoming generation. Also mention the ways by which deforestation can be checked. (Learning by Experiment)

COMPUTATIONAL SKILLS

- Write a powerpoint presentation on the topic 'Biodiversity Now and twenty years ago'. (Problem Solving)

Force and Pressure

9

- Force and its Effects
- Pressure and its application in daily life
- Kinds of forces
- Atmospheric pressure

FORCE AND ITS EFFECTS

Force is defined as a push or a pull. We often use the word force in everyday life for a push, pull, stretch and squeeze. The following examples will make it clear :

1. To open the door from inside, we push it.
2. To move a pile of rubbish, we push it with a broom.
3. To move a lawn mower, a gardener pulls it.
4. To move a cart, a horse or a bull pulls it.

When force is applied on a body, it can have the following effects:

1. A Force Can Move a Body Originally at Rest

The body at rest is also called a **stationary body**. When force is applied on a body originally at rest, it begins to move, e.g. a toy car originally at rest, when pushed, begins to move. Similarly, when we apply force, we can push a chair, move a bicycle and lift a bucket.

2. Force Can Stop a Moving Body

When a force is applied to a moving body, it can be made to stop, e.g. a moving vehicle can be stopped by applying brakes. The brakes provide the stopping force.

3. A Force Can Make a Moving Body Move Faster

When a force is applied on a moving body in the direction in which the body is already moving, the body begins to move faster, e.g. the speed of the cycle increases when more force is applied on the pedals by the cyclist. The swing moves faster when a person gives a push to it.

4. A Force Can Slow Down a Moving Body

When the force is applied on a moving body in a direction opposite to the direction of motion of the body, the body slows down, e.g. the speed of a moving vehicle is slowed down by applying brakes. The brakes provide a force in a direction opposite to the direction of motion.

5. A Force Can Change the Direction of a Moving Body

If a force is applied on a moving body in a direction other than its direction of motion, the direction of the motion of the body changes, e.g. a player kicks a moving football to change its direction of motion.

6. A Force Can Change the Shape of a Body

When a force is applied on a body which is not free to move, it gets deformed, that is, the shape or size of the body changes, e.g. on pressing a piece of rubber foam, its shape changes.

Thus, from the above discussion, we can say that force is the cause that changes the state of rest or the state of motion of a body or deforms it.

Unit of Force

The standard unit of force is 'newton' denoted by the symbol **N**.

KINDS OF FORCES

Forces can be classified as contact forces and non-contact forces.

Contact forces are those which are applied by actually touching the body, that is, the objects are in physical contact with each other, e.g. muscular force, mechanical force and frictional force.

Non-contact forces are those that act from a distance, that is, they act on the bodies without being touched visibly, e.g. gravitational force, electrostatic force and magnetic force.

Contact Forces

These are:

1. Muscular Force

It is the force exerted by our muscles, e.g. a rickshaw puller uses his leg muscles to move a rickshaw, a labourer uses the muscles of his arms and shoulders to lift the load, a horse uses muscular force to move a cart, etc.

2. Mechanical Force

It is a push or a pull exerted by a machine to move the body or to bring it to rest, e.g. the engine of the car exerts mechanical force to move it.

A machine does not exert a force by itself but it needs energy from other sources to exert a force. The force exerted by wind to move a sailboat is a mechanical force.

3. Frictional Force

It is defined as a force that offers resistance to motion. It comes into existence only when the surfaces of two objects in contact move with respect to each other, e.g., when we stop pedalling a cycle, it gradually slows down and ultimately stops after travelling some distance. A smooth surface exerts lesser force of friction than a rough surface.

4. Elastic Force

It is a force that is produced when:

1. The body is stretched and then released, e.g. a catapult causes a small stone to shoot out when it is stretched and released.
2. The spring is compressed and then released.

Non-Contact Forces

These are :

1. Magnetic Force

The force of attraction or repulsion exerted by magnets is called **magnetic force**. A magnet attracts materials made of iron, cobalt and nickel.

2. Electrostatic Force

It is the force of attraction or repulsion between the two charged bodies.

3. Gravitational Force

It is the force with which the earth pulls all the objects towards its centre, e.g. a ripe fruit falls towards the ground from a tree and water falls down the hills, etc.



Activity

Aim: To demonstrate electrical force

Materials required: A clean comb (free from oil) and some bits of paper

Procedure:

1. Take the clean plastic comb free from oil.
2. Brush your dry hair briskly with it.
3. Now bring this comb near the bits of paper and observe.

Observation: You would observe that the comb would attract the bits of paper.

Conclusion: This happens because an electrical charge is produced in the comb when we brush our dry hair with it. This charge exerts a pull on the bits of paper and attracts them. This charge is electrostatic force.

We have studied about force in detail. When it acts over a large area, only a small effect is produced. On the other hand, when it acts over a small area, its effect gets multiplied. Thus, force acting per unit area of the surface is called pressure.

PRESSURE AND ITS APPLICATIONS IN DAILY LIFE

Force is measured in **newton (N)** and the area in **metre square (m^2)**. So, the unit of pressure is **newton/metre² (N/m^2)**.

$$\text{Pressure} = \frac{\text{force}}{\text{area}}$$

This unit of pressure (N/m^2) is given a new name **pascal (Pa)** in honour of French mathematician and physicist Blaise Pascal who made important discoveries in the study of pressure. Hence, the standard unit of pressure is **pascal**.

$$1 \text{ pascal} = 1 \text{ N/m}^2$$

It is clear from the above formula that pressure depends on two factors:

Force applied: If the force increases, the pressure also increases.

Area over which the force acts: If the area over which the force acts decreases, the pressure increases.

For example— The pointed end of a nail exerts more pressure and thus helps in easy piercing.

Applications of Pressure in our Daily Life

1. Heavy trucks have six to eight wheels in order to increase the area of contact on which their weight acts. As a result, the pressure decreases.
2. It is easy to pierce with a sharp needle.
3. The tools used for cutting like knives, blades, etc have sharp edges, to reduce the area of contact.
4. A camel can easily walk on the sand due to its broad feet. As the area of contact increases, less pressure is exerted.
5. School bags have wide straps made up of thick cloth. This increases the area of contact and puts less pressure on the shoulders of the children.
6. The wheels of army tanks are provided with a wide steel belt over the wheels to increase the area of contact. This in turn reduces the pressure.
7. Foundations of high rise buildings are kept broad and wide so that they exert less pressure on the ground and do not sink due to high pressure of the building.
8. The coolie carries load on the turban.

Pressure Exerted by Liquids

The pressure a fluid exerts on an object in the fluid is applied in all directions. That is because the particles that make up the fluid can move in any direction.

1. Liquids exert a downward pressure. Let us understand it with the help of some activities.



Activity

Aim: To show that liquids exert a downward pressure

Materials required: A test-tube having both of its ends open and a deflated balloon

Procedure:

1. Take the test-tube open at both the ends.
2. Tie the mouth of the deflated balloon at its lower end and pour water into it and observe.

Observation: The balloon bulges downwards.

Conclusion: Thus, liquids exert pressure downwards, i.e., they exert the pressure on the base of the container in which they are kept.

2. Liquid exerts pressure side ways.



Activity

Aim: To show that liquids exert pressure sideways

Materials required: A vessel with tap-like opening and a deflated balloon

Procedure:

1. Take the vessel with a tap-like opening on one of its side.
2. Tie the balloon to that opening.
3. Fill the vessel with water and observe.

Observation: You will find that the balloon bulges outwards.

Conclusion: This shows that liquids exert pressure sideways, i.e. on the sides of the container in which they are kept.

3. Liquids exert pressure in all directions.
4. The pressure exerted by liquids varies with depth.

Measuring the Pressure Exerted by Liquids

The pressure exerted by liquids can be measured by an instrument called the **manometer**. It is easy to make a manometer at home as well.

Applications of the Pressure Exerted by Liquids

1. The wall of the dam is made thicker at the bottom so that it can withstand a greater pressure as we know that the water pressure increases with depth.
2. Water supply tanks are placed at high levels in the buildings such as on terrace or on the top of the buildings. This is because when tanks are placed at high levels, the pressure of water would be large enough to force the water to reach the taps of the houses.

Pressure Exerted by a Gas

Gases too exert pressure on the walls of the container. For example: air in the bicycle tube comes out when it has a puncture.

ATMOSPHERIC PRESSURE

It is the pressure exerted by the air, on the surface of the earth. The earth is surrounded by a layer of air called the **atmosphere**. We live at the bottom of this layer. The air above us presses down on us with a force that is equal to the force exerted by a mass of 1 kg on every square centimetre. This is called the **atmospheric pressure**.

Measuring Atmospheric Pressure

Atmospheric pressure is measured by an instrument called a **barometer**.

Air pressure is measured by the force exerted per unit area of surface.

$$\text{Air Pressure} = \frac{\text{Weight of air column (in newton)}}{\text{area (in metre}^2\text{)}}$$

Uses of Atmospheric Pressure

Atmospheric pressure is used in many of the things the we do.

1. **Slipping drinks using straw** : When we drink a liquid with a straw, while sucking, our lungs expand and air in the drinking straw goes into them. The air pressure in the straw decreases and the atmospheric pressure acting on the surface of the liquid then forces it to move inside the straw and into our mouth some other applications of atmospheric pressure in our daily life are.
2. Flushing toilets
3. Breathing
4. Our body shape is sustained due to air pressure.
5. Nose bleeding as we go to hilly areas.
6. Blowing birthday balloons.
7. Due to atmospheric pressure movement of clouds also takes place.

◀ REWIND

- Force is a push/pull/stretch or a squeeze.
- Force can have various effects on a body. It can move or stop a body, can increase or decrease the speed of the body or can even change its shape.

- In nature, there are various kinds of forces. These can be classified as contact and non-contact forces.
- The standard unit for measuring force is newton.
- Pressure is defined as the force per unit area.
- Pressure on a surface depends on the area over which it acts.
- The standard unit of pressure is pascal.
- The liquid pressure is measured by an instrument called the manometer.
- The pressure exerted by the atmosphere (air) is called the atmospheric pressure.
- The atmospheric pressure is measured by an instrument called the barometer.
- Both the pressure exerted by liquids and atmospheric pressure find various applications in our daily life.

ASSIGNMENTS

COGNITIVE SKILLS

(Learning by Doing)

1 Tick (✓) the correct option.

- | | | | |
|---|-------------------|------------------|----------------|
| 1. The unit of force is | (a) Kg | (b) g | (c) N |
| 2. A body falls downwards because of _____ force. | (a) Gravitational | (b) Muscular | (c) Frictional |
| 3. Pressure is defined as | (a) Area/Force | (b) Area × Force | (c) Force/Area |
| 4. The standard unit of pressure is | (a) Newton | (b) Pascal | (c) Hertz |
| 5. The pressure exerted by liquids can be measured by | (a) Barometer | (b) Speedometer | (c) Manometer |

2 Fill in the blanks.

- The pressure on a surface can be increased by _____ the area of contact.
- The _____ is measured by an instrument called barometer.
- The _____ exert maximum pressure at the bottom.
- _____ and _____ are made to wear special suits when they go to space or in deep sea, respectively.

3 State whether the following statements are 'True' or 'False'.

- The frictional force acts in the direction of the motion of a body.
- Gravitational force is a non-contact force.
- At the same depth, liquids exert same pressure in all directions.

4. We can get crushed by the atmospheric pressure.
5. Force is measured in Newton.

4 Match the column A with the column B.

Column A

1. Moving a pile of rubbish
2. Moving a lawn mower
3. Stretching a spring balance
4. Pressing a lump of wax
5. Frictional force

Column B

- (a) force as a stretch
- (b) force as a squeeze
- (c) contact force
- (d) force as a pull
- (e) force as a push

5 Answer in a few words only.

1. In order to move a pram, we have to apply force. In this case, which type of force is applied, a push or pull?
2. What will a magnet attract out of these-wood shavings, iron nails, erasers?
3. What kind of force is exerted by animals?
4. The pressure exerted by liquids increases with depth. State whether the statement is right or wrong?
5. Name two factors on which the pressure depends.
6. Write the S.I. Unit of pressure and force.

6 Answer in one or two sentences.

1. Give two examples wherein force acts as a squeeze.
2. How do we measure atmospheric pressure?
3. How is ink filled in the fountain pen?
4. Why do knives and blades have sharp edges?
5. Which out of these is a non-contact force-frictional, magnetic, muscular, elastic? Give reasons for your answer.

7 Define the following terms.

- | | | |
|-----------|-------------------|-------------------------|
| 1. Force | 2. Pressure | 3. Atmospheric pressure |
| 4. Pascal | 5. Contact forces | 6. Gravitational force |

8 Answer in one or two paragraphs.

1. List the effects of force explaining each on briefly.
2. Write some applications of pressure in our daily life.
3. Explain the relation between pressure, force and area.

4. Give an activity to show that liquids exert pressure on the sides of container in which they are kept.
5. Write some applications of atmospheric pressure in our daily life.

9

Differentiate between.

1. Contact and non-contact forces
2. Barometer and manometer
3. Elastic and electrostatic force
4. Mechanical and muscular force
5. Gravitational and magnetic force

NEP QUERY



COMMUNICATION SKILLS

- ♦ Sea divers and astronauts wear special suits. Why they do so? Discuss in your class.

(Discussion)

PROJECT WORK

- ♦ Make and use a spring balance to measure force.

(Learning by Experiment)

Reproduction in Animals

10

- Reproduction
- Fertilisation
- Asexual reproduction
- Types of fertilisation
- Sexual reproduction in humans
- Viviparous and oviparous animals

REPRODUCTION

Reproduction is an important process responsible for the continuity of life on the planet Earth. Reproduction is defined as the process of producing the offspring of one's own kind. It helps in increasing the population of a particular species, generation after generation. This is the fundamental feature of the existence of all life forms on the Earth.

There are two modes of reproduction:

1. Asexual Reproduction
2. Sexual Reproduction

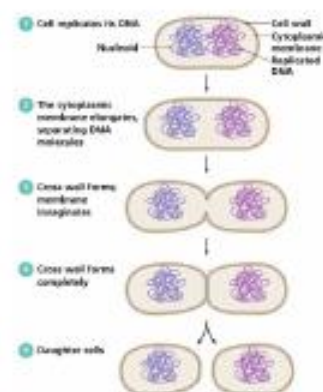
Asexual Reproduction	Sexual Reproduction
1. Involves single parent.	Involves two parents.
2. No production of gametes (sex cells).	Male and female gametes (sex cells) are produced.
3. It produces offsprings that are identical to the parent.	It produces offsprings that are different from parents.
4. Two or more than two offsprings are produced.	One or more than one offsprings are produced.
Example— Hydra, algae etc.	Example— Human beings, animals and plants.

ASEXUAL REPRODUCTION

It is the simplest mode of reproduction in which only one sex is involved. There are two forms of asexual reproduction:

1. Binary Fission

In binary fission, the parent cell divides into two equal individual cells. The nucleus divides into two parts; thus each part receives a nucleus and is followed by division of cytoplasm into two parts. A constriction develops in the body of the mature unicellular organism which depends to form two identical daughter cells. Each daughter cell finally grows into a full size cell. It occurs in unicellular organism like amoeba, paramecium etc.

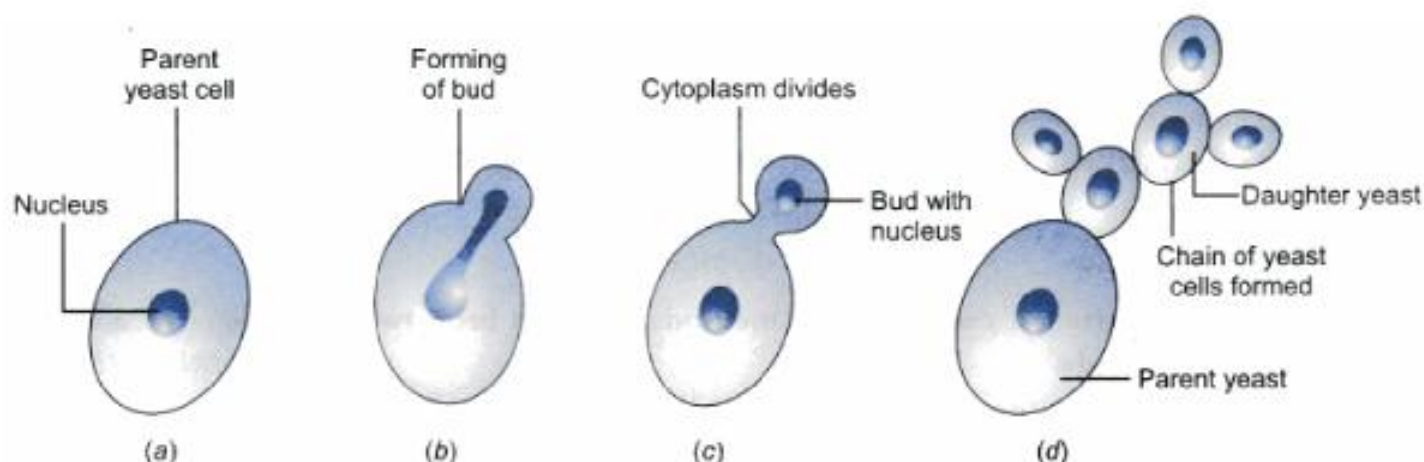


2. Budding

This kind of reproduction can be seen in hydra and yeast. Budding is the process of reproduction in which new individuals develop as outgrowths from a single parent. These outgrowths are referred to as buds. These buds slowly enlarge and separate themselves from the parent's body to grow into a new young organism.



(a) Single parent (b) Development of a bud (c) Growth of the bud (d) Detachment of the bud to form a new young organism



3. Cloning

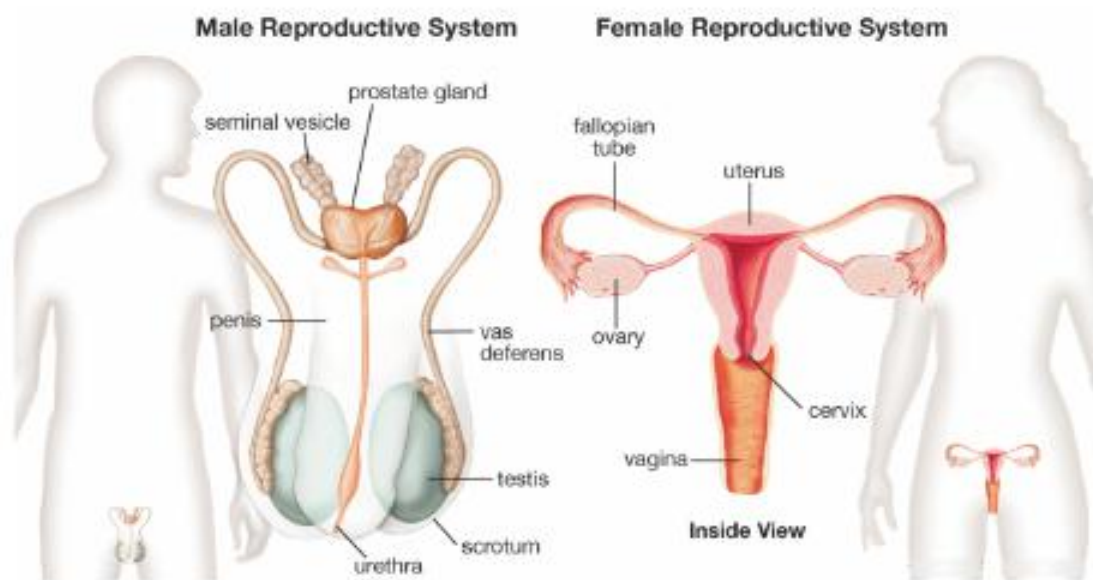
Cloning is defined as the production of an identical cell or any living organism by asexual means. First cloning was successfully done by Ian Wilmut and his colleagues at the Roslin Institute in Edinburgh, Scotland. They cloned the sheep named Dolly on July 5, 1996. Embryo cloning is the production of human embryos for use in research. The goal of this process is to make stem cells that can be used to treat diseases like cancer. Stem cells can be used to generate virtually any type of specialized cell in the human body.

SEXUAL REPRODUCTION IN HUMANS

In humans, reproduction is done most commonly by way of sexual reproduction. In this method, two parents participate in producing their offsprings, one is a male, and the other a female.

In simple words, sexual reproduction takes place by way of fusion of the reproductive cells. These reproductive cells (sex cells) are often termed as gametes. There are two types of gametes produced by both the parents. The ones produced by the reproductive organ of the male parent are known as the **male gametes** or the **sperms**, and the ones produced by the reproductive organ of the female parent are called the **female gametes** or the **ova** (egg).

Once the male gamete and the female gamete unite with each other, they form a structure called the **zygote (Fertilised egg)**. This happens when the nuclei of the sperm and the egg unite to form a single nucleus.



Just after the process of fertilisation, the zygote grows and further divides repeatedly. A ball of cells is formed due to the repeated division of the zygote. These cells later unite to form different organs and tissues in the body, and the structure so developed is termed as an embryo. Later, the embryo further develops and thus an individual comes into being.

Male Reproductive System

The sexual maturity in human males is attained around the age of 14 to 15 years.

Male reproductive system consists of the following organs:

Testis (Plural, testes)

These are two oval-shaped endocrine glands present in the males. Testes are made up of several coiled tubes that produce millions of male gametes called **sperms**. These sperms are small in size and consist of a head, a middle piece and a tail.

Vas deferens (Sperm ducts)

This is a narrow tube joining the testis and urethra. Various glands produce different fluids inside this tube, which mix with sperm and form a fluid called **semen**. The semen is carried to the urethra by the muscular action of the walls of vas deferens.

Urethra

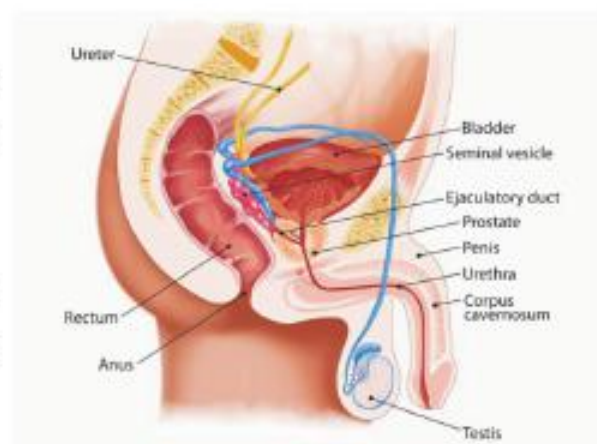
A narrow tube that runs through the penis to the outside is called urethra. The urethra, at different times, carries both semen and urine.

Sex Glands

Sex glands produce a liquid that contain nutrient for the sperms. This liquid together with the sperms form semen.

Penis

Urethra opens into a muscular structure called penis. It is penis that transfers the semen, and with it the sperms into the female body.



Female Reproductive System

Female reproductive system has the following female reproductive organs:

Ovary

Ovaries are the two almond-shaped endocrine glands present in the female body. These ovaries produce female gametes called **ova (singular, ovum)**. The process of release of egg or ovum is called **ovulation**.

Fallopian tubes (Oviducts)

These tubes are meant for carrying ova released by the ovary for fertilisation.

Uterus

The fallopian tubes open into a wide, thick walled muscular chamber called the **uterus or womb**. It is the organ inside which the embryo grows and develops.

Vagina

The uterus communicates outside through a muscular passage called **vagina**. During sexual intercourse, this body part receives the penis.

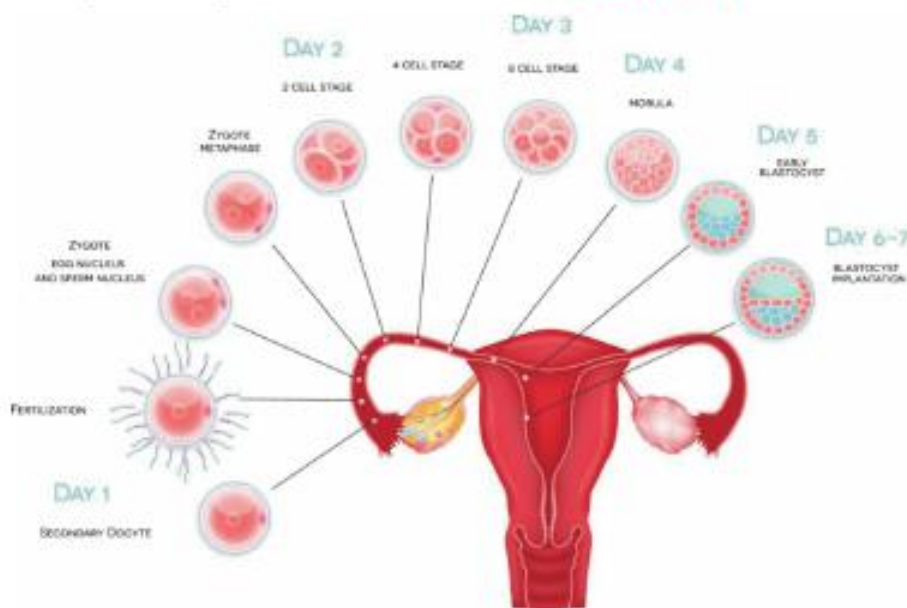
Females start producing ovum at the age of 11-12 years. At this age, they start attaining sexual maturity.



FERTILISATION

The process of fertilisation in humans take place inside the female's body in the oviduct. During this process the nuclei of the sperm and the egg fuse to form a single nucleus. As a result, the fertilised egg (zygote) is formed. The fertilised egg continues its journey down the oviduct into the utreus. The zygote implaints itself in the uterus wall and develops into human embryo.

If the ovum is not fertilised by a sperm, it is expelled from the utreus along with the lining of uterus and blood through the vagina. This is known as **menstruation**.

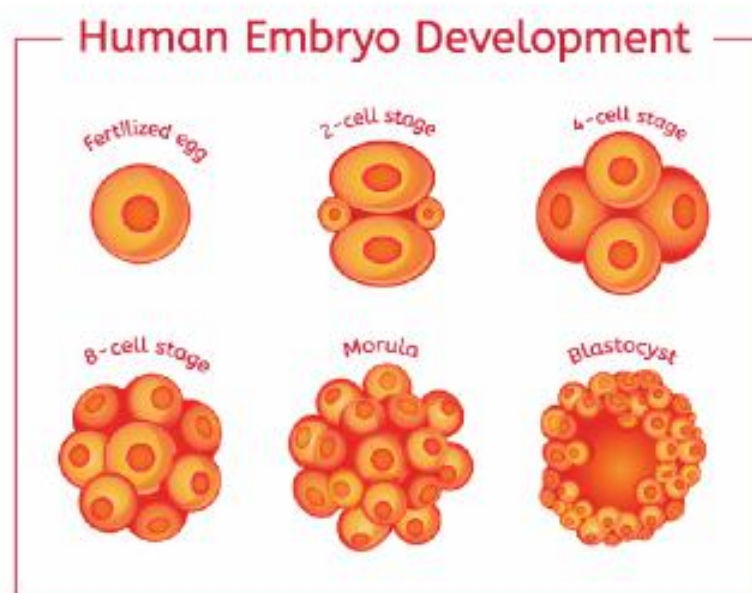


Development of an Embryo

Once the process of fertilization is over, a zygote gets formed. This zygote further develops to form an embryo. However, this happens when the zygote begins to divide repeatedly and forms different groups of cells. These groups of cells further develop and take the shape of different tissues and organs. This structure, where these developments take place, is called the embryo.

The embryo attaches itself to the wall of the uterus and grows further to form the baby.

The stage of the embryo's development in which the development and emergence of the various body parts such as the hands, the feet, the eyes, etc can be identified, is called the **foetus**.



TYPES OF FERTILISATION

Fertilisation is of two types:

1. Internal Fertilisation

In this type of fertilisation, the fusion of male and female gametes takes place inside the female body. Examples of organisms that reproduce through internal fertilisation are insects, birds and mammals like cows, cats and human beings.

2. External Fertilisation

In external fertilisation, fusion of male and female gametes takes place outside the female body. Examples of organisms that reproduce through external fertilisation are: frog, fish and starfish. Let us discuss the example of frogs.

During rainy season, frogs move to slow-flowing streams of water and the ponds. When male and female come in close contact with each other, female frog lays hundreds of eggs. Frog's eggs are very delicate and are not protected by any shell like hen's eggs. There is a jelly layer that holds these eggs together and protects them. Male frogs then deposit sperms over these eggs. The sperms swim in water with the help of a long tail and hence come in contact with the female eggs. This ultimately results in external fertilisation.

VIVIPAROUS AND OVIPAROUS ANIMALS

Animals can be divided into two types based on how they give birth to their young ones.

Animals that give birth to young ones are called **viviparous animals**. Mammals like dog, cat, horse and human beings are the examples of viviparous animals.

Animals that lay eggs and hatch them into offsprings are known as oviparous animals. The examples of oviparous animals are snakes, butterflies, frogs and birds.

Metamorphosis

The young ones of mammals and newly hatched birds look similar to their parents. However, there are some insects and animals whose young ones look quite different from the adults and undergo a series of changes after birth. The features of these young ones are completely different from the adults.



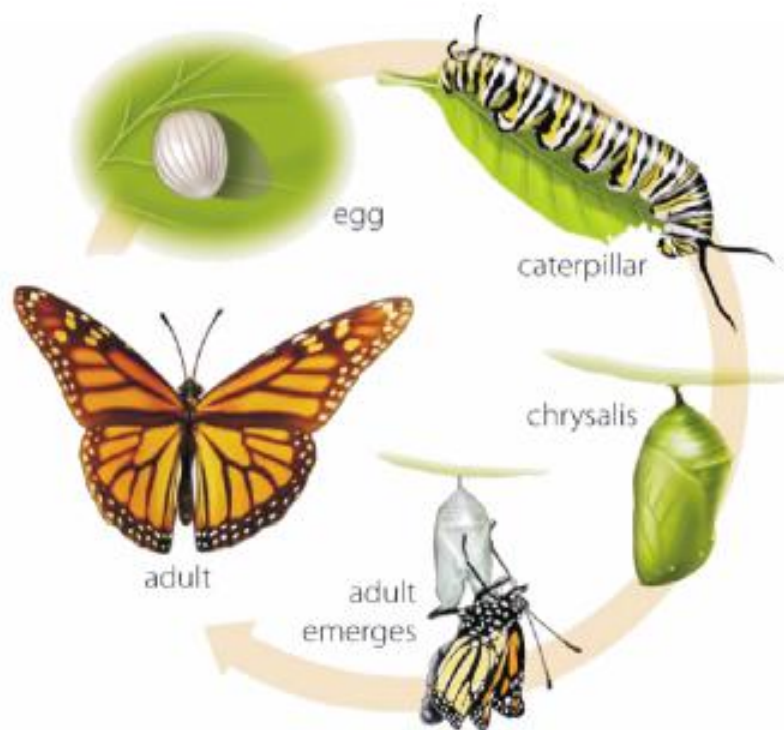
A biological process in which larva transforms into an adult involving sudden and abrupt changes in the body of the animal during the life cycle of an invertebrate or amphibian is called **metamorphosis**.

Metamorphosis in Butterfly

The life cycle of a butterfly undergoes the following stages:

Egg → Caterpillar → Pupa → Adult

The various changes that occur during the life cycle of a butterfly are shown in the figure given below.



In this case, the larva (caterpillar) and pupa (chrysalis) possess completely different features from the adult butterfly. This transformation of a larva into an adult through a series of drastic changes is called metamorphosis in butterfly.

◀ REWIND

- Two modes of reproduction are (i) Sexual reproduction and (ii) Asexual reproduction.
- In hydra, new individuals develop from buds and an amoeba reproduces by dividing itself into two.
- The reproductive organs in male include testis, sperm ducts, and penis.
- The reproductive organs in the female include ovaries, oviducts, and uterus.
- The ovaries produce female gametes called ova, and the testis produce male gametes called sperms.
- Ovum and sperm fuse together to form a fertilised egg called zygote.
- Fertilisation is of two types internal and external.
- The zygote divides repeatedly to give rise to an embryo.
- The embryo gets embedded in the wall of the uterus for further development.
- Embryo develops to form foetus, a stage when all the body parts are developed.
- The phase beginning right from fertilisation till the time a baby is born is called Gestation period. It is nine months in humans.
- When the male sperm and the female egg are fused in a laboratory, it is called Artificial Fertilisation.



ASSIGNMENTS

COGNITIVE SKILLS

1

Tick (✓) the correct option.

(Learning by Doing)

1. Internal fertilization occurs

(a) Inside the female body
(b) Inside the male body
(c) Outside the male body
2. Which of the following is a viviparous animal?

(a) Frog
(b) Duck
(c) Human being
3. The fusion of the egg with the sperm is called

(a) Fertilisation
(b) Parturition
(c) Budding
4. A tadpole develops into an adult frog by which of the following processes?

(a) Binary fission
(b) Fertilization
(c) Metamorphosis
5. The reproductive organs responsible for the release of sperms is

(a) Testis
(b) Ovary
(c) Sperm duct
6. Which one of the following structures is formed when male and female reproductive cells fuse?

(a) Zygote
(b) Sperm
(c) Ovum

2 Fill in the blanks.

1. Offsprings look _____ to parents in asexual reproduction.
2. The _____ repeatedly divides to form an embryo.
3. _____ is an oviparous animal.
4. Hydra multiplies by _____.
5. Male gamete is called _____.

3 Match the columns (Double matching).

Column A

Column B

Column C

- | | | | | |
|------------|-------|---|-----|----------------------------|
| 1. testis | (i) | part where the growth of embryo takes place | (a) | male reproductive system |
| 2. ovary | (ii) | carries both semen and urine | (b) | female reproductive system |
| 3. uterus | (iii) | sperms | (c) | female gamete |
| 4. urethra | (iv) | ova | (d) | male gamete |

4 State whether the following statements are 'True' or 'False'.

1. Each sperm is a single cellular structure.
2. An embryo is a multicellular structure.
3. Binary fission is a method of sexual reproduction.
4. Starfish reproduces by external fertilisation.
5. Gametes are formed during asexual reproduction.
6. Oviparous animals give birth to young ones.
7. A zygote is formed as a result of fertilisation.

5 Complete the following analogy.

1. Female : Ova :: Male : _____
2. Birds : Oviparous :: _____ : Viviparous
3. Hydra: Budding :: _____ : Binary fission
4. Cow: Internal fertilization :: _____ : External fertilization
5. Hydra : Asexual reproduction :: _____ : Sexual reproduction

6 Answer in a few words only.

1. Name the process of producing the young ones of one's own kind.
2. Name the process of production of an identical cell or any living organism by asexual means.
3. Name the process of the union of the male and female gametes.
4. Name two oval-shaped endocrine glands present in the males.
5. Name tube meant for carrying ova released by the ovary for fertilisation.

6. Name the method in which the male sperm and the female egg are fused in a laboratory.
7. Name the phase beginning right from fertilization till the time a baby is born.

7

Answer in one or two sentences.

1. Describe the importance of reproduction in brief.
2. Differentiate between the following terms.
(a) Zygote and embryo (b) Internal and external fertilisation
(c) Oviparous and viviparous animals (d) Asexual and sexual reproduction
3. What is meant by foetus?
4. Define the following terms:
(a) Zygote (b) Gametes (c) Ovulation
5. What do you understand by cloning?

8

Answer in one or two paragraphs.

1. Define fertilisation. What is the process of development of an embryo?
2. Define asexual reproduction. Describe the different methods of asexual reproduction in animals.
3. Explain the reproductive system of human males with the help of the labelled diagram.
4. Explain the reproductive system of a human female with the help of the labelled diagram.
5. What is metamorphosis? How does it take place in a butterfly?

NEP QUERY



COMMUNICATION SKILLS

- Conduct a group discussion with your classmate about 'The National and International efforts towards disabled person'.

(Discussion)

Friction

11

- Friction and its causes
- Way of reducing and increasing friction
- Types of Friction
- Advantages of Friction

FRICION AND ITS CAUSES

There are some forces that act on objects by direct or indirect physical contact. Friction is one such force. It is defined as a force which opposes motion. In daily life, we have to move bodies from one place to another by pushing or pulling them. Friction is a contact force that tends to stop a moving object or stops a stationary object from moving.

Friction is a very complicated phenomenon. The force of friction acts along the two surfaces in contact that opposes the motion of one body over the other. Thus, frictional force comes into play when two surfaces are in physical contact with each other.

Let us understand friction with the help of an activity.



Activity

Aim: To experience friction

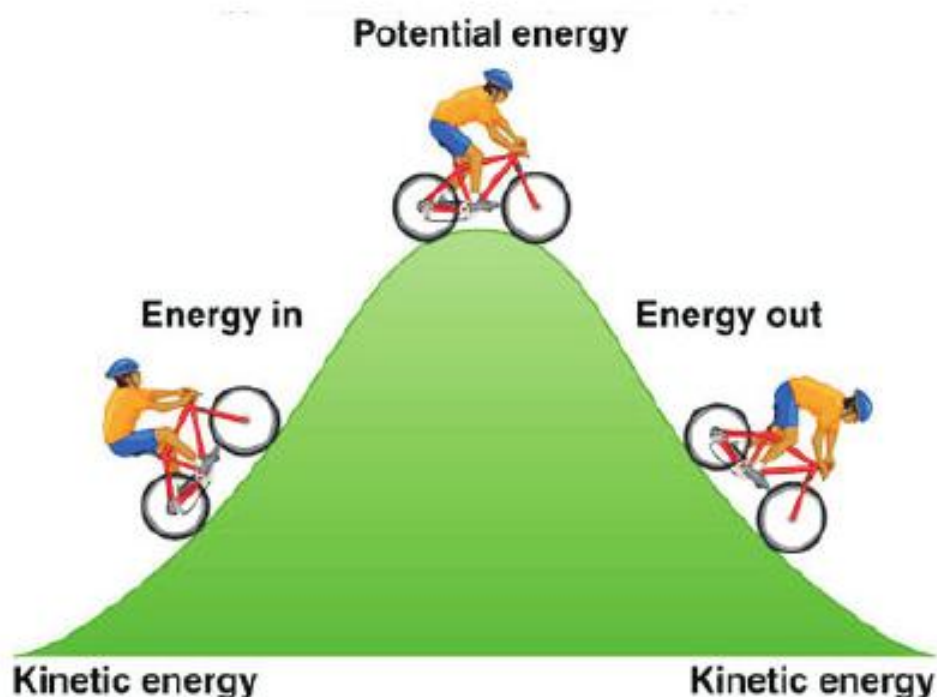
Materials required: A ball

Procedure: Take the ball and roll it on the ground

Observation: You will observe that it gradually slows down after travelling a certain distance and finally stops.

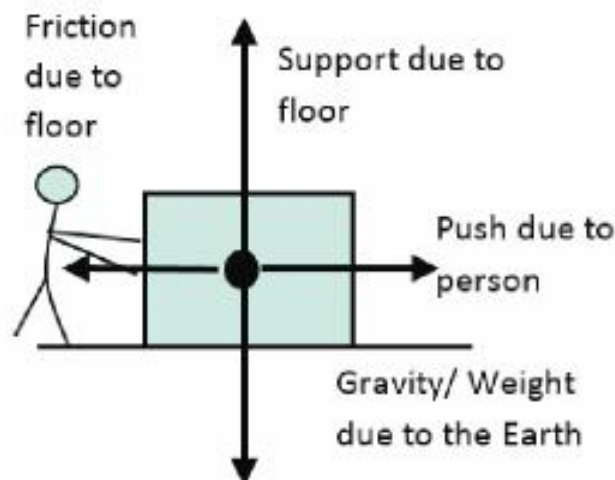
Conclusion: Again, this is due to the force of friction between the ball and the ground that opposes the motion of the ball and brings it to stop.

Similarly, if we stop pedalling a bicycle, it slows down and after travelling a certain distance, it stops because of force of friction between the bicycle tyres and the road. To start the bicycle, we have to exert a force on the pedals greater than the force of friction between the tyres and the road. Friction always acts in a direction opposite to the direction of motion.



Causes of Friction

The force of friction arises due to interlocking of the irregular projections on the two surfaces. One theory propagates that no solid surface is perfectly smooth. Any surface may appear to be very smooth but when seen under microscope, it has many irregularities. However, the degree of roughness varies from surface to surface. These irregularities are like hills and valleys. When two such surfaces slide over each other, there is resistance of motion, i.e., friction. Friction is caused due to these irregularities found on the two surfaces in contact with each other. Due to the interlocking of the irregularities of the two surfaces, friction is produced.



FACTORS AFFECTING FRICTION

The force of friction depends upon the following factors

- (a) nature of surface
- (b) weight of the body

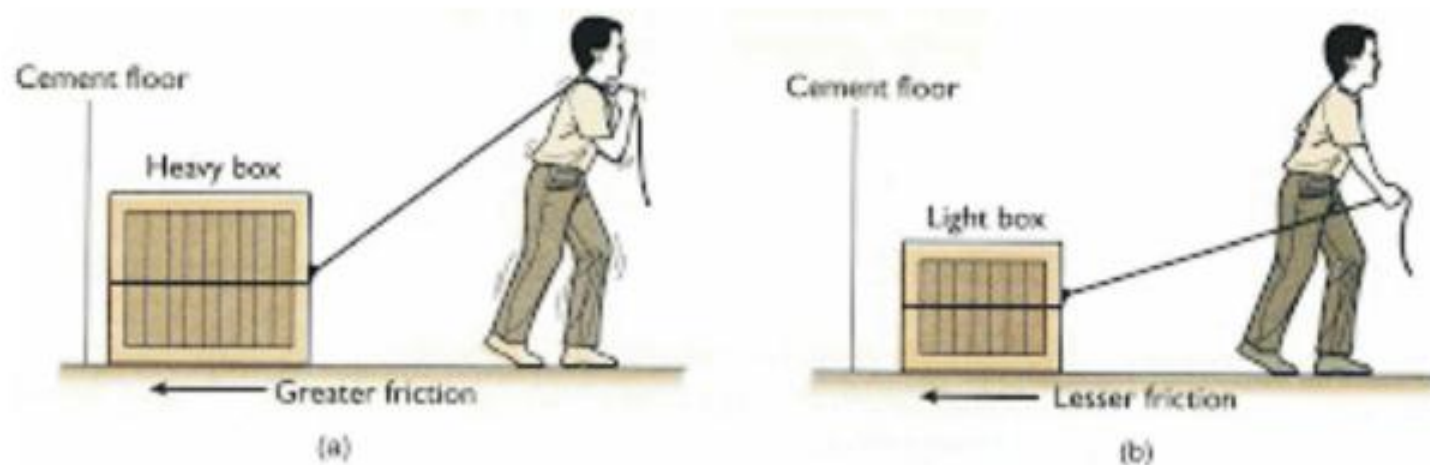
(a) Friction Depends upon the Smoothness or roughness of the Surfaces in Contact

It is a common experience that if a bicycle moves on a rough road, it covers a short distance after we stop pedalling. But if the bicycle moves on a smooth road, it covers a longer distance

before it stops. This clearly indicates that the friction is more when the surface is rough and friction is less when the surface is smooth.

(b) Friction Depends on the Weight of the Body

It is our common experience that if we push a box to move, we have to exert some force. But if we put another box on top of the previous box and push both the boxes together, we would have to exert much greater force. This shows that greater the weight of the body, the greater is the force of friction.



Nature of Frictional force

It is important to note that force of friction is self adjusting. It automatically adjusts itself to the applied force. Force of friction increases proportionately with increase in applied force. However, force of friction cannot increase infinitely. It always has a maximum limit. When the applied force is increased further, the body begins to move in the direction of the applied force.

TYPES OF FRICTION

The frictional force is of three types:

- (a) **Static friction** is defined as the friction existing between the two surfaces in contact when there is no relative motion between them.
- (b) **Sliding friction** is defined as the friction existing between two surfaces wherein one surface slides over another surface.
- (c) **Rolling friction** is defined as the friction that exists between the two surfaces wherein one object rolls over the surface of another object.

It has been proved experimentally that

Static friction > Sliding friction > Rolling friction

Friction in Liquids and Gases

Liquids and gases also exert friction on solid bodies moving in either of them. The frictional

force exerted by liquids or gases is called **drag**. Liquids exert less force of friction as compared to solids. Also, the gases exert the least force of friction as compared to solids and liquids.

Streamlined shape is one such shape around which a fluid such as the liquid or gas can flow easily. Such a shape offers the least friction. The nature has provided streamlined body to fish so that they experience minimum friction in water. Even a swimmer tries to streamline his body while swimming so as to face minimum friction from water. The boats and ships are also designed in such a way that they experience least friction while moving through water. An aeroplane also has a streamlined body so that it experiences least resistance through air.

WAYS OF REDUCING FRICTION

Sometimes, it is required to reduce friction because friction decreases the speed of a moving body, produces heat and also causes wear and tear. The following ways are helpful in reducing friction:

1. By Making the Surfaces Smooth

On polishing the surfaces, the irregularities (in the form of hills and valleys) of the surfaces are covered making them smoother. Roads are cemented to make them smooth.

2. By use of Lubricants

A lubricant is a substance that reduces friction, e.g., oil, grease oil, talcum powder, etc.

3. Using Ball Bearings

Ball bearing is a hollow circular device containing small metal balls that are fitted around the moving part of a machine. The ball bearings reduce the friction by converting sliding friction into rolling friction.

4. By Streamlining

We have also studied that bodies of boats, aeroplanes, etc are designed in a particular way to reduce friction against air and water. Such bodies are called streamlined bodies.

5. By using Rollers

Heavy objects like suitcases are provided with small rollers or wheels to reduce friction.

Ways of Increasing Friction

It can be increased in the following ways—

1. By making surfaces rough, sometimes sand is sprinkled on the surfaces to make it rough.
2. By using the dry surfaces as these have more friction while the wet surfaces have less friction.
3. Increasing the weight of a moving body increases the friction as it makes the body have a better grip on the other surface.

ADVANTAGES EFFECTS OF FRICTION (ADVANTAGES)

Most of our daily activities of life depend on the existence of friction.

1. Friction helps us to walk on roads. If there were no friction. We would slip on icy roads or polished smooth surfaces due to less or no friction.
2. Vehicles are able to move on the road due to friction. The tyres get a better grip on the road due to increased friction.
3. Friction helps to support a ladder inclined on a wall. This enables them to have better grip on the ground preventing them from slipping.
4. We are able to write on paper due to friction existing between the tip of pencil or pen and paper.
5. When we apply brakes, a vehicle stops due to friction.
6. It would not be possible to sit on a chair without friction.
7. The matchstick burns due to friction between the surface and the head of the matchstick.
8. The screws and nails are held together tightly due to force of friction.

◀ REWIND

- Friction is a force which opposes motion.
- Friction always acts in a direction opposite to the direction of motion.
- Friction is caused due to the interlocking of irregular projections on the two surfaces.
- The force of friction depends on the
 - (a) weight of the body
 - (b) kind or texture of surface
- Rolling friction is less than sliding friction.
- Friction has various disadvantages, for example, it produces heat, causes wear and tear and opposes motion.
- There are various ways to reduce friction such as streamlining, using ball bearings and lubricants or making surfaces smooth.
- Friction can be increased by increasing the weight of a moving body, using dry surfaces and making surfaces rough.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

- Friction
 - Helps in motion
 - Opposes the motion
 - Acts in the direction of motion
- Friction is reduced by making the surface
 - Smooth
 - Glossy
 - Both (a) and (b)
- Streamlining a body
 - Increases friction
 - Reduces friction
 - Makes friction zero
- Friction
 - Causes wear and tear
 - Produces heat
 - Both (a) and (b)
- Friction is increased by
 - Making the surface smooth
 - Making the surface wet
 - Increasing the weight of a moving body

2 Fill in the blanks.

- We use ball bearings to _____ friction.
- Friction _____ motion.
- Lubricants are used to _____ friction.
- Friction causes _____ of moving parts of machine.
- The frictional force exerted by liquids and gases is called _____.

3 Match the words/statements in column A with that in column B.

Column A

- Friction
- Lubricants
- Rough and dry surfaces
- Drag
- Streamlined body

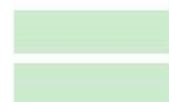
Column B

- increase friction
- frictional force exerted by fluids
- fish
- reduce friction
- opposes motion

4 State whether the following statements are 'True' or 'False'.

- Friction is a force that opposes motion.
- Force of friction depends upon the area of contact.
- Rolling friction is less than sliding friction.

4. Use of ball bearings increases friction.
5. Friction is a necessary evil.



5 Answer in a few words only.

1. Friction always acts in a direction similar to or opposite to the direction of motion.
2. Does friction depend upon the weight of body?
3. Which is more rolling friction or sliding friction?
4. Which parts of the machine gradually wear out?
5. In a streamlined body, the front portion is sharp or blunt.

6 Answer in one or two sentences.

1. "Frictional force is self adjusting". Explain it.
2. List the two factors on which the force of friction depends.
3. Why is it necessary to reduce friction sometimes?
4. Why are ball bearings used in wheels and axles?
5. Would we be able to walk if friction were absent?

7 Define the following terms.

- | | | | |
|---------------------|------------------|---------------------|--------------|
| 1. Frictional force | 2. Drag | 3. Rolling friction | 4. Lubricant |
| 5. Streamlining | 6. Ball bearings | 7. Rollers | |

8 Answer in one or two paragraphs.

1. Briefly explain the causes of friction.
2. List the advantages of friction explaining in brief.
3. Explain a few ways of reducing and increasing friction.

NEP QUERY



COMMUNICATION SKILLS

- ◆ Conduct a debate in your classroom about "Why is the invention of wheel considered as an important step of development in the history of science?" (Discussion)

COMPUTATIONAL SKILLS

- ◆ Visit a factory and gather information about how wear and tear is caused in machines and what measures are taken to keep them in good working conditions. Make a PowerPoint Presentation. (Problem Solving)

- How is a sound produced
- Structure and working of a human ear
- Characteristics of vibrations
- Noise pollution
- Characteristics of sound

Sound is defined as a form of energy that travels through a medium in the form of vibrations. When these vibrations reach a person's ears, he is able to hear the sound. Both living and non-living things produce sound. However, each sound is special to the object producing it, e.g. chirping of birds, ticking of clocks, ringing of bells, murmur of water, etc. While some sounds are pleasant, some are unpleasant. Some can be heard from a distance and some cannot be heard from a distance.

HOW IS A SOUND PRODUCED?

A sound can be produced in the following ways:

1. By beating: a drum or tabla or hitting a hammer.
2. By rubbing: playing a violin or brushing a broom across the floor.
3. By blowing: playing a flute or trumpet or whistling of a pressure cooker.
4. By plucking: playing a guitar or sitar or when an arrow is twanged.

Thus, sound is produced when something moves. This movement that produces sound in rapid to and fro motion is called **vibration**. Thus, sound is a form of vibration. The vibrating body disturbs air and produces sound waves. When the object stops vibrating, we do not hear the sound.

CHARACTERISTIC OF VIBRATIONS

A vibration has three important characteristics. They are frequency, amplitude and time period. They determine the characteristics of the sound produced.

1. Frequency

It is defined as the number of vibrations made by a vibrating body in one second. Its unit is **hertz**, e.g. if a body makes 600 vibrations in one second, its frequency is 600 Hz.

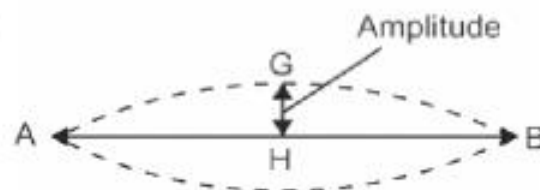
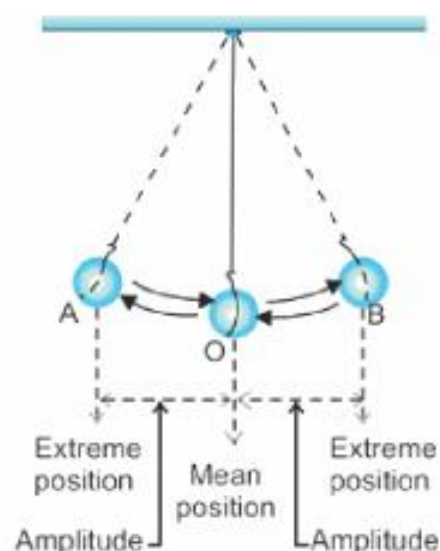


fig 12.1

2. Amplitude

It is defined as the maximum extent to which a vibrating body can vibrate. For example: in the given figure (fig 12.1) if AB is the vibrating string, GH is the amplitude of vibration.



3. Time period

It is defined as the time taken by a vibrating body for one complete vibration.

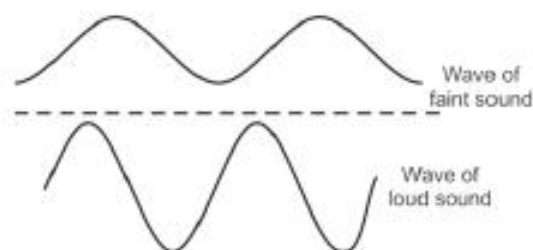
CHARACTERISTICS OF SOUND

Sound has three important characteristics. These are loudness, pitch and quality.

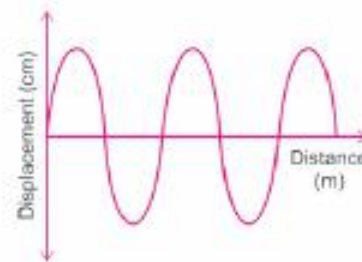
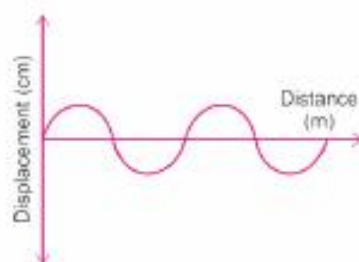
1. Loudness

It is a characteristic of sound that enables us to distinguish a loud sound from low sound. We hear different sounds such as the horn of a vehicle, chirping of birds, the sound of different musical instruments, barking of a dog, etc. We can easily recognise the voice of our friend or relative, because they have different loudness. The loudness of a sound depends on three factors amplitude, area of the vibrating body and distance of the listener from the source of sound.

(a) Amplitude : The greater the amplitude, the louder is the sound.



Similarly, if you strike the drum gently, a faint sound is heard. But if you strike it hard, a loud sound is heard. Loudness of sound is directly proportional to the square of the amplitude of the vibration producing the sound. If amplitude doubles, the loudness becomes four times.

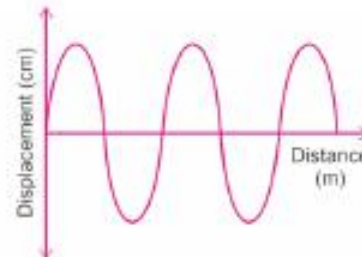
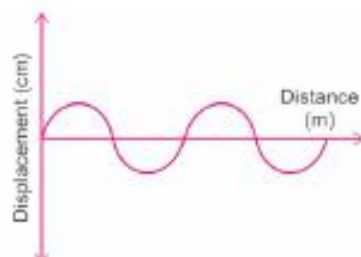


(b) Area of vibrating body : The more the area, the louder is the sound, e.g., a big drum produces louder sound than a smaller drum.

(c) Distance between the listener and the source of sound : The nearer is the listener to the source of sound, the louder he can hear the sound.

2. Pitch

It is a characteristic of sound that enables us to distinguish a shrill sound. The pitch of a sound depends on the frequency, a sound of high frequency has a high pitch while the sound of low frequency has a low pitch e.g., the voice of a girl is shriller than that of a boy because it is of a higher pitch. The voice of a child is shriller than that of an adult.



3. Quality

It is defined as the characteristic of sound by virtue of which we can differentiate between the two sounds of same pitch and loudness produced by two different sources, e.g. the sound produced by veena and the sound produced by a flute. We can recognise people even on the telephone by their voice as each one has a voice with a particular quality.

Audible sound

We can hear the sounds of frequencies in the range from 20 hertz to 20,000 hertz. We cannot hear sounds of frequencies below 20 Hz and above 20,000 Hz. The range of frequencies from 20 Hz to 20,000 Hz is called the **audible range** for the normal human ear.

Ultrasonic and Infrasonics

Sounds of frequencies higher than 20,000 Hz are called the **ultrasonic**. Some animals like dogs, bats, monkeys, deer, leopards, etc can hear ultrasonic sound.

Sounds of frequency lower than 20 Hz are called the **infrasonic**. We cannot hear infrasonic. The pendulum of a clock makes one vibration in 2 seconds. It is an infrasonic sound which we cannot hear.

STRUCTURE AND WORKING OF A HUMAN EAR

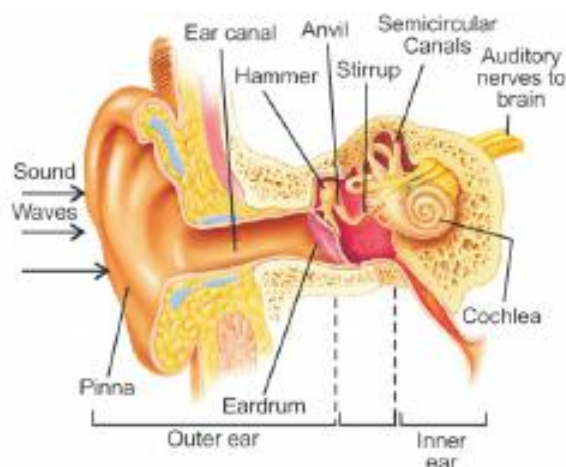
The ear is the sense organ for hearing. The sound waves reach our ears and we hear the sound. The three parts of the ear are outer ear, middle ear and inner ear.

The outer ear, consists of pinna, ear canal and ear drum.

The middle ear consists of a cavity with three interlocked bones namely hammer, anvil and stirrup.

The inner ear consists of a semi-circular organ of hearing called **cochlea**. It is filled with fluid and has many hair cells inside it.

The auditory nerve is also present in the inner ear.



Working of the Human Ear

1. Pinna collects the vibrations from the surroundings and sends them to the ear canal.
2. The vibrations reach the ear drum present at the end of the ear canal. The ear drum starts vibrating as soon as the vibrations strike it.
3. The three bones of the middle ear also start vibrating with the ear drum. These three bones amplify the vibrations received from the ear drum and send them to the inner ear.
4. The cochlea (in the inner ear) receives the amplified vibrations and converts them into nerve impulses. The tiny hair and semi-circular canals get stimulated due to these vibrations. As a result, the nerve impulses are sent to the brain via the auditory nerve present in the inner ear.
5. The brain decodes the nerve impulses and we are able to hear the sound.

Medium of Propagation of Sound

Sound cannot travel through a vacuum. It needs a medium to travel such as the solid, liquid or gas.

Speed of Sound

The speed of sound is maximum in solids, less in liquids and least in gases. You must have noticed that in the rainy season, the sound of the thunder is heard much later than its lightning is seen. The reason is that the light travels much faster than the sound.

NOISE POLLUTION

It is defined as too much noise in our surroundings that causes discomfort and disturbance of any kind.

The loudness of sound is measured in decibels (dB).

A sound of about 90 dB and above produced for a long time causes noise pollution.

Noise pollution causes various health hazards such as : headache, deafness, earache and many more.

Ways to control noise pollution

1. Automobiles should be fitted with silencers and soft horns. Do not blow horns unnecessarily.
2. Use of loudspeakers should be banned.
3. The music system, radio and television should be played at a low volume.
4. Avoid bursting crackers as it causes not only noise pollution but also air pollution.
5. Machines should be well maintained so as to produce minimum noise and people working in noisy conditions should wear ear plugs.
6. Airports, factories and railway stations should be away from the residential areas.

◀ REWIND

- Sound is produced due to vibrations.
- Vibrations have three important features, they are frequency, amplitude and time period.
- A human ear can hear sounds with frequency between 20 Hz and 20,000 Hz.
- Sound needs a medium such as solid, liquid or gas to travel. It cannot travel through a vacuum.
- Sound waves travel the fastest in solids, comparatively slower in liquids and the slowest in gases.
- Sudden and irregular vibrations produce noise whereas regular sounds produce music.
- Sudden unwanted noise in the environment that causes discomfort is called noise pollution. It has various ill effects and hence, it must be controlled.

🔍 ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

- | | | | |
|---|--------------------------|---------------------|--------------------------|
| 1. The maximum displacement of the vibrating body from its resting position is called its | | | |
| (a) Frequency | ☐ | (b) Amplitude | ☐ |
| | | (c) Pitch | ☐ |
| 2. We can distinguish a shrill sound from a flat sound by | | | |
| (a) Amplitude | ☐ | (b) Loudness | ☐ |
| | | (c) Pitch | ☐ |
| 3. We can hear sounds of frequency range between | | | |
| (a) 20 to 20,000 Hz | ☐ | (b) 50 to 50,000 Hz | ☐ |
| | | (c) 50 to 100 Hz | ☐ |
| 4. The three bones hammer, anvil and stirrup are present in the | | | |
| (a) Outer ear | ☐ | (b) Inner ear | ☐ |
| | | (c) Middle ear | ☐ |

5. Sound cannot travel in

(a) Solids

(b) Liquids

(c) Vacuum

2 Fill in the blanks.

1. Too much of noise and disturbance in the environment causes _____.
2. The sound waves travel the _____ in gases.
3. Sound is produced when a body _____.
4. Veena, guitar and violin are _____ instruments.
5. The most important source of sound in humans is _____.

3 State whether the following statements are 'True' or 'False'.

1. The loudness of sound depends on the frequency.
2. We cannot hear the sound of frequency 50,000 Hz.
3. Sound travels the fastest in vacuum.
4. Sudden and irregular vibrations produce music.
5. Sound is a form of energy.

4 Match the words/statements in column A with the column B.

Column A

1. Amplitude
2. Pitch
3. Frequency
4. Noise
5. Tabla

Column B

- (a) hertz
- (b) unpleasant sound
- (c) percussion instrument
- (d) frequency
- (e) loudness

5 Define the following terms.

1. Sound
2. Vibrations
3. Audible sound
4. Noise pollution
5. Loudness of sound

6 Differentiate between.

1. Noise and music
2. Shrill sound and hoarse sound
3. Pinna and cochlea
4. Ultrasonics and infrasonics

7 Answer in a few words only.

1. In which form does the sound travel?
2. Can we always see the vibrations?
3. How can the lower vibrations be produced?

4. Which of these will produce a louder sound a big drum or a small drum?
5. Who has the longer vocal cords men or women?

8 Answer in one or two sentences.

1. List three characteristics of sound.
2. What is the role of the pinna?
3. How is noise produced?

9 Answer in one or two paragraphs.

1. List the various ways in which sound can be produced.
2. Give the characteristics of vibrations explaining each one in brief.
3. Explain the different characteristics of sound.
4. Explain the functioning of a human ear with a labelled diagram.
5. Explain the effects and ways to minimise noise pollution.

NEP QUERY



OBSERVING SKILLS

- Visit a music shop nearby and find out about the different musical instruments available in that shop. (Emotional Thinking)

COMMUNICATION SKILLS

- Conduct a group discussion 'What is hearing loss' in your class. (Discussion)

PROJECT WORK

- Make a collage on the topic 'Causes and effects of noise pollution'. (Learning by Experiment)

COMPUTATIONAL SKILLS

- Prepare a PowerPoint Presentation (PPT) on the topic 'Echo and its uses'. (Problem Solving)

Chemical Effects of Electric Current

13

- Conduct of electric current through liquids
- Electroplating
- Electrolysis

All substances are made up of atoms as an atom is the smallest unit of a substance. An atom is further constituted of electrons, protons and neutrons.

In normal conditions, an atom is neutral, i.e. it does not possess any type of net charge in it. This is because electrons are negatively charged, whereas protons are positively charged. The number of electrons and protons is equal in an atom, so there is no net charge in an atom.

If number of electrons exceeds the number of protons, the substance will acquire a negative charge and if protons become more than electrons, the substance will become positively charged. If these two substances are connected in the presence of a driving force, electrons will flow from the negatively charged body to the positively charged body, and an electric current will flow from the positive end towards the negative end.

The substances that allow the electric current to pass through them are called **conductors**, while those which do not allow the electric current to pass through them are called **insulators**.

Electricity can produce various types of effects. For example: when it flows through a bulb, it produces light and on passing through a heater, it produces heat. Similarly, if an electric current brings about a chemical change in a substance, it is called **chemical effect of electric current**.

CONDUCTION OF ELECTRIC CURRENT THROUGH LIQUIDS

Like solids, liquids may also conduct electricity through them. But all the liquids are not conductors. The current passing through liquids is comparatively weak, so we can replace the bulb with LED (Light Emitting Diode), because a bulb requires more current to glow. LED lights up in the presence of even a small amount of current. Two wires are connected to LED, one wire is longer than the other and is always connected to the positive end, whereas the smaller one to the negative end.

Pure water is a bad conductor of electricity. That's why distilled water does not conduct electricity. Tap water has some salts dissolved in it. That's why it conducts electricity. Acids and bases are also capable of conduction of electricity. That's why lemon juice and soap solution also conduct

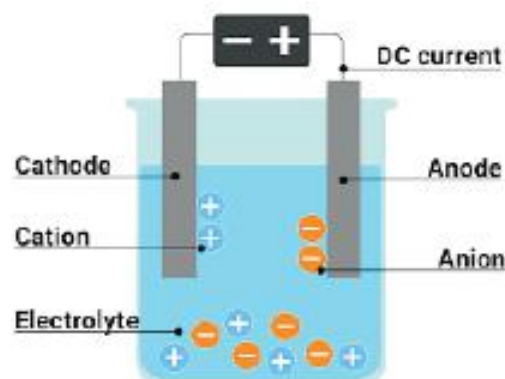
electricity and make the LED glow. But you will observe that sugar solution is not capable of conducting current while salt solution is. This is because on passing electricity salt dissociates into charged ions while sugar does not.

FACT

Salts cannot conduct electricity in a solid state. But when these are dissolved in water or some other solvent, they break up into their ions which are capable of conducting an electric current.

ELECTROLYSIS

The word 'electrolysis' is made up of two words: **electro** + **lysis**, lysis means 'breaking down'. So, electrolysis means a process of breaking down of a substance by passing an electric current through it in its dissolved or molten state. The dissolved or molten state of a substance that is being decomposed is called electrolyte. The container in which electrolysis is brought about is called an electrolytic cell. An electrolytic cell consists of two electrodes and an electrolyte. An electrode is a conductor that allows electric current to pass from one medium to another using the graphite or metal plates or rods. Let's perform electrolysis of an acidified water.



The positively charged electrode is known as anode and the negatively charged electrode is called **cathode**. Positive ions, i.e. **cation** move from anode towards negative electrode, i.e. cathode and negative ions, i.e. **anions** move from cathode towards positive electrode, i.e. **anode**. So, in electrolysis of water, H_2 gas is found near cathode and O_2 near anode.

As, in a water molecule, i.e. H_2O , the amount of hydrogen is double than oxygen the bubbles observed around cathode are more than that around anode.

Any substance that is made from ions can be electrolysed in this way.

Some electrolytes and ions obtained by their decomposition are as follows:

S. No.	Name of electrolyte	Formula	On cathode	On anode
1.	Sodium chloride	$NaCl$	Na^+	Cl^-
2.	Copper sulphate	$CuSO_4$	Cu^{++}	SO_4^{--}
3.	Silver chloride	$AgCl$	Ag^+	Cl^-
4.	Zinc nitrate	$Zn(NO_3)_2$	Zn^{++}	NO_3^-

5.	Potassium Iodide	KI	K^+	I^-
6.	Calcium hydroxide	$Ca(OH)_2$	Ca^{++}	OH^-

ELECTROPLATING

Electroplating is one of the various applications of electrolysis. In this method, a metal object is coated with a thin layer of another metal.

The process of coating of a metal object with another metal with the help of electrolysis is called **electroplating**.

The process of electroplating is done by using an electrolytic cell (Voltameter). The object to be electroplated is used as cathode (Negative electrode) and connected to the negative terminal of a battery or D.C. source. The metal object that is to be coated/deposited over is used as anode (Positive electrode) and it is connected to the positive terminal of the battery. The electrolyte used for electroplating should be a water soluble salt of the metal (Current conductive material) to be coated/deposited. For example, if a zinc plate is to be electroplated with copper metal, we use copper as an anode and the zinc plate as cathode. The electrolyte used in this case, is copper sulphate solution. We can perform electroplating easily in our lab.

Uses of Electroplating

1. Electroplating is used to protect the objects of more reactive metals from rusting. For example: iron objects are coated with less reactive zinc to prevent them from rusting.
2. Chromium is also electroplated over car parts, taps, handles, etc as it corrodes less and also give the objects an attractive appearance.
3. Containers used to store food are made up of iron. As iron is a reactive metal and can spoil the food by reacting with it, they are coated with tin which is comparatively a less reactive metal.
4. Gold and silver are very expensive metals, but they do not corrode easily and also look beautiful. So, the jewellery made up of other metals and alloys such as brass is electroplated with gold and silver to make it attractive and affordable.

◀ REWIND

- Some substances like metals can conduct electricity through them. They are called conductors.
- Some substances cannot conduct electricity through them. They are called insulators.
- Most liquids are conductors of electricity due to the presence of ions in them.
- An LED is more efficient device than a bulb. It requires less current to glow.
- Ions are formed due to addition or elimination of electrons from the atoms.

- When a substance is broken down into its constituents by passing electricity through it, the process is called electrolysis.
- In electrolysis of a substance, cations get deposited on cathode and anions on anode.
- Water can be electrolysed to H_2 and O_2 gases.
- Electroplating is an application of electrolysis.
- A thin layer of a metal is deposited over an object of another metal by electrolysis to make it more attractive and protected. This is known as electroplating.



ASSIGNMENTS

COGNITIVE SKILLS

1

Tick (✓) the correct option.

(Learning by Doing)

- An atom is

(a) Positively charged	☐	(b) Negatively charged	☐	(c) Neutral	☐
------------------------	--------------------------	------------------------	--------------------------	-------------	--------------------------
- An ion is

(a) Positively charged particle	☐	(b) Negatively charged particle	☐		☐
(c) Either (a) or (b)					☐
- Which of the following is a conductor of electric current?

(a) Wood	☐	(b) Iron	☐	(c) Plastic	☐
----------	--------------------------	----------	--------------------------	-------------	--------------------------
- What can be the effect of electricity?

(a) Production of light	☐	(b) Production of heat	☐		☐
(c) Both of these					☐
- Which of the following ions will deposit on cathode?

(i) Ca^{++}		(ii) Cl		(iii) K^+	
(a) All (i), (ii), (iii),	☐	(b) (i) and (ii)	☐	(c) (i) and (iii)	☐
- Which of the following metals is used in electroplating?

(a) Iron	☐	(b) Tin	☐	(c) Sodium	☐
----------	--------------------------	---------	--------------------------	------------	--------------------------

2

Fill in the blanks.

- Metal objects are _____ to prevent them from corrosion.
- Metals are the _____ of electricity.
- During electroplating, the object to be electroplated is always made _____.
- Water, on electrolysis produces _____ and _____.
- Sodium and chloride ions will be obtained by the electrolysis of _____.

- Longer wire of LED is always connected to _____ terminal.
- Electrons flow from _____ to _____.

3

Recognise the following.

- I am an electrode, positive ions gather around me.
- I am a substance which is broken down by electricity.
- I am a charged species, negative ions get attracted towards me.
- I am very liberal, I allow electricity to pass through me.
- I am a metal that is used to make anode if electroplating of silver is to be done.

4

State whether the following statements are 'True' or 'False'.

- Cathode is a positively charged electrode.
- Cation is positively charged ion.
- Electricity flows from negative to positive terminal.
- Iron is plated over tin.
- An atom is neutral in normal conditions.
- LED needs comparatively less current to glow.

5

Answer in a few words only.

- Which form of water conducts electricity, pure or impure?
- Why do acids conduct electricity?
- What is the full form of LED?
- Which of the following will conduct electricity?
tap water, distilled water, vinegar, sugar solution, salt solution
- What name is given to the process of breaking down of a substance through electricity?
- Which charges do the cathode and anode possess?
- An electrolyte produces K^+ and I^- ions on electrolysis. Write the formula of electrolyte.
- Name an application of electrolysis.

6

Answer in one or two sentences.

- Define electrolysis.
- What are conductors and insulators?
- What is electroplating?
- Why and how are iron objects prevented from corrosion?
- Why are we advised not to touch any electrical appliances with wet hands?
- Why are more bubbles seen near cathode in electrolysis of water?
- Define electrolyte and electrolytic cell.
- Write the equations of electrolysis of water.

9. Give two uses of electroplating.
10. How does a metal get deposited over other during electroplating?

7

Answer in one or two paragraphs.

1. Explain the process of electrolysis of sodium chloride solution. Also draw a diagram.
2. Explain how you will electroplate an iron rod with silver.

NEP QUERY



CRITICAL THINKING

- ♦ Eight words from the chapter are hidden in the following word mesh. Locate them vertically and horizontally.

(Self-Awareness)

A	C	T	O	T	M	U	O	E	L	O	M
E	B	C	R	E	N	C	P	Y	Z	P	R
C	A	U	M	C	C	A	N	I	O	N	N
N	N	R	A	A	H	T	R	Q	T	O	S
P	D	R	N	R	Q	H	H	I	V	E	D
E	L	E	C	T	R	O	L	Y	S	I	S
S	E	N	A	N	O	D	E	N	T	C	V
T	D	T	T	P	A	E	Q	O	M	P	N
U	R	Y	I	T	E	R	T	G	U	V	D
R	E	C	O	N	D	U	C	T	O	R	S
V	Q	U	N	G	R	U	S	Q	C	B	L

COMMUNICATION SKILLS

- ♦ "Fuse wire is a safety device or not". Conduct a group discussion on this topic.

(Discussion)

PROJECT WORK

- ♦ Find out various uses of electrolysis. Collect pictures regarding it. Paste the pictures and write about each in your scrapbook.

(Learning by Experiment)

COMPUTATIONAL SKILLS

- ♦ Make a PowerPoint Presentation (PPT) about electroplating.

(Problem Solving)

- Reflection of light
- The human eye
- Formation of an Image
- Visually Challenged People
- Dispersion of light

We have studied about light in the previous classes. We are familiar with the importance of light. We can see objects around us when there is light. Nothing can be seen in the dark. If we enter a dark room, we will not be able to see anything till we switch on torch or light a candle. In daylight, we do not need anything else to see objects around us.

Thus, light is a form of energy. Light itself is not visible. However, in the presence of light, other objects become visible. When light reaches our eyes after striking an object, we are able to see an object. Most of the objects reflect the light falling on them. When this reflected light reaches our eyes, we are able to see the object. Light may therefore be defined as the external physical cause that affects the eye to produce the sensation of vision.

REFLECTION OF LIGHT

If you throw a ball on the ground, you will notice that it rebounds after striking the ground. Similarly, when the light falls on a shiny surface such as the mirror, it returns back into the same medium after striking the surface.

The bouncing back of light into the same medium after it falls on a surface is called **reflection**.

Let us understand the process of reflection. When the light falls on a surface, a portion of it is sent back into the same medium and the rest is either absorbed or transmitted by the medium. The surface which sends back the light falling on it into the same medium is called a **mirror**. A mirror has shining smooth surface due to which a greater part of light is thrown back.

Different objects reflect light to different extent. When a light falls on an object, a part of light is reflected. If the object is opaque, the remaining part of light is absorbed by the object. If the object is transparent, the remaining part of light passes through the object. Thus, all objects reflect light but the extent may vary.

Terms Related to Reflection of Light

1. **Incident ray**: The ray of light striking the surface is called an **incident ray**.

2. **Point of incidence:** The point on the surface at which the incident ray strikes is called the point of incidence.
3. **Reflected ray:** The ray of light which is returned back into the same medium after the incident ray strikes the surface is called the **reflected ray**.
4. **Normal:** The perpendicular drawn to the reflecting surface at the point of incidence is called the normal.
5. **Angle of incidence:** The angle between the incident ray and the normal is called the angle of incidence. It is represented by the letter '*i*'.
6. **Angle of reflection:** The angle between the reflected ray and the normal is called the angle of reflection. It is represented by the letter '*r*'.

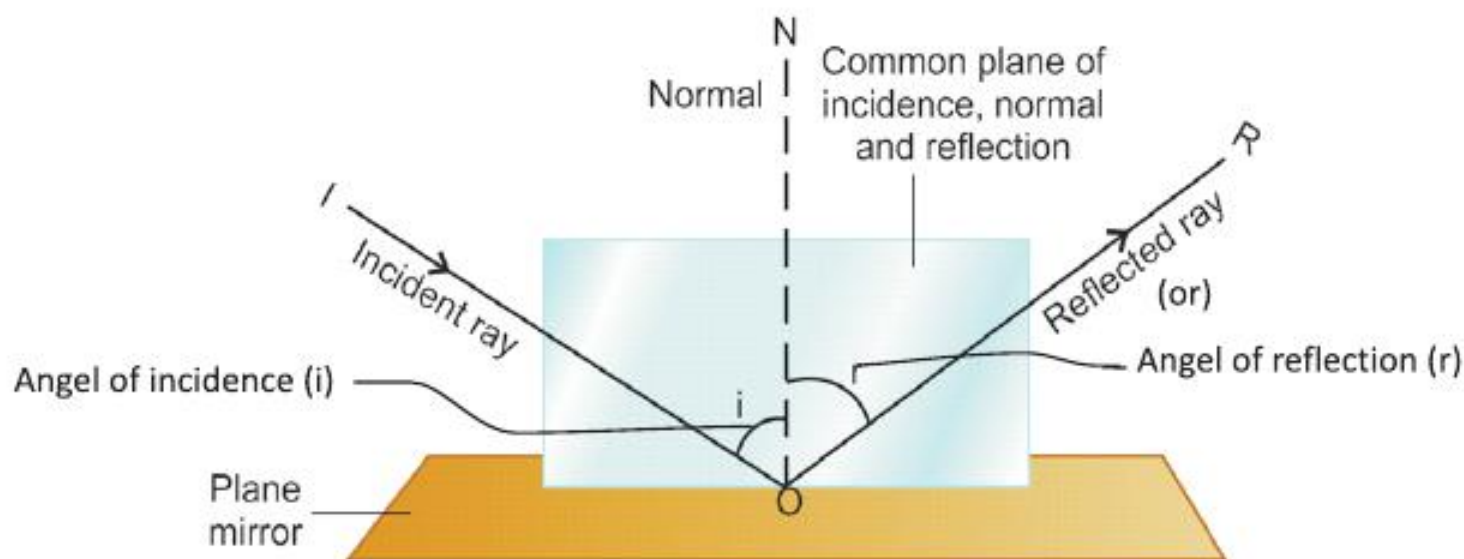
The figure given below shows the labelled diagram of reflection of light from a plane mirror. MM' is the plane mirror, IO is the incident ray, O is the point of incidence, OR the reflected ray. ON is the normal, ION is the angle of incidence, RON is the angle of reflection.

Laws of Reflection

The reflection of light ray at a surface obeys the following two laws of reflection:

1. The angle of incidence is equal to the angle of reflection.

$$\angle \text{ION} = \angle \text{RON} \quad \text{or} \quad \angle i = \angle r$$

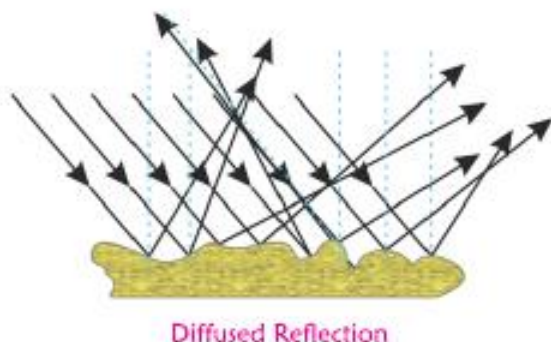
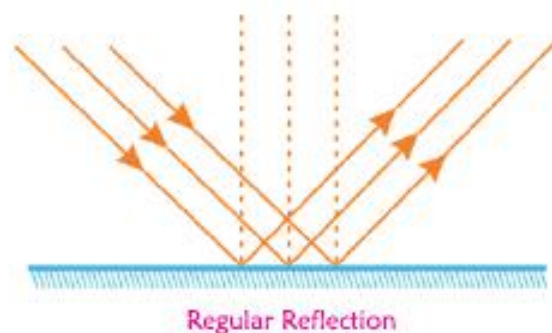


2. The incident ray, the reflected ray and the normal all lie in the same plane.

Types of Reflection

There are two types of reflection (A) regular and (B) diffused. The type of reflection depends upon the nature of the reflecting surface.

(A) **Regular reflection:** It is defined as the reflection of light from a smooth and polished surface such as the plane mirror.



(B) **Diffused or irregular reflection:** It is defined as the reflection of light from an uneven surface. When a beam of light falls on a surface that is not smooth and polished such as a wall, wood, paper, etc; the different portions of the surface reflect light in different directions.

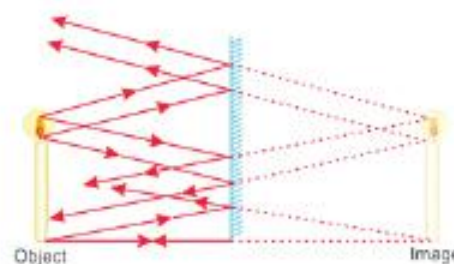
Table 16.1: Differences between regular reflection and diffused reflection

S.No.	Points of difference	Regular reflection	Diffused reflection
1.	Types of reflecting surface	Smooth, highly polished and shiny.	Rough and irregular like a wall, wood, etc.
2.	Reflected rays	The reflected beam of light is parallel.	The reflected beam of light is not parallel.
3.	Direction of reflected rays	All the reflected rays are in the same direction.	All the reflected rays spread in different directions.
4.	Image	Image is formed.	Image is not formed.

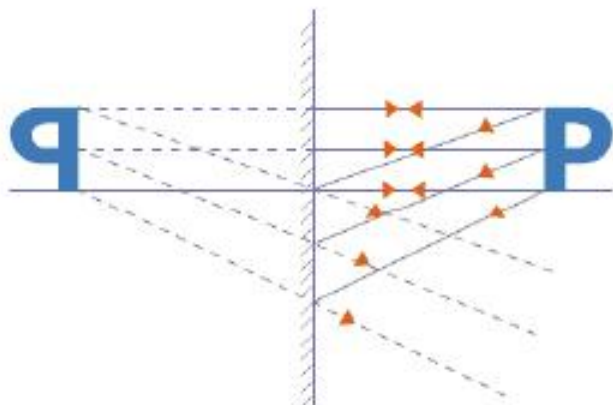
FORMATION OF AN IMAGE

We have already studied about the image formed by a plane mirror. Let us quickly recall the characteristics of an image formed in a plane mirror.

1. It is upright or erect.
2. It is virtual.
3. It is of same size as the object.
4. It is as far behind the mirror as the object is in front of it.



5. It is laterally inverted that is there is interchange of left and right sides in the object and image, e.g. if you hold a playcard with the word 'DEN' written on it and stand in front of the mirror, you will find that all the letters of the word 'DEN' is in reverse order. It is also because of the phenomenon of lateral inversion.



Kaleidoscope

It is a device based on the principle of multiple reflections. It consists of three strips of mirror M_1 , M_2 and M_3 inclined at an angle of 60° to each other and enclosed in a cylindrical tube. One end of this tube is closed by means of a piece of milky translucent paper while the other end is closed by a piece of cardboard with a hole at its centre. Several bits of multicoloured glass pieces are loosely enclosed between the three inclined mirrors.

Beautiful coloured patterns are seen when viewed through the hole of the cardboard along the axis of the tube. If we turn the tube, the glass pieces arrange differently and we see new designs.

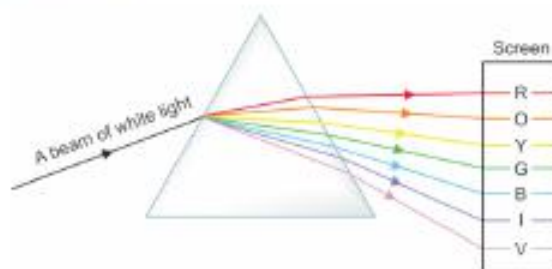
Refraction of Light

Refraction occurs because light travels at different speeds through different transparent medium. Light travels at 3×10^8 m/s in vacuum or air. But when it pass through a medium like glass or water, it slows down.

Due to change in speed its path changes and so it bends. The phenomenon of bending of light when it travels from one medium to another is called refraction of light.

DISPERSION OF LIGHT

It is defined as the phenomenon of splitting of white light into different colours on passing through a transparent medium such as a prism. Sir Issac Newton was the first scientist to study that if a white light is passed through a prism, it splits into different colours. Newton allowed the sunlight to enter a dark room



through a small hole in a window and placed a prism in the path of light.

He received the light coming out of the prism on a white screen. He found the screen to be coloured with the seven colours of rainbow. He saw seven colours in the following order (from below):

Violet, Indigo, Blue, Green, Yellow, Orange and Red

(V) (I) (B) (G) (Y) (O) (R)

The order of colours can be remembered by the acronym VIBGYOR. This coloured band is called the **spectrum of white light**. Thus, spectrum is the coloured band obtained on a screen on passing the white light through a prism. From this experiment, Newton concluded that the white light is a mixture of seven colours. The prism does not produce colours but simply separates the colours that already exist in the white light.

Newton also argued that seven colours could produce white light if they could be somehow recombined. He demonstrated it by the following experiment.

He took a circular disc of cardboard and painted on it the seven colours (V, I, B, G, Y, O, R) in sequence. On rotating the disc at a high speed, it appeared white.

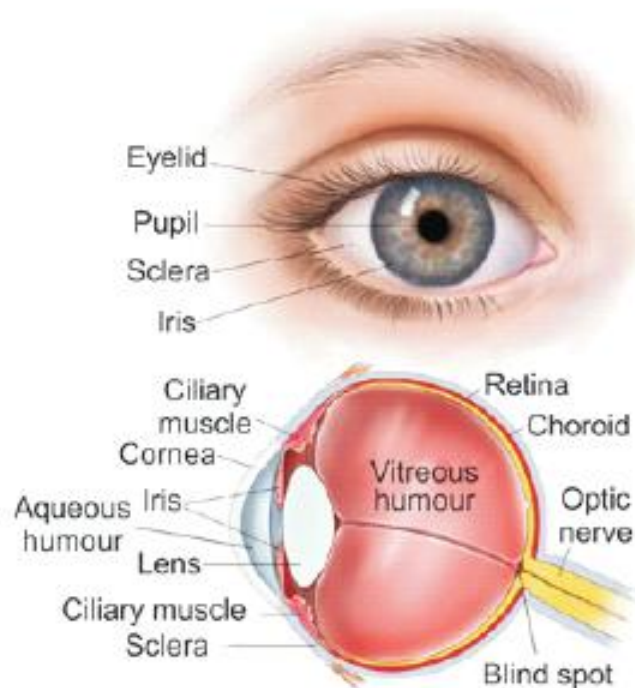
The rainbow that appears in the sky just after the rain when the sun shines is due to dispersion of light.

THE HUMAN EYE

The human eye is an important sense organ that enables us to see the world around us. It is nearly spherical in shape with a slight bulge in the front part.

The bulging transparent part is called the **cornea**. It protects the eye.

Behind the cornea is a diaphragm called the **iris**. The iris has a hole at the centre called the **pupil**. The pupil allows the light to enter the eye whereas the iris controls the amount of light entering the eye. Iris makes the size of the pupil smaller or bigger. In dim light, the size of pupil increases to allow more light to enter the eye. In bright light, the size of the pupil decreases so as to reduce the amount of light entering the eye.



Behind the iris, there is convex lens called the **eye lens**. It helps to focus the light passing through the pupil into the eye. This helps to focus the image on retina, by bending the light rays.

The lens is held in position by a pair of **ciliary muscles**. They change the focal length of the eye lens.

The space between the cornea and lens is filled with a liquid called the **aqueous humour**.

The inner back surface of the eye is called the **retina**. It serves as the screen where image is formed by the eye lens. It is sensitive to light and has receptors called rods and cones. Both these respond to light and generate impulses that the brain is able to read. Then the brain sends the message back to us telling us what we have seen.

The space between the lens and the retina is filled with a liquid called the **vitreous humour**.

The sensation of the image formed on the retina is carried to the brain by the optic nerve. Thus, it connects the eye to the brain.

When we look at the object, the eye lens forms an inverted, real, diminished image on the retina. The optic nerve gives the sensation of the image to the brain as an erect image.

The white part of the eye that we can see is called the **sclera**. It is filled with clear watery fluid.

Blind Spot (Scotoma)

It is a portion on the retina where the nerve fibres enter the optic nerve. There are no rods and cones present in this area. Hence, the images falling on this portion cannot be seen. This spot is called the **blind spot**.

Power of Accommodation

The human eye is able to see the distant as well as the nearby objects. The ability to focus the eye on the nearby and distant objects is called the **power of accommodation**.

The image of a distant object is formed on the retina when the ciliary muscles are completely relaxed. The focal length of eye lens in this condition is more. But the image of a nearby object is formed on the retina when the ciliary muscles contract and decrease the focal length of the eye lens.

Near Point and Least Distance of Distinct Vision

There is a limit to the power of accommodation of the eye. An object placed very close to the eye is not clearly visible. The point nearest to the eye at which an object is clearly visible is called the **near point of the eye**. The distance of the near point from the eye is called the **least distance of distinct vision**. For a normal eye, the least distance of distinct vision is 25 cm.

Defects of Vision

A normal eye can see all the objects situated at a distance greater than 25 cm because of its power of accommodation. When the eye is unable to accommodate for different distances, it is said to have some defects. The most common defects of the eye are—

Short-sightedness or Myopia

A person suffering from myopia can see only the nearby objects clearly but cannot see the distant objects clearly.

This defect can be corrected by using the spectacles having concave lenses.

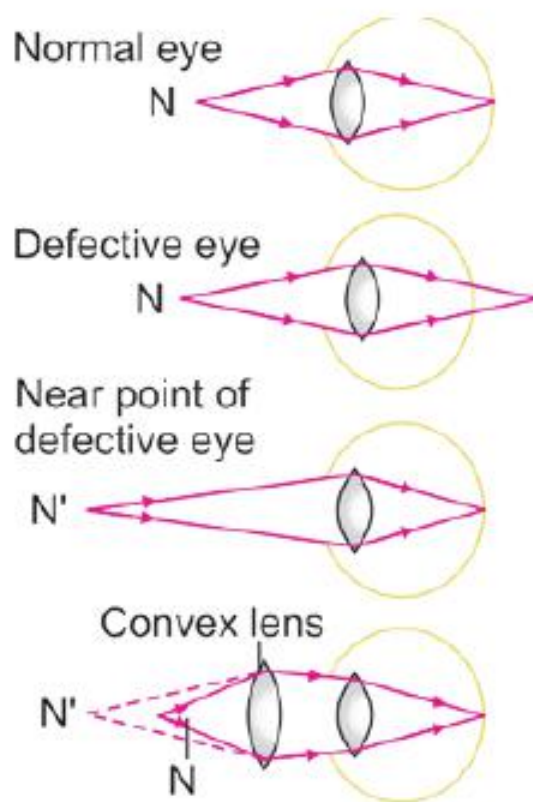
Long-sightedness or Hypermetropia

A person suffering from hypermetropia can see only the distant objects clearly but cannot see the nearby objects clearly.

This defect can be corrected by using the spectacles having convex lenses.

Cataract

It is a condition wherein the lens becomes cloudy or opaque. Due to this, the vision becomes dim or blurred. It is very common among elderly people. It is because the light rays cannot pass as the lens is cloudy. This can be corrected by surgical operation. The affected eye lens is removed and a new eye lens is implanted in its place.



VISUALLY CHALLENGED PEOPLE

These are the people who are unable to see, that is, they are blind. Blindness may be congenital (by birth) or acquired (during lifetime). Acquired blindness can occur due to cataract, glaucoma or some accident or injury.

Visually challenged people make good use of other senses to cope with their problem.

Facilities for the Blind

1. **Braille System:** It is a system available for the blind to enable them to read. It is a system of representing characters by raised dots. Both letters and numerals are represented in this way. This technique was invented by Louis Braille, who himself became blind in his childhood. Various books and periodicals are available for the blind.
2. **Cane:** The blind people use a white cane with a red tip. This helps them to detect obstacles

on the way and they can move independently.

3. **Other devices:** These include gadgets like talking computer, calculator, clock and other devices. These days personalised blind-friendly computers are available. They have special keyboards based on the Braille system. Special scanners and printers are also available.

◀ REWIND

- Light is defined as an external physical cause that affects the eye to produce sensation of vision.
- The bouncing back of light into the same medium after it falls on a surface is called reflection.
- Reflection is of two types regular and diffused.
- Laws of reflection are as follows :
 - (a) The angle of incidence is equal to the angle of reflection.
 - (b) The incident ray, reflected ray and normal all lie in the same plane.
- The image formed by a plane mirror is virtual, erect, same size as that of the object and laterally inverted.
- Multiple images are formed when two plane mirrors are kept at a certain angle.
- Kaleidoscope is a device based on the principle of multiple reflections.
- Dispersion is defined as the phenomenon of splitting up of white light into different colours on passing through a transparent medium.
- The human eye is an important sense organ for sight.
- We must take good care of eyes.
- Myopia and hypermetropia are two defects of vision. Cataract occurs due to age and can be rectified.
- People who are unable to see are called the visually challenged. There are many facilities available for them these days.

🔍 ASSIGNMENTS !

COGNITIVE SKILLS

1 Tick (✓) the correct option.

(Learning by Doing)

1. Light is a form of

(a) Heat

(b) Energy

(c) Mass

2. Angle of incidence is _____ angle of reflection.

(a) Equal to

(b) Not equal to

(c) More than

3. The image formed in the plane mirror is
 (a) Virtual (b) Erect (c) both (a) & (b)
4. In a human eye, the image is formed on
 (a) Pupil (b) Cornea (c) Retina
5. For a normal eye, the least distance of distinct vision is
 (a) 15 cm (b) 20 cm (c) 25 cm

2 Fill in the blanks.

1. A defect of vision occurring commonly in elderly people is _____.
2. On passing white light through a prism, the coloured band obtained is called the _____.
3. The receptors present on retina are called the _____ and _____.
4. For _____ reflection to take place, smooth and highly polished surface is required.
5. The space between cornea and lens is filled with a liquid called the _____.

3 Match the words/statements in column A with the column B.

Column A

1. Normal
2. Diffused reflection
3. Two plane mirrors kept side by side
4. Coloured disc
5. Ciliary muscles

Column B

- (a) one image
- (b) Newton
- (c) range the focal length
- (d) uneven surface
- (e) perpendicular drawn to reflecting surface

4 State whether the following statements are 'True' or 'False'.

1. Braille system is available for the blind.
2. A person suffering from hypermetropia cannot see distant objects clearly.
3. The image is formed on the retina of the eye.
4. The lens carries the electrical signals to the brain.
5. Kaleidoscope is a device based on multiple reflections.

5 Define the following terms.

1. Braille
2. Spectrum
3. Blind spot
4. Reflection
5. Light

6 Differentiate between

1. Incident ray and reflected ray
2. Persistence of vision and power of accommodation
3. Regular reflection and irregular reflection
4. Near point of the eye and least distance of distinct vision

5. Myopia and hypermetropia

7

Answer in a few words only.

1. Name the person who invented the technique of Braille.
2. Name the defect of eye because of which a person can see only nearby objects clearly.
3. Name the portion of the retina where the nerve fibres enter the optic nerve.
4. Name a transparent medium that enables us to get a spectrum on screen.
5. Name the phenomenon in which there is interchange of the left and the right side.

8

Answer in one or two sentences.

1. How is a cane useful for the blind?
2. What is near point of the eye?
3. What is sclera?
4. How are we able to see the objects?
5. State the two laws of reflection.

9

Answer in one or two paragraphs.

1. Draw a labelled diagram of reflection of light clearly defining all the terms related to it.
2. Explain the type of image formed in a plane mirror.
3. Explain how multiple reflections are obtained.
4. Explain the working of an eye. Also list down the ways of taking care of eyes properly.
5. Write a note on the defects of vision explaining each with a diagram.

NEP QUERY



CRITICAL THINKING

- The thin oil films on water appear coloured. Explain why.

(Self-Awareness)

PROJECT WORK

- Deliver a speech in a seminar on the topic 'Blind, yet successful'. Collect information about people who were born blind or became blind due to illness/accident yet gained fame due to their notable work in various fields.

(Learning by Experiment)

COMPUTATIONAL SKILLS

- Make a power point presentation (PPT) on the topic 'Organs of humans'.

(Problem Solving)

Pollution of Air and Water

15

- Air pollution and its effects
- The Ganga Action Plan

- Water pollution
- Portable water

AIR POLLUTION AND ITS EFFECTS

Our atmosphere contains complex mixture of gases that are essential for supporting life on the earth. The earth's atmosphere contains 78% of nitrogen, 21% of oxygen and the remaining is occupied by various other gases such as carbon dioxide, argon, ozone, water vapours, etc. All these gases play a vital role in the process of photosynthesis and nitrogen fixation. Air pollution is defined as the accumulation of harmful substances or increase in the quantity of any gas in the air beyond the normal level, which affects the human life and other living matter.

The gaseous air pollutants include carbon monoxide, oxides of nitrogen and sulphur, fumes of acid, etc. The solid pollutants include soot (the small unburnt particles released on burning of fossil fuels), dust particles, lead, asbestos, etc.

Sources of Air Pollution

There are two main sources of air pollution.

1. Natural sources of air pollution
2. Man-made sources of air pollution

1. Natural Sources of Air Pollution

Some of the natural sources contributing to air pollution are as follows:

- Dust, especially from the areas of very less vegetation is the main factor causing air pollution.
- Emission of methane by animals like cattle while digesting their food, also adds to air pollution.
- Emission of Radon gas within the earth's crust in radioactive decay.
- Emission of sulphur, chlorine and ash particulates from volcanic activity.

2. Man-made Sources of Air Pollution

Following are the human driven activities causing air pollution:

- Burning fuels like wood, cowdung cake, coal, kerosene etc release smoke, soot, CO_2 , CO, SO_2

etc which is the major factor causing air pollution.

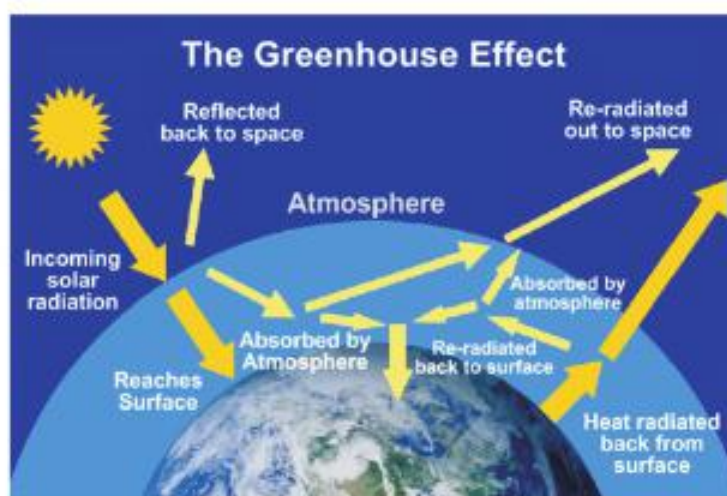
- Exhaust gases emitted by motor vehicles pollute the air.
- Emission of chlorofluorocarbons (CFCs) from the refrigerators and air-conditioners adds to air pollution.
- Smoking also emits many harmful substances like carbon monoxide and nicotine in the atmosphere.
- Nuclear power plants pollute air by releasing radioactive radiations
- Deforestation pollutes air by increasing the amount of carbon dioxide in it.

Harmful Effects of Air Pollution

It has adverse effects on Environment.

Global Warming

The sun's rays strike the earth's surface making it warm. When the sun's rays fall on the surface of the earth, a part of them gets reflected and a part of the radiations get absorbed. This heat is trapped on the earth's surface by the greenhouse gases such as carbon dioxide and methane. These trapped radiations further warm the earth. This phenomenon of trapping of the sun's radiation by the gases present on the earth's surface is called the **greenhouse effect**.



The accumulation of greenhouse gases such as carbon dioxide, carbon monoxide, etc has increased in the atmosphere due to air pollution over the years. This in turn has increased the average temperature of the earth leading to the corresponding changes in the climate. This phenomenon is called **global warming**.

Global warming has become a major concern worldwide and is causing many adverse effects on the climate. Rising of the sea level is one of its results. In many areas such as the coastal areas, floods are often encountered due to the imbalance caused in the climate. It also affects rainfall, agriculture, forests, etc.

Depletion of Ozone Layer

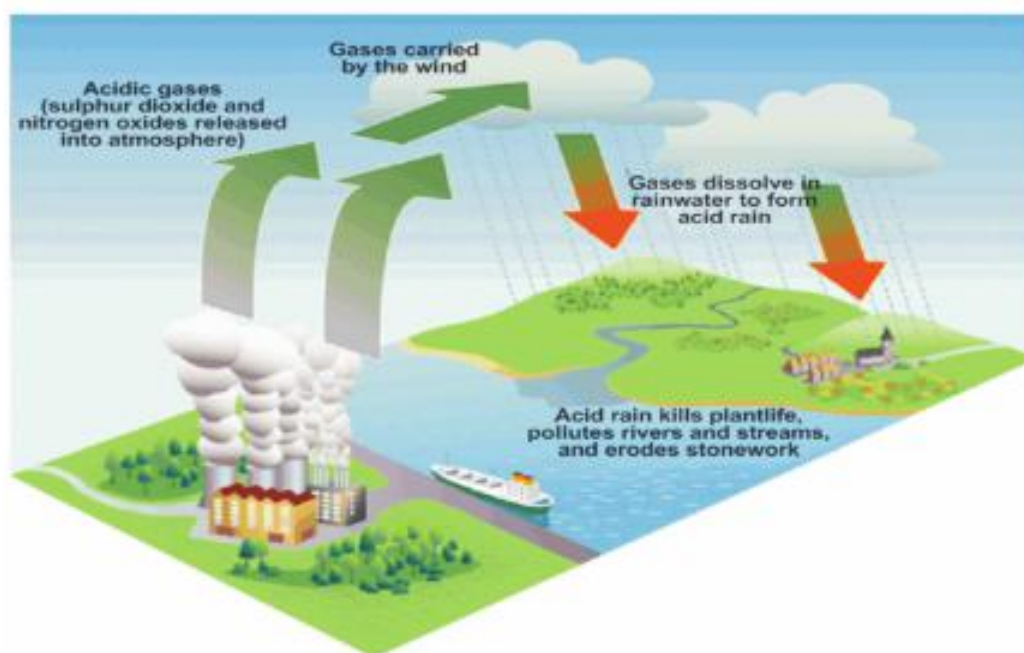
Ozone (O₃) layer can be found at 15 to 35 km above the earth's surface in the lower portion of stratosphere (second major layer of the earth's atmosphere). The ozone layer is like a blanket

on the earth's atmosphere as it prevents us from the harmful ultraviolet rays of the sun. In September 2004, a hole in the ozone layer was discovered right above Antarctica.

The continuous emission of chlorofluorocarbon (CFCs) is majorly responsible for the depletion of the ozone layer. They are also used as propellants in aerosol spray cans.

Acid Rain

It is caused by the oxides of nitrogen (NO , NO_2) emitted from automobiles and sulphur (SO_2) emitted from burning of fossils. They mix with the falling rain and result in the formation of nitric acid and sulphuric acid respectively. These acids come down with rain, making the rainwater acidic. Such rain is called **acid rain**. It causes extensive harm to buildings and monuments. The world famous Taj Mahal is losing its colour due to acid rain.



How to Prevent Air Pollution?

Various strategies that can be implemented to minimise air pollution are as under:

- More and more trees should be planted.
- Use of automobiles should be lessened.
- Use of eco-friendly fuels like CNG (Compressed Natural Gas) should be encouraged.
- The number of industrial units should be reduced to minimum.
- Use of renewable sources of energy like wind, solar and hydropower energy should be encouraged.

WATER POLLUTION

Water is a precious resource necessary for every living being. Rivers, springs, lakes, oceans, seas and ground water are the main sources of water. It is the basic requirement of various human

activities. The contamination of water bodies with unwanted and harmful substances is called **water pollution**. These days, almost all the water bodies in India are polluted and it has become a matter of concern for everybody.

Sources of Water Pollution

1. Natural calamities like tsunamis, earthquakes, storms, volcanic eruptions, etc also lead to water pollution.
2. The discharge of chemical wastes like paints, leather, rubber, textile dyes, paper and drugs from industries into water bodies also leads to water pollution.
3. The leakage of oil during shipping and drilling operations leads to water pollution as the surface of water gets covered with a layer of oil.
4. Fertilisers and manure present in agricultural runoffs pollute the nearby water bodies.
5. The after-washing residue containing detergent solution causes a lot of pollution in the water bodies in which it gets discharged.
6. Certain bio-wastes include sewage, animal excreta, dead bodies and many other harmful things lying in water bodies and untreated wastes from various work houses such as hospitals, food processing industries, etc.

Effects of Water Pollution

Threats to Aquatic life

- Industrial waste water contains poisonous chemicals like fluorides, alkalis, acids, etc. These chemicals are very harmful to aquatic life.
- Oil leakage is one of the factors behind the loss of marine life as it leads to lesser supply of oxygen and sunlight to the aquatic animals.
- Water cleaning processes implemented by using heavy machines at times cause great damage to the aquatic life.
- Agricultural runoffs containing fertilisers act as source of nutrients to weeds and plants such as algae present in the water bodies. These weeds consume most of the dissolved oxygen and become a hindrance for the sunlight to enter the water bodies. Further, when these plants die, more oxygen is utilised by the bacteria to decompose the dead bodies leading to the decrease in oxygen content in water.

This phenomenon of loss of oxygen from water due to excessive growth of plants in the water bodies is known as **eutrophication**. Eutrophication leads to the oxygen starving conditions of the aquatic life.

Threats to Human Beings

- Consumption of polluted water causes many waterborne diseases like dysentery, cholera and so on.
- Pesticides from agriculture runoffs leads to ill effects on the nervous systems of the human beings.
- Consumption of water contaminated with mercury and other chemical compounds causes blurred vision, mental disorders, numbness of lips and limbs.

Prevention of Water Pollution

- Sewage is first to be treated in specifically designed sewage treatment plants. Then only should it be allowed to enter the water bodies.
- Waste water from industries should be properly treated before being discharged into the water bodies.
- Eco-friendly pesticides and fertilisers should be used which don't cause water pollution.
- Garbage should not be thrown in drains. Activities like bathing and washing of clothes should be avoided near the water bodies.

THE GANGA ACTION PLAN

The Ganga is one of the most sacred rivers of India. But it has been polluted for many years. People living near the Ganga river throw large quantities of garbage dead bodies, flowers, untreated sewage, polythene etc into it. This had made the Ganga one of the ten most endangered rivers in the world. To prevent river the Ganga from pollution, the Ganga Action Plan was launched in India in 1985.

The main aim of this project is to reduce the pollution level in the river.

This plan aims at —

(i) Solid waste management, (ii) Installation of sewage treatment plant, (iii) installation and development of river front, (iv) Provision of low cost sanitary facilities, (v) and To create public awareness.

POTABLE WATER

Water suitable for drinking is known as potable water only 0.01% of total water available on the earth is fit for human use. Potable water should be transparent, colourless and odourless. It contain enough quantity of dissolved oxygen and salt. It should be free from microbes and harmful chemicals.

Techniques for Purifying Water at Home

- (i) **Filtration** : Passing water containing suspended impurities through a fine muslin cloth

or substances having fine pores allowing only water to escape (pass) through it, is called **filtration**.

(ii) **Boiling** : Boiling water for 10 to 15 minutes, make it free from germs.

(iii) **Reverse Osmosis** is the method of water purification in which water is passed through a semipermeable membrane by applying pressure. This technique is used in bottled drinking water.

(iv) Chemicals like potassium permanganate or bleaching power (chlorine) are often used to purify ground water.

Purification of Water in Water Treatment Plants

Before supplying water to our homes, water is purified in water treatment plants. It included three steps :-

(i) **Sedimentation** : In this process, impure water is allowed to stand undisturbed for a few hours in settling tank, as a result insoluble impurities such as mud and sand settles down at the bottom of the tank as sediments.

(ii) **Filtration** : Water from sedimentation tank is allowed to pass through filter beds which further removes insoluble impurities from water. The filter beds contain different layers of charcoal, gravels and sand.

(iii) **Disinfection** : After filtration, water is treated with chlorine gas which makes it free from germs and safe for human consumption.

Now, this water is supplied to our homes through pipelines.

◀ REWIND

- Pollution leads to the degradation of the earth's environment and its inhabitants.
- The gaseous air pollutants include carbon monoxide, oxides of nitrogen and sulphur, fumes of acid, etc. The solid pollutants include, dust particles, lead, asbestos, etc.
- Natural sources and man-made sources are the two main sources of air pollution.
- Air pollution causes adverse effects on the plants, animals and environment.
- Some of the major sources of water pollution are natural sources, industrial wastes, agricultural runoffs, release of synthetic detergents and the presence of oxygen demanding bio-wastes in the water bodies.
- Water pollution adversely affects the aquatic life in many ways.
- Some of the ways to purify water at home are boiling and filtration, reverse osmosis, disinfection and chemical treatment.
- Three main steps are followed during the purification of water at water treatment plants. They are sedimentation, filtration and disinfection.



ASSIGNMENTS

COGNITIVE SKILLS

1

Tick (✓) the correct option.

(Learning by Doing)

- What is the percentage of nitrogen present in the atmosphere?
 (a) 78%  (b) 21%  (c) 5% 
- Ozone layer saves us from
 (a) Infrared radiations  (b) Visible radiations  (c) Ultraviolet radiations 
- Which of the following gases is a greenhouse gas?
 (a) Carbon dioxide  (b) Oxygen  (c) Nitrogen 
- Which of the following gases is released within the earth's crust in radioactive decay?
 (a) Methane  (b) Nitrogen  (c) Radon 
- Which of the following things release chlorofluorocarbon?
 (a) Refrigerator  (b) Industrial waste  (c) Agriculture runoffs 
- Which of the following are harmful to human beings?
 (a) The Suspended Particulate Matter (SPM) 
 (b) Chlorofluorocarbon (CFC) 
 (c) Both (a) and (b) 

2

Fill in the blanks by choosing the correct alternative from the brackets.

- Agents or substances that pollute the water or air are called _____.
- Smoking releases a harmful substance called _____ in the atmosphere.
- The Ganga Action Plan was launched in _____.
- In reverse osmosis, water is passed through a _____ membrane.
- _____ leads to the oxygen starving conditions of the aquatic life.
- The process of removal of insoluble impurities by passing the water through the filter bed is called _____.

3

Match the words/statements in column A with the column B.

Column A

- Eutrophication
- Disinfection
- Sedimentation
- Potable water
- Soot

Column B

- small unburnt particles released on burning of fossil fuels
- oxygen starvation condition of aquatic life
- water safe for drinking
- done to make water free from germs
- settling of impurities at the bottom of the tank

4 State whether the following statements are 'True' or 'False'.

1. The ozone layer is like a blanket on the earth's atmosphere.
2. Asbestos causes air pollution.
3. Greenhouse effect is always harmful for the earth.
4. Compressed natural gas is an eco-friendly fuel.
5. Chlorine is preferred more to ozone for disinfecting water.

5 Answer in a few words only.

1. Name two solid and two gaseous pollutants.
2. Name two diseases caused by air pollution.
3. What percentage of total water present on the earth is fit for human use?
4. Name the plan launched by the government for controlling pollution in the river Ganga.
5. In which layer of atmosphere, ozone (O_3) is found?
6. Name two greenhouse gases.
7. Write full form of CNG.

6 Answer in one or two sentences.

1. Define the following terms.
(a) Pollution (b) Eutrophication (c) Greenhouse effect (d) Chlorination
2. Describe the process of occurrence of acid rain in brief.
3. What do you mean by SPM? Why are they considered harmful for human health?
4. What is potable water? Name the techniques for purifying water.
5. Define Global Warming. What are its harmful effects?

7 Answer in one or two paragraphs.

1. What is air pollution? What are the major sources of air pollution?
2. Describe the effects of air pollution on plants and human beings in detail.
3. Write a note on 'yellowing of the Taj Mahal'.
4. Water pollution is a threat to life. Explain.
5. Explain how polluted water affects the aquatic life?
6. How is the river Ganga being affected by water pollution? What steps have been taken to tackle this problem?

NEP QUERY



COMPUTATIONAL SKILLS

- Divide the area where you live in different sectors industrial sector, residential sector and commercial sector. Now, do a survey of all these areas and note down the kinds of pollution that you observe in these sectors. Find out the reasons behind the pollution in all these sectors.

(Problem Solving)



1 Tick (✓) the correct one :

1. Which of the following microorganism causes influenza?
(a) Bacteria (b) Virus (c) protozoa
2. Rhizobium lines ins the roots of
(a) Barley (b) gram (c) mango
3. Thy symbol of Chlorine is
(a) C (b) Cl (c) Cr
4. Royan is obtained from
(a) Wood pulp (b) Plastic (c) Plants
5. Natural gas mainly consists of
(a) Methane (b) Ethane (c) Propane

2 Fill in the blanks :

1. _____ is used for the large scale production of alcohol, wine and beer.
2. _____ is a traditional method of irrigation.
3. Fossil fuels are _____ natural resources.
4. Synthetic fibres do not absorb _____ .
5. _____ does not give flame an combustion.

3 Write T for true and F for false :

1. Sugar is used as a food preservative.
2. Gram is a rabi crop.
3. Anything that burns is a fuel.
4. CO₂ is required for burning.
5. New cells are formed on their own.

4 Answer the following questions :

1. How does a vaccine work?
2. Name the implements used for broadcasting.
3. Define malleability.
4. Explain the properties of plastics.
5. What is calorific value?
6. How is coal formed.
7. What do you understand by cell division?



1 Tick (✓) the correct one :

1. In which year was forest conservation act passed?
(a) 1986 ☐ (b) 1980 ☐ (c) 1991 ☐
2. The standard unit of pressure is
(a) Newton ☐ (b) Pascal ☐ (c) Hertz ☐
3. Which of the following is a viviparous animal?
(a) Frog ☐ (b) Duck ☐ (c) Human being ☐
4. Streamlining a body
(a) increases friction ☐ (b) reduces friction ☐ (c) makes friction zero ☐
5. Sound cannot travel in
(a) Solids ☐ (b) Liquids ☐ (c) Vacuum ☐

2 Fill in the blanks :

1. _____ look like plants but do not have chlorophyll in them.
2. Hydra multiplies by _____.
3. Friction _____ motion.
4. Sound is produced when a body _____.
5. Electron flow from _____ to _____.

3 Define the following :

- | | |
|-----------------|------------------------|
| 1. Ecosystem | 6. Contact force |
| 2. Biodiversity | 7. Drag |
| 3. Zygote | 8. Rollers |
| 4. Ovulation | 9. Pollution |
| 5. Force | 10. Green house effect |

4 Answer the following questions :

1. Name a few national parks and the animals they protect.
2. What do you understand by cloning?
3. How is ink filled in the fountain pen?
4. Briefly explain the causes of friction.
5. What is the role of pinna?
6. What is the full form of LED?
7. Differentiate between myopia and hypermetropia.
8. Water pollution is a threat to life? Explain.